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SURFACE VEHICLE STANDARD

J2945™/1	APR2020

Issued 2016-03
Revised 2020-04

Superseding J2945/1 MAR2016

On-Board System Requirements for V2V Safety Communications

RATIONALE

The SAE J2945/1 standard provides the functional and performance requirements for V2V safety communications between light vehicles based on the basic safety message. The purpose of the revision is to incorporate clarifications and corrections discovered through implementation and validation testing.

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1. SCOPE

This standard specifies the system requirements for an on-board vehicle-to-vehicle (V2V) safety communications system for light vehicles[1], including standards profiles, functional requirements, and performance requirements. The system is capable of transmitting and receiving the SAE J2735-defined basic safety message (BSM) [1] over a dedicated short range communications (DSRC) wireless communications link as defined in the Institute of Electrical and Electronics Engineers (IEEE) 1609 suite and IEEE 802.11 standards [2] to [6].

1.1 Purpose

This standard addresses the on-board system needs for ensuring that the exchange of BSMs in V2V safety communications provides the desired interoperability and data integrity to support the performance of the envisioned safety applications.

2. REFERENCES

2.1 Applicable Documents

The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the latest issue of SAE publications shall apply.

2.1.1 SAE Publications

Available from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, Tel: 877-606-7323 (inside USA and Canada) or +1 724-776-4970 (outside USA), www.sae.org.

[1] SAE J2735 Dedicated Short Range Communications (DSRC) Message Set Dictionary

2.1.2 IEEE Publications

Available from IEEE Operations Center, 445 and 501 Hoes Lane, Piscataway, NJ 08854-4141, Tel: 732-981-0060, www.ieee.org.

[2] IEEE Std 802.11-2012 Standard for LAN/MAN - Specific Requirements Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications

[3] IEEE Std 1609.2-2016 IEEE Standard for Wireless Access in Vehicular Environments - Security Services for Applications and Management Messages (as amended by IEEE Std 1609.2a™-2017 and IEEE Std 1609.2b™-2019)

[4] IEEE Std 1609.3-2016 IEEE Standard for Wireless Access in Vehicular Environments (WAVE) - Networking Services

[5] IEEE Std 1609.4-2016 IEEE Standard for Wireless Access in Vehicular Environments (WAVE) - Multi-Channel Operation

[6] IEEE Std 1609.12-2016 IEEE Standard for Wireless Access in Vehicular Environments (WAVE) - Identifier Allocations

[1] See 3.1 for the definition of light vehicle. Other vehicle classes and trailers will be addressed in future revisions of this standard, or in other standards within the SAE J2945 family of standards. These revisions or additional standards are expected to be compatible with the requirements of this standard and may define additional capabilities beyond the requirements for light vehicles.

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2.2 Related Publications

The following publications are provided for information purposes only and are not a required part of this SAE Technical Report.

[7] Federal Communications Commission (FCC) 47 Code of Federal Regulations (CFR) Parts 0, 1, 2, and 95 amendments for Dedicated Short Range Communications Services and Mobile Service for Dedicated Short Range Communications of Intelligent Transportation Service in the 5.850-5.925 GHz Band (5.9 GHz Band). http://www.gpo.gov/fdsys/pkg/FR-2006-09-07/pdf/E6-14795.pdf.

[8] Mitigation Strategies for Design Exceptions. Federal Highway Administration, October 15, 2014. http://safety.fhwa.dot.gov/geometric/pubs/mitigationstrategies/chapter3/3_lanewidth.cfm.

[9] CAMP Vehicle Safety Communications Security Studies: Study 3 Final Report: Definition of Communication Protocols between SCMS Components and Specification of the Components Pseudonym Certificate Authority, Registration Authority, and Linkage Authority: National Highway Traffic Safety Administration (Cooperative Agreement Number DTFH61-01-X-00014). July 31, 2014.

[10] CAMP Vehicle Safety Communications Security Studies: Study 1: Security Credential Management System (DTFH61-01-X-00014): National Highway Traffic Safety Administration. July 31, 2014.

[11] National Highway Traffic Safety Administration, ”Vehicle Safety Communications - Applications (VSC-A) Final Report,” DOT HS 811 492A, September 2011. http://www.nhtsa.gov/DOT/NHTSA/NVS/Crash%20Avoidance/Technical%20Publications/2011/811492A.pdf.

[12] National Highway Traffic Safety Administration: Vehicle-to-Vehicle Safety System and Vehicle Build for Safety Pilot (V2V-SP) Final Report, Volume 1 of 2, Driver Acceptance Clinics: National Highway Traffic Safety Administration (Cooperative Agreement Number DTFH61-01-X-00014). April 10, 2014. https://www.regulations.gov/contentStreamer?documentId=NHTSA-2014-0022-0042&attachmentNumber=1&contentType=pdf.

[13] Vehicle Safety Communications - Applications Final Report: Appendix Volume 1 System Design and Objective Test,” DOT HS 811 492B, September 2011. http://www.nhtsa.gov/DOT/NHTSA/NVS/Crash%20Avoidance/Technical%20Publications/2011/811492B.pdf.

[14] Vehicle-to-Vehicle Safety System Light Vehicle Builds and Model Deployment Support (V2V-MD): Test Plan and Test Procedures for Vehicle Awareness Devices and Aftermarket Safety Devices: National Highway Traffic Safety Administration (Cooperative Agreement Number DTFH61-01-X-00014). March 4, 2014.

[15] ITU-R TF.460-6: Standard-Frequency and Time-Signal Emissions.

[16] FIPS PUB 140-2: Federal Information Processing Standards Publication: Security Requirements for Cryptographic Modules.

[17] H. Krishnan, A. Weimerskirch, “Verify-on-Demand - A Practical and Scalable Approach for Broadcast Authentication in Vehicle-to-Vehicle Communication,” SAE Int. J. of Passeng. Cars - Mech. Syst., June 2011, 4:536-546.

[18] Interoperability Issues of Vehicle-to-Vehicle Based Safety Systems Project: Phase 1 Final Report: National Highway Traffic Safety Administration (Cooperative Agreement Number DTFH61-01-X-00014). September 12, 2014.

[19] Interoperability Issues of Vehicle-to-Vehicle Based Safety Systems Project: Phase 2 Final Report Volume 1: Scalability for V2V Safety Development: National Highway Traffic Safety Administration (Cooperative Agreement Number DTFH61-01-X-00014). February 24, 2015.

[20] Interoperability Issues of Vehicle-to-Vehicle Based Safety Systems Project: Phase 2 Final Report Volume 2: Scalability for V2V Safety Analysis: National Highway Traffic Safety Administration (Cooperative Agreement Number DTFH61-01-X-00014). March 31, 2015.

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3. TERMS AND DEFINITIONS

3.1 Definitions

For the purposes of this standard, the following definitions apply.

3.1.1 CRASH

A collision between two or more vehicles.

3.1.2 CRITICAL EVENT FLAG

The hard braking, ABS, traction control, and stability control event flags within DE_VehicleEventFlags.

3.1.3 CRITICAL EVENT CONDITION

When an event that corresponds to a critical event flag occurs.

3.1.4 HARD BRAKING

A light vehicle is decelerating at a level greater than 0.4 g.

3.1.5 LATENCY

The delay from an event occurrence to the desired outcome.

3.1.6 LAYER 2 ADDRESS

The DSRC radio subsystem MAC address, which is from the locally administered MAC address space [5].

3.1.7 LIGHT VEHICLE

A class 2 or class 3 vehicle as defined by FHWA, excluding ambulances, law enforcement vehicles, fire department vehicles, and construction vehicles.
(http://onlinemanuals.txdot.gov/txdotmanuals/tri/vehicle_classification_using_fhwa_13category_scheme.htm)

3.1.8 PACKET COLLISION

When two or more transmissions overlap in time at a potential receiver, causing the receiver to fail to interpret the content of any of the transmissions.

3.1.9 SECURITY CREDENTIAL MANAGEMENT SYSTEM

The public key infrastructure that issues certificates and manages other security functions.

3.2 Abbreviations and Acronyms

The abbreviations and acronyms cited below are terms used in this standard.

ABS Antilock Brake System

ACR Adjacent Channel Rejection

AES Advanced Encryption Standard

ASN.1 Abstract Syntax Notation One

BSM Basic Safety Message

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BSS	Basic Service Set
BSW	Blind Spot Warning
CA	Certificate Authority
CAMP	Crash Avoidance Metrics Partnership
CAN	Controller Area Network
CBP	Channel Busy Percentage
CCH	Control Channel
CCM	Counter Mode with Cipher Block Chaining Message Authentication Code
CFR	Code of Federal Regulations
CLW	Control Loss Warning
CME	Certificate Management Entity
CPR	Certificate Provisioning Request
CRL	Certificate Revocation List
CRLG	Certificate Revocation List Generator
DCM	Device Configuration Manager
DE	Data Element
DF	Data Frame
DNS	Domain Name Services
DOT	Department of Transportation
DSRC	Dedicated Short Range Communications
DTI	Distance to Intersection
DVI	Driver Vehicle Interface
ECDSA	Elliptic Curve Digital Signature Algorithm
ECIES	Elliptic Curve Integrated Encryption Scheme
ECU	Electronic Control Unit
EDCA	Enhanced Distributed Channel Access
EEBL	Emergency Electronic Brake Lights
EGNOS	European Geostationary Navigation Overlay Service
FCC	Federal Communications Commission
FCW	Forward Crash Warning

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FHWA	Federal Highway Administration
FIPS	Federal Information Processing Standards
GHz	Gigahertz
GNSS	Global Navigation Satellite System
HCF	Hybrid Coordination Function
HV	Host Vehicle
Hz	Hertz
ICA	Intermediate Certificate Authority
IEEE	Institute of Electrical and Electronics Engineers
IMA	Intersection Movement Assist
IPv6	Internet Protocol Version 6
ITT	Inter-Transmit Time
LCW	Lane Change Warning
LOS	Line of Sight
LTA	Left Turn Assist
MA	Misbehavior Authority
MAC	Medium Access Control
MD	Model Deployment
MHz	Megahertz
MIB	Management Information Base
MLME	MAC Sublayer Management Entity
MPR	Minimum Performance Requirements
NHTSA	National Highway Traffic Safety Administration
OBE	On-Board Equipment
OCB	Outside the Context of a BSS
OFDM	Orthogonal Frequency Division Multiplexing
OTA	Over the Air
PCA	Pseudonym Certificate Authority
PER	Packet Error Ratio
PH	Path History

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PHY	Physical Layer
PICS	Protocol Implementation Conformance Statement
PLME	Physical Layer Management Entity
PP	Path Prediction
PPS	Pulse per Second
PSDU	Physical Layer Convergence Procedure Service Data Unit
PSID	Provider Service ID
QPSK	Quadrature Phase Shift Keying
RA	Registration Authority
RF	Radio Frequency
RSE	Roadside Equipment
RP	Radiated Power
RV	Remote Vehicle
SAP	Service Access Point
SBAS	Satellite Based Augmentation System
SCH	Service Channel
SCMS	Security Credential Management System
SDEE	Secure Data Exchange Entity
STA	Station
3D	Three-Dimensional
TSF	Time Synchronization Function
TTC	Time-to-Collision[2]
TTI	Time-to-Intersection
Tx	Transmit
UPER	Unaligned Packed Encoding Rules
URL	Uniform Resource Locator
UTC	Universal Coordinated Time
V2V	Vehicle-to-Vehicle

[2] The TTC acronym uses collision instead of crash to be consistent with other industry work.

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V2V-SE Vehicle-to-Vehicle Systems Engineering and Vehicle Integration Research for Deployment (Project)

VOD Verify on Demand

VSA Vendor Specific Action

VSC-A Vehicle Safety Communication - Applications

VSC 3 Vehicle Safety Communications 3 (Consortium)

WAAS Wide Area Augmentation System

WAVE Wireless Access in Vehicular Environments

WGS World Geodetic System

WME WAVE Management Entity

WSM WAVE Short Message

WSA WAVE Service Advertisement

WSMP WAVE Short Message Protocol

3.3 Requirement Numbering Convention

Each requirement in this standard is tagged with a requirement number of the form:

<Subsection Number>-V2V-< Category Abbreviation>-<Subcategory Abbreviation>-<Number>

For example, if the requirement number is 6.5.2-V2V-SECPRIV-BSMSIGN-005, the subsection is 6.5.2, the category is Security and Privacy, the subcategory is BSM signing, and it is requirement number 5 in the subcategory. Table 1 identifies the sections in this standard and the corresponding subsections and abbreviations that are used. The abbreviation is also in parentheses following each section heading in this standard. The requirement numbering convention applies to both mandatory and optional features.

Table 1 - Requirement numbering abbreviations

Section	Subsection	Category	Category Abbreviation	Subcategory	Subcategory Abbreviation
6.1	6.1.1	Standards Profiles	STD	IEEE 802.11	802.11
	6.1.2			IEEE 1609.2	1609.2
	6.1.3			IEEE 1609.3	1609.3
	6.1.4			IEEE 1609.4	1609.4
	6.1.5			IEEE 1609.12	1609.12
	6.1.6			SAE J2735	SAE J2735
6.2	6.2.1	Positioning and Timing	POSTIM	Position Determination	POSDETER
	6.2.2			Wide Area Augmentation System	WAAS
	6.2.3			Coordinate System and Reference	COORDSYSREF
	6.2.4			System Time Coordination	SYSTIMCOORD
6.3	6.3.1	BSM Transmission	BSMTX	BSM Contents	BSMCONT

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Section	Subsection	Category	Category Abbreviation	Subcategory	Subcategory Abbreviation
	6.3.2			Channel and Data Rate	CHDATARATE
	6.3.3			First BSM after Startup and Generation Timing	GENTIM
	6.3.4			User Priority and EDCA Settings	UPEDCA
	6.3.5			Minimum Transmission Criteria	MINTX
	6.3.6			Data Element Accuracy	DATAACC
	6.3.7			Data Persistency	DATAPERSIST
	6.3.8			BSM Scheduling and Congestion Control	CONGCTRL
6.4	6.4.1	RF Performance	RFPERF	DSRC Transmit Power Accuracy and Radiated Transmit Power	DSRCTX
	6.4.2			DSRC Receive Sensitivity	DSRCRXSENS
6.5	6.5.1	Security and Privacy	SECPRIV	ID Randomization	IDRAND
	6.5.2			BSM Signing	BSMSIGN
	6.5.3			Certificate Change	CERTCHG
	6.5.4			BSM Verification	BSMVERIFY
	6.5.5			Certificate Revocation	CERTREV
6.6	6.6.1	Security Management	SECMGMT	Bootstrap: Enrollment and Initialization Processing	ENINIT
	6.6.2			Certificate Loading	CERTLOAD
	6.6.3			Certificate Storage	CERTSTORE
	6.6.4			CRL Loading	CRLLOAD
	6.6.5			Secure Hardware	SECHW

4. V2V SAFETY SYSTEMS CONCEPT OF OPERATIONS AND SYSTEM DESCRIPTION

This section provides a high-level description of the V2V safety concept of operations and system description. Section 4.1 provides an overview of the system, and 4.2 provides the system description for V2V safety features.

4.1 V2V System Overview

V2V safety communications are designed to exchange basic safety information among vehicles for driver assistance by supporting detection of imminent crash threats and alerting the driver. V2V communications use dedicated short range communications (DSRC) radios to transmit BSMs that include a subset of the available data frames and elements in SAE J2735 [1]. On-board safety applications use the information about the host vehicle (HV) and remote vehicles (RVs) to detect potential crash threats and alert the driver. Messages can be used for additional purposes, but only the scenarios described herein were used to develop this standard. For the purposes of the crash scenarios described herein, the HV and RV terminology is used to identify which vehicle is receiving and acting on BSMs (HV), and the set of vehicles from which BSMs are being received (RVs).

V2V communications can enable improved safety system effectiveness by complementing or providing an alternative to self-contained sensors such as radar, lidar, or camera systems. V2V communications provide the vehicle and driver with 360-degree awareness and can detect potential threats at a greater distance than other types of sensors, as well as detecting potential threats to some degree even under non-line-of-sight or low visibility conditions. This enables the driver to receive alerts earlier and have more time to take action to avoid crashes.

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Because vehicles need to trust messages from each other, security is essential to protect messages from attacks such as spoofing, alteration, or replay that could cause false alerts or suppress true alerts. In addition, driver privacy is protected appropriately, so the system does not disclose identifying information about the driver, or allow for easy tracking. All BSMs are sent with a signature that enables the receiving device to verify the message. Broadcast information that could potentially be used to identify and track drivers is anonymous and randomized, and other system security measures are also incorporated to protect privacy appropriately.

Figure 1 illustrates the components of the V2V system and its interfaces. An infrastructure-based security credential management system (SCMS) is responsible for generating and delivering the security certificates that are used in the message verification process. The SCMS can also revoke certificates that cannot be trusted (e.g., the associated device may have been tampered with or is misbehaving) by placing them on a certificate revocation list (CRL) that the SCMS distributes to all systems. Section 6.6 in this standard describes the SCMS interface.

The V2V on-board equipment (OBE), which is the on-board vehicle-to-vehicle (V2V) safety communications system defined in this standard (hereafter referred to as the system), typically consists of multiple subsystem components, which may be discrete or integrated depending on the implementation. Figure 1 illustrates the following subsystems within the system:

- DSRC Radio Subsystem: Transmits and receives BSMs. In this standard, a DSRC radio subsystem is assumed to be a single-channel-at-a-time device. The OBE can include one or more DSRC radio subsystems and still comply with this standard.

- Positioning Subsystem: The subsystem that includes a global navigation satellite system (GNSS) receiver and provides vehicle position, heading, speed, and time information. The system may augment and enhance postioning using additional information and components, which are not shown in Figure 1. Examples of these are speed data from the CAN bus, dead reckoning sensors, and optical/camera based systems.

- OBE Control Processor Electronic Control Unit (ECU): Executes software that generates BSMs for transmission according to the requirements in this standard.

- Antennas: Support radio frequency (RF) links for the DSRC radio and GNSS receiver. A second diversity antenna for the DSRC Radio Subsystem is recommended to improve performance. GNSS and DSRC antennas may be integrated (dual band).

Systems communicate amongst themselves using the DSRC radio subsystem as an interface. The system (OBE) can interface to a safety application ECU that detects threats and issues alerts through a driver-vehicle interface (DVI). The DVI can provide visual, audio, and/or haptic alerts. The OBE can also interface with the vehicle controller area network (CAN) bus to obtain vehicle status information. The safety application ECU, CAN bus, and DVI are outside the scope of this standard.

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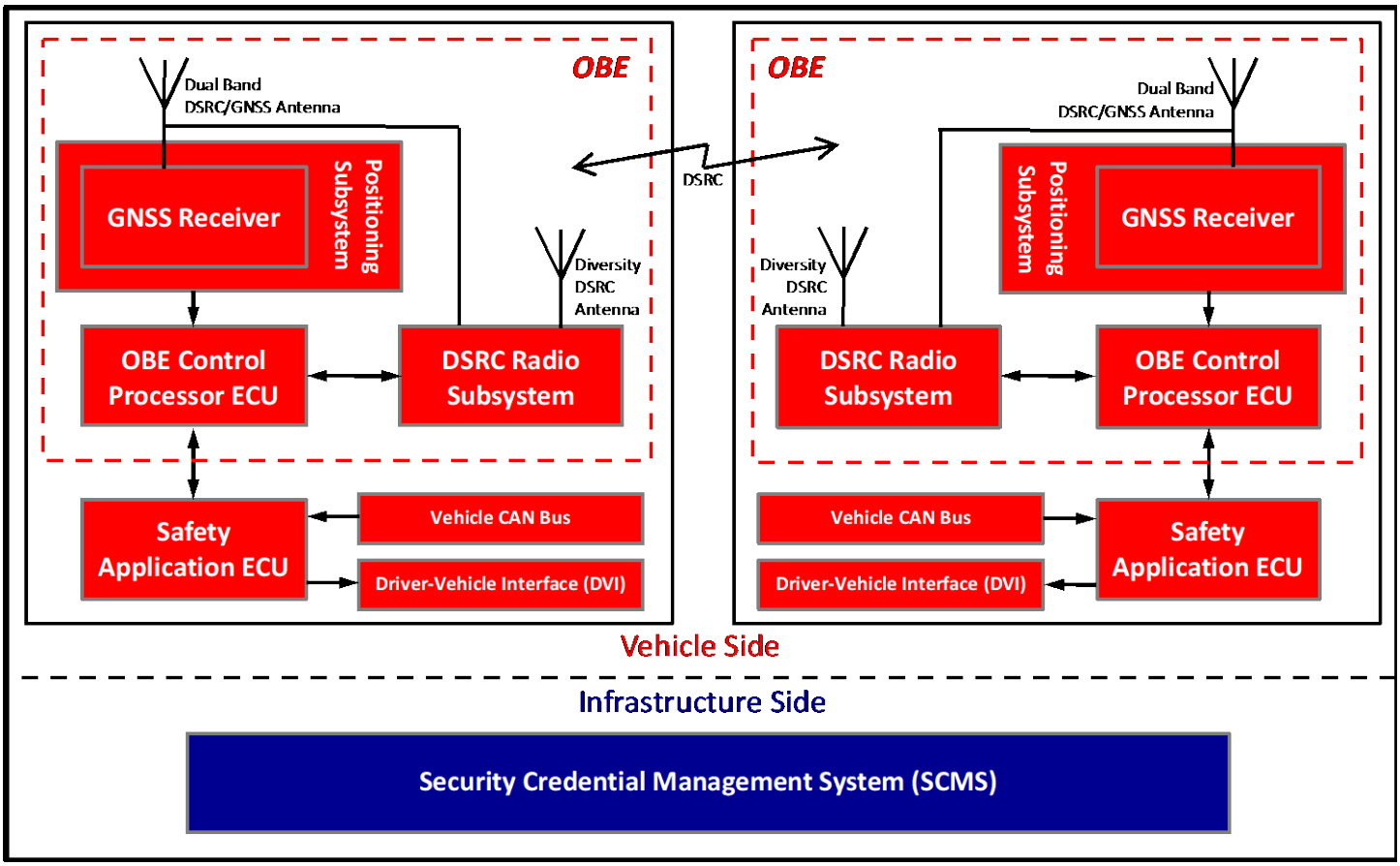

Figure 1 - On-board V2V system

4.2 V2V Safety Features

4.2.1 Critical Crash Scenarios for V2V Safety

The set of crash scenarios that could be addressed by the system were initially analyzed and documented in the Crash Avoidance Metrics Partnership (CAMP) Vehicle Safety Communications-Applications (VSC-A) project [11]. Table 2 lists the seven scenarios selected based on a composite ranking of crash frequency, crash cost, and functional years lost[3]. Scenario 7 was subsequently added to address left-turn-across-path intersection crashes during the CAMP VSC 3 driver acceptance clinics project [12].

[3] Functional years lost is a composite measure of crash severity. It measures the loss of an individual's productive time due to injuries sustained in a crash.

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Table 2 - Selected crash-imminent scenarios

Crash Imminent Scenario	Crash Category		
	Frequency	Cost	Functional Years Lost
Lead vehicle stopped	✓	✓	✓
Control loss without prior vehicle action	✓	✓	✓
Vehicle(s) turning at non-signalized junctions	✓	✓	
Straight crossing paths at non-signalized junctions			✓
Lead vehicle decelerating	✓	✓	
Vehicle(s) changing lanes - same direction	✓		
Left turn across path - opposite direction			

✓ Denotes top five ranking for the crash category.

4.2.2 Mapping Between Critical Crash Scenarios and the Selection of V2V Safety Applications

In [11] and [12], V2V safety applications were developed to address the selected scenarios. Table 3 illustrates the mapping between the crash-imminent scenarios identified in Table 2 and the list of safety applications. These safety applications are defined and discussed in more detail below. The decision thresholds used to determine when to warn the driver are not specified in this standard and are left to the implementer.

Table 3 - Crash-imminent scenario to V2V safety application mapping

Safety Applications / Crash Scenarios	EEBL	FCW	BSW/ LCW	IMA	LTA	CLW
Lead vehicle stopped		✓				
Control loss without prior vehicle action						✓
Vehicle(s) turning at non-signalized junctions				✓	✓	
Straight crossing paths at non-signalized junctions				✓		
Lead vehicle decelerating	✓	✓				
Vehicle(s) changing lanes - same direction			✓			
Left turn across path - opposite direction					✓	

4.2.3 Emergency Electronic Brake Lights (EEBL)

4.2.3.1 Definition

The EEBL safety application warns the driver of the HV in the case of a hard-braking event by an RV that is ahead and in the same lane or an adjacent lane. The RV broadcasts a hard-braking event in the BSM upon a hard braking maneuver. Upon receiving such event information, the HV determines the relevance of the event to its own travel path and provides a warning to the driver, if appropriate.

4.2.3.2 EEBL Use Case Scenario

a. EEBL: Abruptly Slowing RV (Figure 2)

- The HV follows a moving RV-2, which in turn follows RV-1 that abruptly brakes hard. RV-2 may or may not be equipped with V2V communications, but RV-1 is equipped with V2V capability.

- The HV may receive a warning from the EEBL feature when RV-1 applies brakes and decelerates at a level greater than the threshold that corresponds to hard braking (see the definition in 3.1).

- The timing of the warning is expected to be set such that the driver of the HV can avoid a rear-end crash with the vehicle (RV-2) in front.

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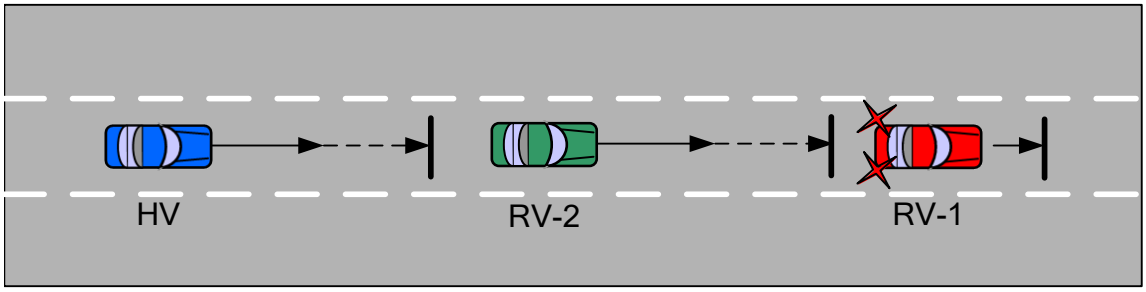

Figure 2 - EEBL - abruptly slowing RV

4.2.3.3 EEBL Feature Systems Description

The EEBL feature warns the driver of the HV in the case of a hard-braking event by an RV that is ahead and in the same lane or in an adjacent lane. The relevant RV zones for the EEBL feature are illustrated in Figure 3. The EEBL feature is expected to function in both straight and curved roadway geometries.

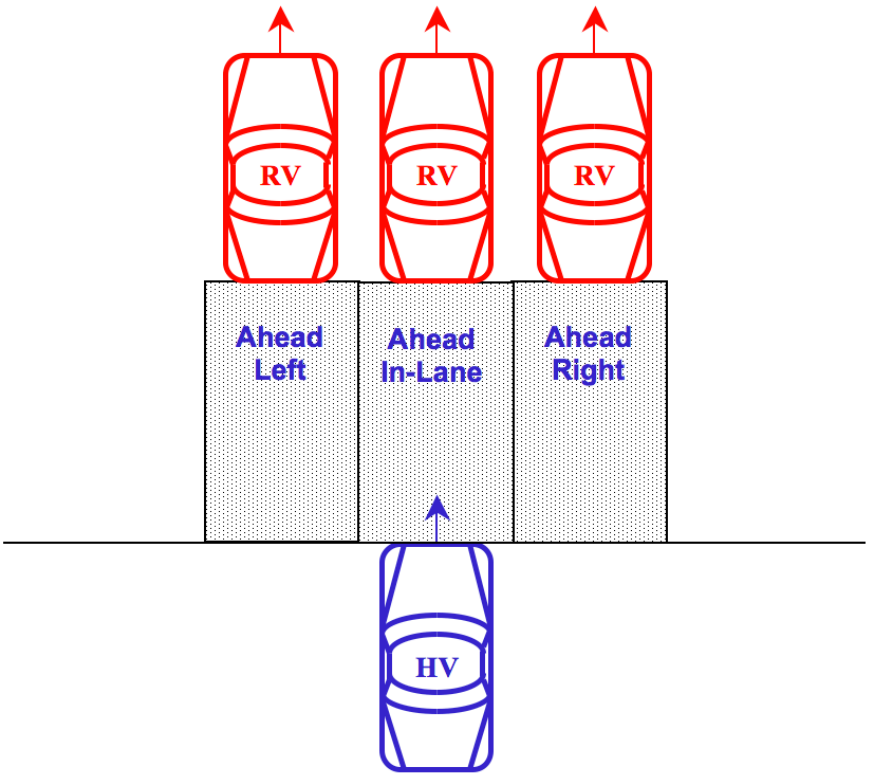

Figure 3 - Relevant RV zones for the EEBL feature

EEBL performs the following operations:

- The RV includes a hard-braking event in the broadcasted BSM during a hard-braking maneuver.

Upon receiving such event information, the HV performs the following operations:

- Determines which, if any, RVs have reported a hard-braking event.
- For each RV that has reported a hard-braking event and is classified as “ahead in-lane,” “ahead left,” or “ahead right,” determines if the distance between the vehicles is less than an implementation-specific threshold value.

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- Calculates the EEBL threat levels among all RVs identified above, determines the principal threat, and sets the appropriate threat status.
- Provides a warning to the driver via a DVI.

4.2.4 Forward Crash Warning (FCW)

4.2.4.1 Definition

The FCW safety application warns the driver of the HV in the case of an impending rear-end crash with an RV directly ahead in the same lane and direction of travel. The FCW is intended to help drivers avoid or mitigate rear-end vehicle crashes in the forward path of travel.

4.2.4.2 FCW Use Case Scenarios

a. FCW: Stopped RV in Same Lane (Figure 4)

- The HV approaches RV-1, which is stopped in the same lane as the HV.
- The HV receives a warning from the FCW feature when there is imminent danger of a rear-end crash with the stopped RV-1 in its lane of travel.
- The timing of the warning is expected to be set such that the driver of the HV can avoid a rear-end crash with the stopped RV-1.

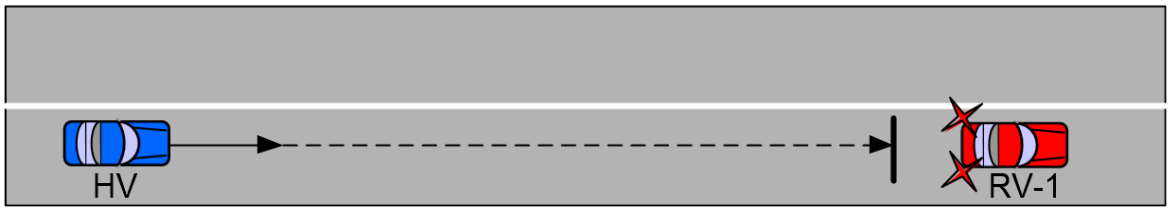

Figure 4 - FCW - stopped RV in same lane

b. FCW: Stopped RV in Adjacent Lane (Figure 5)

- The HV approaches RV-1, which is stopped in the lane adjacent to the HV.
- The driver of the HV does not receive a warning from the FCW feature since there is no imminent danger of a rear-end crash.

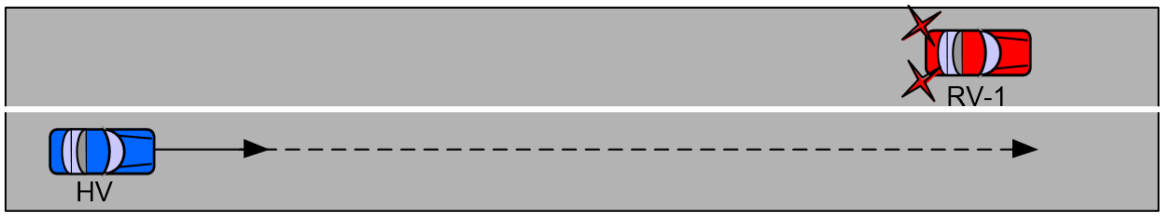

Figure 5 - FCW - stopped RV in adjacent lane

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c. FCW: Slower-Moving or Decelerating RV in Same Lane (Figure 6)

- The HV approaches RV-1, which is moving slower and/or decelerating in the same lane as the HV.
- The HV driver receives a warning from the FCW feature when there is imminent danger of a rear-end crash with the slow-moving RV-1 in its lane of travel.
- The timing of the warning is expected to be set such that the driver can avoid a rear-end crash with the slow-moving RV-1.

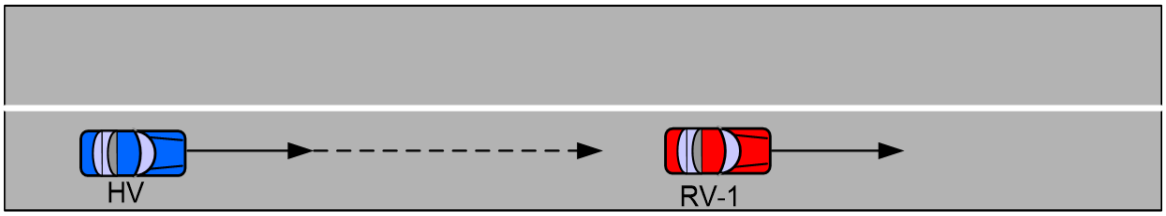

Figure 6 - FCW - slow-moving RV in same lane

d. FCW: Stopped and Obstructed RV (Figure 7)

- The HV follows a moving RV-2, which in turn approaches RV-1 that is stopped in the same lane. RV-2 may or may not be equipped with V2V communications, but RV-1 is equipped with V2V capability.
- RV-2 makes a lane change to avoid the stopped RV-1.
- The HV driver receives a warning from the FCW feature when there is imminent danger of a rear-end crash with the stopped RV-1 in its lane of travel.
- The timing of the warning is expected to be set such that the driver of the HV can avoid a rear-end crash with the stopped vehicle RV-1.

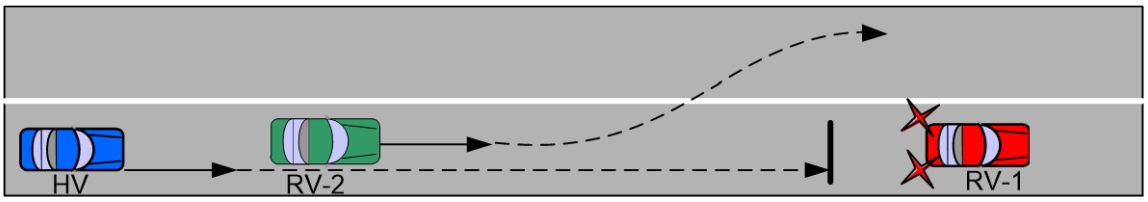

Figure 7 - FCW - stopped and obstructed RV

4.2.4.3 FCW Feature Systems Description

The FCW warns the driver of the HV when there is imminent danger of a rear-end crash with a remote lead vehicle in its lane of travel. The FCW feature is expected to function in both straight and curved roadway geometries. The relevant RV zones for the FCW feature are illustrated in Figure 8.

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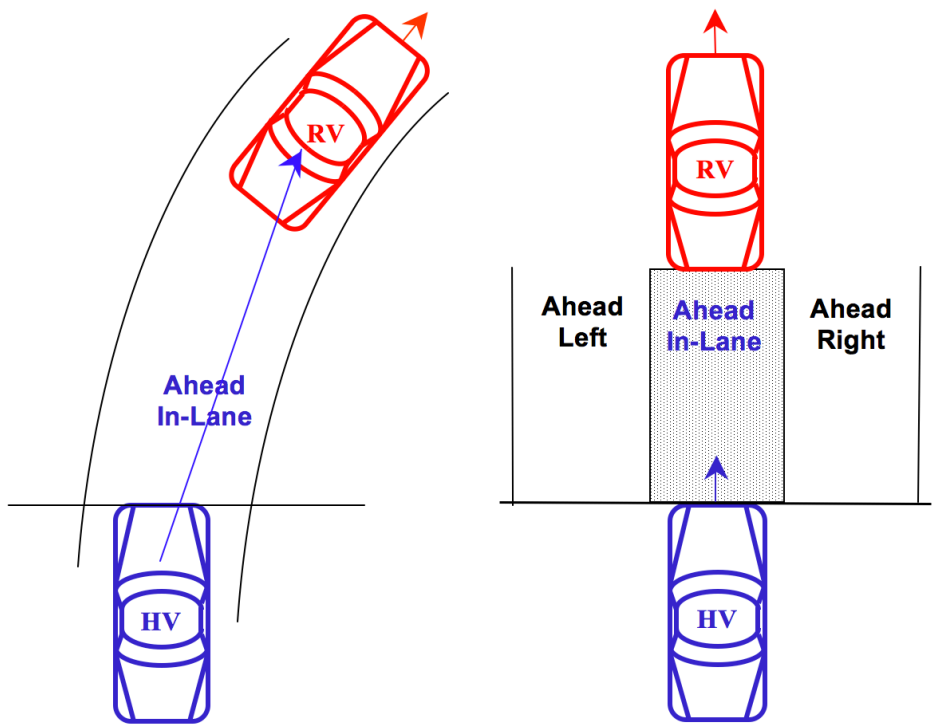

Figure 8 - Relevant RV zones for FCW feature

FCW performs the following operations:

- Analyzes received BSMs from each of the RVs and determines which of the RVs are classified as “ahead in-lane” to determine if the HV is at risk of being involved in a rear-end crash with an RV located in the same lane of travel.
- Determines which, if any, RVs classified as “ahead in-lane” are within a range threshold.
- Calculates time to crash (TTC) and/or crash avoidance range for each “ahead in-lane” RV to determine potential forward crash threats.
- Identifies the principal threat, if at least one RV is determined to be a threat.
- Provides a warning to the driver via a DVI.

4.2.5 Blind Spot Warning/Lane Change Warning (BSW/LCW)

4.2.5.1 Definition

The BSW/LCW safety application warns the driver of the HV during a lane change attempt if the blind-spot zone into which the HV intends to move is, or will soon be, occupied by another vehicle traveling in the same direction. Moreover, the application may also provide advisory information that is intended to inform the driver of the HV that a vehicle in an adjacent lane is positioned in a blind-spot zone of the HV when a lane change is not being attempted.

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4.2.5.2 BSW/LCW Use Case Scenarios

a. BSW/LCW: RV in Blind-Spot Zone (Figure 9)

- The HV drives in its lane while RV-1 is alongside the HV within the blind-spot zone.
- The HV driver may receive an advisory warning from the BSW/LCW feature indicating the presence of RV-1 in the blind-spot zone.
- The HV driver receives a warning if the HV detects the driver's intent to change lanes into the lane occupied by RV-1 (for example, through the use of the turn signal).
- The timing of the warning is expected to be set such that the driver of the HV can avoid a crash with RV-1 in the adjacent lane.

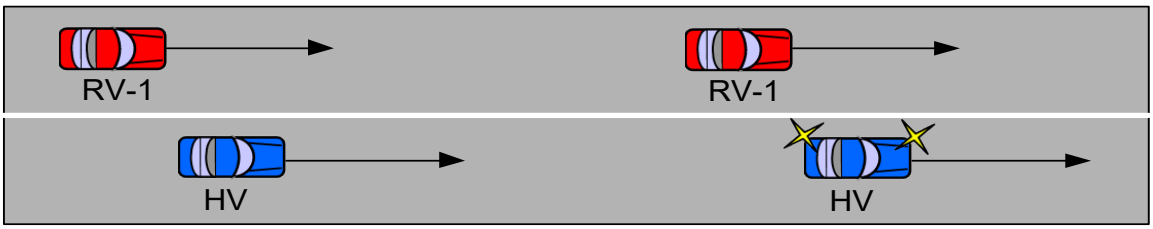

Figure 9 - BSW - RV in blind-spot zone

b. BSW/LCW: Approaching RV in Adjacent Lane (Figure 10)

- The HV drives in its lane. A faster moving RV-1 traveling in the same direction in an adjacent lane will soon occupy the HV's blind-spot zone.
- The HV driver may receive an advisory warning from the BSW/LCW feature anticipating the presence of RV-1 in the blind-spot zone.
- The HV driver receives a warning if the HV detects the driver's intent to change lanes into the lane that will soon be occupied by RV-1.
- The timing of the warning is expected to be set such that the driver of the HV can avoid a crash with a faster-moving RV-1 in the adjacent lane.

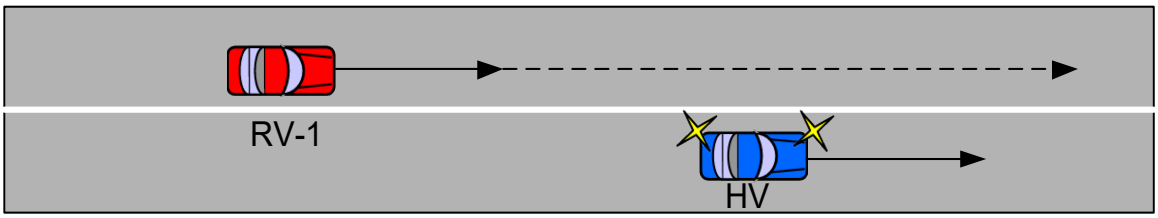

Figure 10 - LCW - approaching RV in adjacent lane

4.2.5.3 BSW/LCW Feature Systems Description

The BSW/LCW safety application warns the driver of the HV during an attempted lane change if the blind-spot zone into which the HV intends to move is, or will soon be, occupied by another vehicle traveling in the same direction. The relevant RV zones to the BSW/LCW feature are illustrated in Figure 11. The BSW/LCW feature is expected to function in straight and curved roadway geometries.

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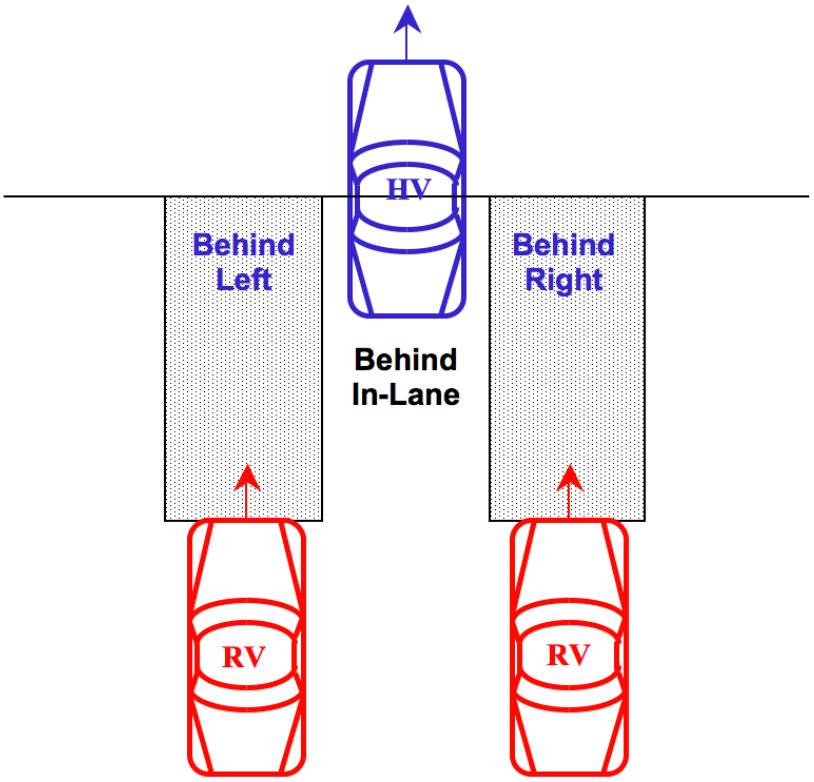

Figure 11 - Relevant RV zones for BSW/LCW feature

BSW/LCW performs the following operations:

- Determines which RVs have been classified as behind in the adjacent left lane or behind in the adjacent right lane relative to the HV.
- Evaluates the position of each RV relative to the HV to determine if that RV is currently positioned within the HV's blind-spot zone or if that RV is predicted to soon be within the HV's blind-spot zone.
- Sets the threat status corresponding to each side (left and right) to advise the HV driver if an RV is or will be located in the corresponding left or right blind spot.
- Sets the threat status to warn the HV driver of the current or predicted presence of a threat in an adjacent lane during an attempted lane-change maneuver.
- Provides an advisory or warning to the driver via a DVI.

4.2.6 Intersection Movement Assist (IMA)

4.2.6.1 Definition

The IMA safety application warns the driver of an HV when it is not safe to enter an intersection due to a crash possibility with RVs.

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4.2.6.2 IMA Use Case Scenarios

a. IMA: Stopped HV at Intersection (Figure 12)

- The HV is stopped at an intersection and visibility may be limited by the presence of RV-2. RV-2 may or may not be equipped with V2V communications, which has no impact on the scenario, but RV-1 is equipped with V2V capability.
- RV-1 approaches the intersection from the left or right of the HV.
- The HV driver receives a warning from the IMA feature indicating that a crash is predicted with RV-1 if the HV begins to enter the intersection.
- The timing of the warning is expected to be set such that the driver of the HV can avoid a crash with the approaching RV-1.

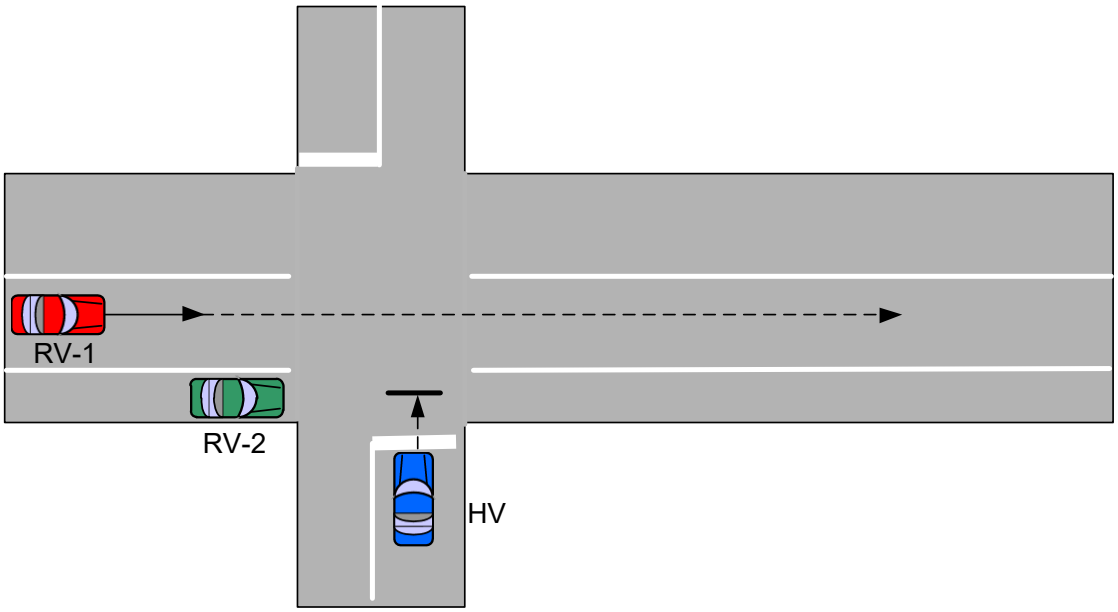

Figure 12 - IMA - stopped HV at the intersection

b. IMA: Both Vehicles Approaching Intersection (Figure 13)

- The HV approaches the intersection and visibility may be limited by the presence of RV-2. RV-2 may or may not be equipped with V2V communications, but RV-1 is equipped with V2V capability.
- RV-1 approaches the intersection from the left or right of the HV.
- The HV driver receives a warning from the IMA feature indicating that a conflict is predicted with RV-1 if the HV tries to enter the intersection.
- The timing of the warning is expected to be set such that the driver of the HV can avoid a crash with the approaching vehicle, RV-1.

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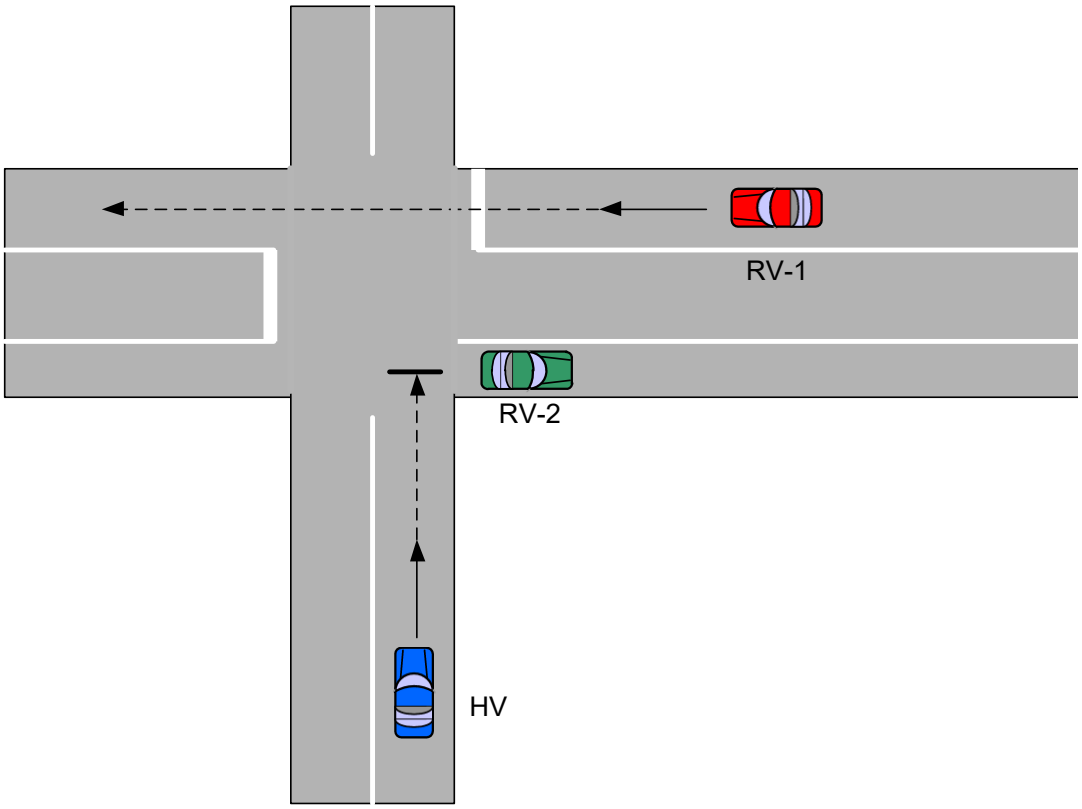

Figure 13 - IMA - both vehicles approaching intersection

4.2.6.3 IMA Feature Systems Description

IMA warns the driver of the HV when there is imminent danger of a crash with a remote vehicle that is approaching the same intersection. The relevant RV zones for the IMA feature are illustrated in Figure 14.

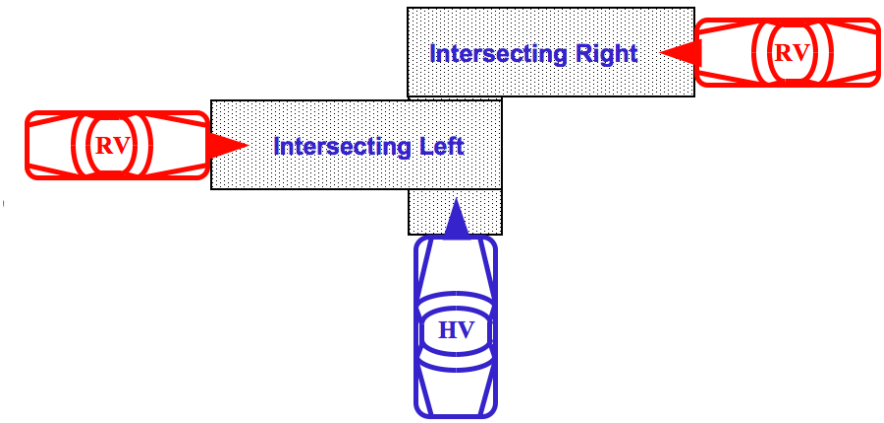

Figure 14 - Relevant RV zones for IMA feature

IMA performs the following operations:

- Analyzes received BSMs from RVs approaching the intersection and determines which of the RVs are classified as “intersecting left” or “intersecting right.”
- Determines which, if any, RVs classified as “intersecting left” or “intersecting-right” are within a lateral range threshold.

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- Calculates time-to-intersection (TTI) and distance-to-intersection (DTI) for each “intersecting left” or “intersecting right” RV to determine if the HV is at risk of being involved in a crash with an RV traveling toward the same intersection.
- Identifies the principal threat, if at least one RV is determined to be a threat.
- Provides a warning to the driver via a DVI.

4.2.7 Left Turn Assist (LTA)

4.2.7.1 Definition

The LTA safety application warns the driver of an HV that, due to oncoming traffic, it may not be safe to proceed when attempting a left turn.

4.2.7.2 LTA Use Case Scenarios

a. LTA: Left Turn Across Path (Figure 15)

- The HV approaches an intersection to make a left turn and visibility may be limited or obstructed by RV-2. RV-2 may or may not be equipped with V2V communications, but RV-1 is equipped with V2V capability.
- RV-1 approaches the intersection from the opposite direction.
- The HV driver receives a warning from the LTA feature when the HV driver attempts a left turn.
- The timing of the warning is expected to be set such that the driver of the HV can avoid a crash with the approaching vehicle, RV-1.

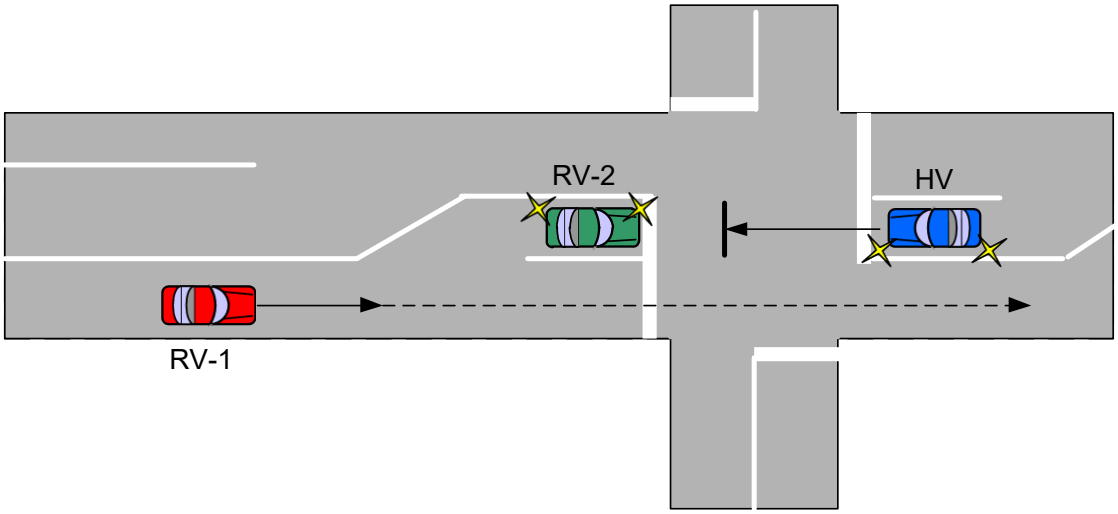

Figure 15 - LTA - left turn across path

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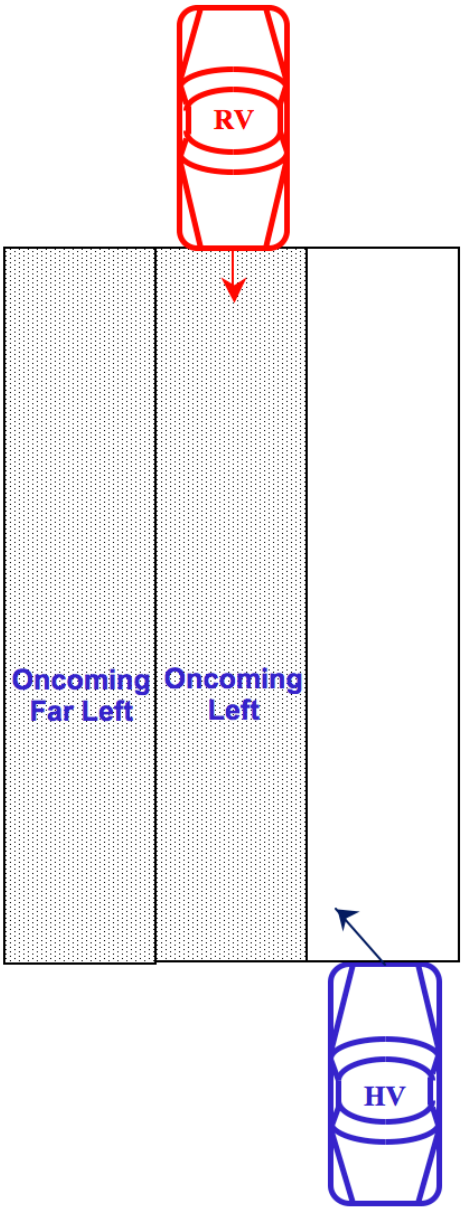

Figure 16 - Relevant RV zones for LTA feature

4.2.7.3 LTA Feature Systems Description

LTA should warn a driver intending to make a left turn across an intersection path when there is imminent danger of a crash with an RV in an oncoming opposite lane of travel. The relevant RV zones for the LTA feature are illustrated in Figure 16.

LTA performs the following operations:

- Analyzes received BSMs from the RVs approaching the intersection and determines which of the RVs are classified as “oncoming left” or “oncoming far left” to determine if the HV is at risk of being involved in an intersecting crash with an RV approaching in an oncoming lane of travel.

- Determines which, if any, RVs classified as “oncoming left” or “oncoming far left” are within a range threshold.

- Calculates the clearance gap for each “oncoming left” or “oncoming far left” RV to determine potential intersecting crash threats.

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- Identifies the principal threat, if at least one RV is determined to be a threat.
- Provides a warning to the driver via a DVI.

4.2.8 Control Loss Warning (CLW)

4.2.8.1 Definition

The CLW safety application warns the driver of the HV in the case of an emergency control loss event (defined as activation of the antilock brake system, traction control system, or stability control system) by an RV traveling in the same or opposite direction. The RV broadcasts control loss event information within the BSM. Upon receiving such event information, the HV determines the relevance of the event and provides a warning to the driver of the HV.

4.2.8.2 CLW Use Case Scenarios

a. CLW: RV Same Direction of Travel (Figure 17)

- The HV follows the RV in the same direction.
- The HV receives BSMs from the RV indicating a control loss event (antilock brake system, traction control system or stability control system active).
- The timing of the corresponding warning to the driver is expected to be such that the driver of the HV can avoid a crash with the RV.

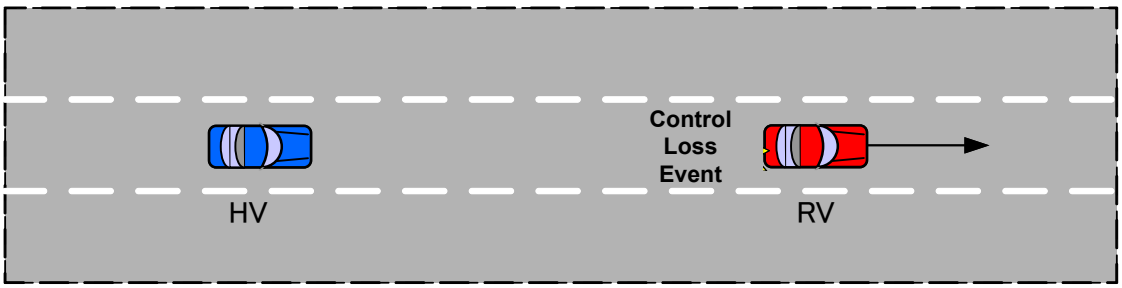

Figure 17 - CLW - RV same direction of travel

b. CLW: RV Traveling in Opposite Direction (Figure 18)

- The RV approaches the HV from the opposite direction.
- The HV receives BSMs from the oncoming RV indicating a control loss event (antilock brake system, traction control system or stability control system activated).
- The timing of the corresponding warning to the driver is expected to be such that the driver of the HV can avoid a crash with the RV.

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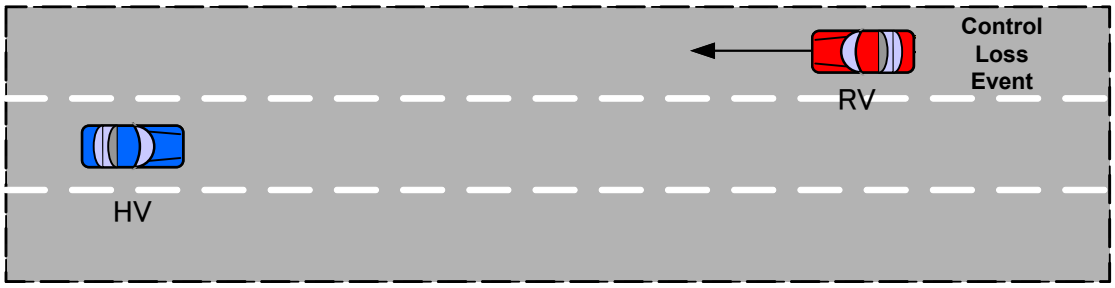

Figure 18 - CLW - RV traveling in opposite direction

4.2.8.3 CLW Feature Systems Description

The CLW safety application warns the driver of the HV in the case of an emergency control loss event by an RV traveling in the same or opposite direction. The relevant zones for the CLW features are illustrated in Figure 19.

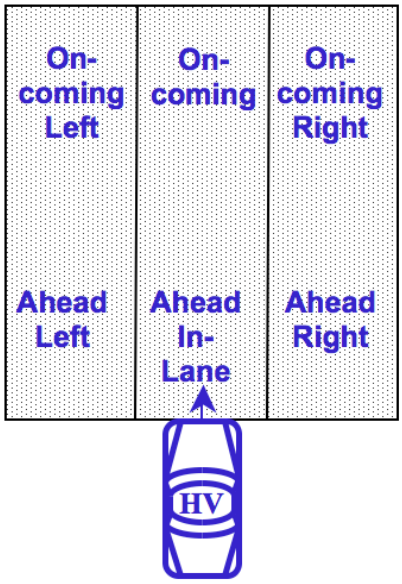

Figure 19 - Relevant RV zones for CLW feature

CLW performs the following operations:

- The RV broadcasts a control loss event in the BSM upon activation of the antilock brake system, traction control system, or stability control system.

Upon receiving such event information, the HV performs the following operations:

- Determines which, if any, RVs have reported a control loss event.
- For each RV that has reported an event and is classified as “ahead in-lane,” “ahead left,” “ahead right,” “oncoming,” “oncoming left,” or “oncoming right,” determines if the range or TTC is less than a threshold value.
- Calculates the CLW threat levels among all RVs identified above, determines the principal threat, and sets the appropriate threat status.
- Provides a warning to the driver via a DVI.

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5. INTERFACE DESCRIPTION

5.1 V2V Over-the-Air Data Description

5.1.1 Basic Safety Message Exchange

The BSM, which is defined in SAE J2735 [1], is the message used for V2V safety communications in this standard. Each vehicle broadcasts (refer to [11]) BSMs to provide neighboring vehicles with trajectory and status information. With a sufficiently high population of V2V-equipped vehicles, the number of BSMs being transmitted can congest the channel, necessitating congestion control procedures.[4] The BSM consists of all data elements listed in Part I and selected data elements and data frames listed in Part II of the SAE J2735 standard. Part I contains the vehicle position, speed, heading, acceleration, transmission, steering-wheel angle, brake, and vehicle-size information.

For the system defined in this standard, the following additional information is transmitted as part of the Part II vehicle safety extension:

- Event flags, which convey the sender's status with respect to safety-related events such as antilock brake system (ABS) activation, stability control activation, and hard braking.
- Path history, which provides a concise representation of the vehicle's recent movement. It consists of a sequence of positions selected to represent the vehicle's path within an allowable error.
- Path prediction, which provides an estimate of the vehicle's future trajectory. The trajectory is represented as a radius of curvature.
- Exterior lights, which provides the vehicle light status, including turn signals.

5.1.2 Positioning

Many V2V safety applications need relative positioning of the HV and RVs with lane-level granularity. For example, a safety application can determine if the HV and an RV are in the same or adjacent lanes. The system includes a GNSS receiver to enable the system to determine its own position and accurate time. The system also maintains its own path history and calculates its path prediction. Each vehicle broadcasts its time-tagged position, heading, speed, and acceleration, plus its path history and path prediction in the BSM. Based on the HV and RV information, the system can calculate the range, range-rate, difference in headings, and relative position between vehicles. The path history and path prediction information are used to estimate the relative lane positioning between the HV and RV.

5.1.3 Security and Privacy

5.1.3.1 Signing and Verification Algorithm

To support trust in message exchange between vehicles, BSM signing and verification are performed using a public key digital signature algorithm. The transmitter computes a signature using an elliptic curve digital signature algorithm (ECDSA) with a private key, and the receiver verifies the signature using the associated certificate.

5.1.3.2 BSM Signature and Certificate Transmission

Each BSM is transmitted with a signature and either a security certificate containing the public key or a certificate digest (hash of the current security certificate). The BSM with a certificate is transmitted approximately every 500 ms (see 6.5.2), and other BSMs are transmitted with a certificate digest to reduce the overall message length. The receiver buffers recently received certificates and is able to identify the certificate corresponding to a received digest.

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[4] Non-BSM transmissions from V2X devices, e.g., from roadside units, may also contribute to congestion on the channel.

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5.1.3.3 BSM Verification

The receiver can verify every message or use a verify on demand (VOD) approach [17] whereby only a subset of BSMs is verified. For example, the receiver may verify only BSMs containing information that would trigger an alert to the driver, or the receiver may verify a subset of messages received to support other functional requirements. Upon selecting a BSM for verification, if the BSM contains a security certificate, the receiver uses the public key in the certificate to perform the verification. If the BSM contains a digest, the receiver can use the digest to identify the corresponding buffered certificate for use in the verification. In certain cases, a system may obtain (learn) certificate information from peer systems over the same interface.

5.1.3.4 SCMS

The SCMS is responsible for generating and distributing security certificates. The SCMS is also responsible for generating and distributing a certificate revocation list (CRL), which contains a list of linkage information that all receiving devices can use to identify non-trustworthy certificates. Certificates and the CRL are stored within the OBE system, and are updated by the SCMS. A well-behaved system does not send a BSM if its linkage information is on the CRL. If a receiver receives a BSM with linkage information on the CRL, the BSM is considered invalid. More information about the CRL is available in 1609.2 [3]. For the initial equipping of a system, it interfaces to a device configuration manager (DCM) to acquire the root certificate authority (CA) certificate (see 6.6.1).

5.1.3.5 Privacy

To protect privacy, the signing security certificate is changed after a variable length of time (for example, every 5 minutes), and fields within the broadcast message (DE_MsgCount, DE_TemporaryID, and the layer 2 address) are randomized whenever the certificate is changed. The SCMS is structured as a set of components, so the component generating the certificates has no knowledge of which certificates are used by a particular device. For a more detailed description of the SCMS and associated security system design and operation, refer to [9] and [10].

5.1.4 Startup and Shutdown

To ensure performance, systems include additional capabilities related to device startup and shutdown. Systems store the last known heading and path history information on shutdown so they can be retrieved for use in BSMs upon the next system device startup. Systems also randomize their message generation schedule upon startup to avoid repeated packet collisions of transmitted BSMs with those transmitted by other systems.

5.1.5 Mapping to the V2V Over-the-Air Data

A generic logic design of the safety features is shown by the flowchart in Figure 20. Table 4 provides a mapping to the V2V over-the-air data.

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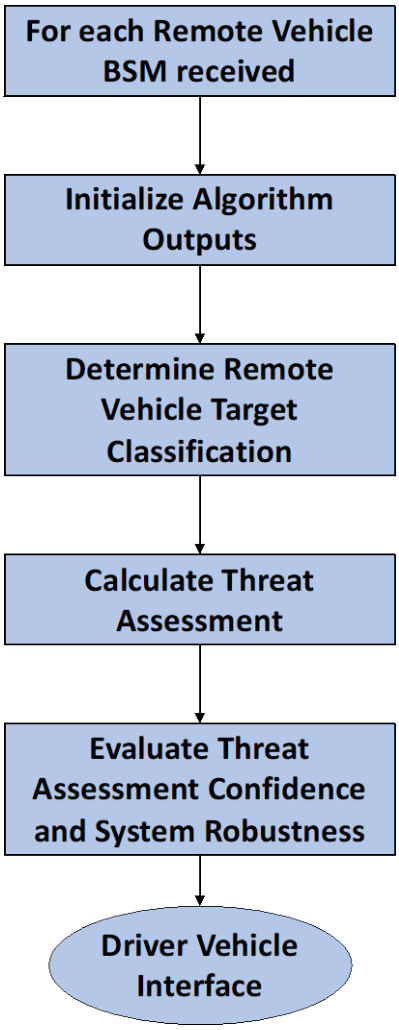

Figure 20 - Safety feature logic

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Table 4 - Mapping crash scenarios to the V2V over-the-air data

V2V Safety Message BSM Contents (refer to SAE J2735 [1])		FCW, BSW/LCW, IMA Stopped, LTA	EEBL, CLW, IMA Moving
DE_DSecond DE_Latitude DE_Longitude DE_Elevation DF_PositionalAccuracy DE_Heading	Relative Road Level Positioning		Required for: • Relative Road Level Target Classification • Threat Assessment • Threat Assessment Confidence and System Robustness
	Relative Lane Level Positioning	Required for: • Relative Lane Level Target Classification • Threat Assessment • Threat Assessment Confidence and System Robustness	
DE_VehicleWidth DF_PathHistory DF_PathPrediction		Required for: • Relative Lane Level Target Classification	Required for: • Relative Road Level Target Classification
DE_Speed DE_TransmissionState DE_Acceleration (Longitudinal) DF_BrakeSystemStatus DE_ExteriorLights DE_VehicleLength		Required for: • Threat Assessment • Threat Assessment Confidence and System Robustness	Required for: • Threat Assessment • Threat Assessment Confidence and System Robustness
DE_SteeringWheelAngle DE_Acceleration (Lateral) DE_Acceleration (Vertical) DE_YawRate		Required for: • Threat Assessment Confidence and System Robustness	Required for: • Threat Assessment Confidence and System Robustness
DE_VehicleEventFlags			Required for (EEBL and CLW only): • Hard-Braking Event Notification • Control-Loss Event Notification

NOTE: The positioning and other related data (e.g., speed, heading) accuracy requirements in this standard are designed to meet the relative lane level positioning needs of the crash scenarios described in the preceeding sections (see 6.3.6.5 through 6.3.6.12).

5.1.6 Objective Tests Conducted for V2V Safety Applications (Informative)

Details and results from testing for the crash scenarios described in this standard can be found in the VSC-A final report, Appendices C-2 and C-3 [13]. Refer to the report for details.

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5.2 System Interfaces

This section provides an overview of the system interfaces. Section 6 provides the detailed requirements for these interfaces.

5.2.1 Vehicle to Vehicle Communications Interface

The system interfaces to other vehicles (systems) by transmitting and receiving BSMs.

- The format and contents of the BSM are compliant with SAE J2735 [1].
- The BSM is transmitted as a wireless access in vehicular environments (WAVE) short message (WSM) using the WAVE short message protocol (WSMP) as defined in IEEE 1609.3 [4].
- BSM security is compliant with IEEE 1609.2 [3].
- The over-the-air medium access control (MAC) and physical layer (PHY) protocol are compliant with IEEE 1609.4 [5] and IEEE 802.11 [2].
- The WSM provider service ID (PSID) is set as specified in IEEE 1609.12 [6].

Section 6.1 of this standard profiles the applicable requirements from each of these standards.

5.2.2 System to SCMS Communications Interface

The system interfaces to the SCMS to request security credential generation, download security credentials, and receive CRLs. Learning security credentials from other systems that interface to the SCMS can also be used. The requirements listed in this standard are based on the following reports, which include dialogs and other SCMS interface details:

- CAMP Vehicle Safety Communications Security Studies: Study 1 - Security Credential Management System [10].
- CAMP Vehicle Safety Communications Security Studies: Study 3 Final Report: Protocols and Components of the SCMS [9].

Section 6.6 of this standard defines the applicable requirements for interfacing to the SCMS.

5.2.3 System to Positioning Subsystem Interface

The system has access to positioning subsystem information to meet the positioning and timing requirements of 6.2.

6. MINIMUM REQUIREMENTS

6.1 Standards Profiles (STD)

The standards compliance subsections below describe the requirements from the corresponding standards necessary to support V2V operation. Additional requirements apply if the system optionally supports the SCMS interface via DSRC-equipped roadside equipment (RSE). If a clause from the referenced standard is not explicitly referenced below, then it and its sub-clauses are considered informative or optional and not required for either V2V or SCMS operation, unless they are implicitly referenced from one of the mandatory clauses of that referenced standard.

6.1.1 IEEE 802.11 (802.11)

This section specifies the requirements from IEEE 802.11 [2] to support V2V and SCMS operation, as described in 6.1. In Table 5, items marked V2V in the “required for” column are always required, and items marked SCMS are required only if interfacing to an SCMS over DSRC is implemented.

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Table 5 - IEEE 802.11 requirements

802.11 Clause	Title (802.11 Clause)	Required For	Requirement
4.3.11	STA Transmission of Data Frames Outside the Context of a BSS	V2V	The DSRC radio subsystem shall operate within the procedures for STA transmission of data frames outside the context of a BSS (dot11OCBActivated=True), as specified. [6.1.1-V2V-STD-802.11-001]
5	MAC Service Definition		
5.1	Overview of MAC Services	V2V	The DSRC radio subsystem shall comply with MAC services with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-002]
5.2	MAC Data Service Specification	V2V	The DSRC radio subsystem shall comply with MAC data service specification features with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-003]
6	Layer Management		
6.3	MLME-SAP Interface		Note: MLME-SAP primitives provide guidance when determining actual requirements. The DSRC radio subsystem includes the specified functionality corresponding to each service primitive, but the method in which it is implemented is implementation specific.
6.3.10, 6.5.2	Reset	V2V	The DSRC radio subsystem shall be capable of changing the layer 2 address during operation. [6.1.1-V2V-STD-802.11-004]
6.3.42	Get TSF Timer	SCMS	If TSF timer capabilities are used for time synchronization, the DSRC radio subsystem shall use the TSF timer as specified. Other methods may also be utilized for enhanced time synchronization. [6.1.1-V2V-STD-802.11-005]
7	PHY Service Specification		
7.1	Scope	V2V	The DSRC radio subsystem shall be capable of PHY services with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-006]
7.2	PHY Functions	V2V	The DSRC radio subsystem shall be capable of PHY services with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-007]
7.3	Detailed PHY Service Specification	V2V	The DSRC radio subsystem shall be capable of PHY services with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-008]
8	Frame Formats		
8.1	General Requirements	V2V	The DSRC radio subsystem shall comply with the frame format general requirements with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-009]
8.2	MAC Frame Formats	V2V	The DSRC radio subsystem shall comply with the MAC frame formats with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-010]
8.3	Format of Individual Frame Types		
8.3.1	Control Frames		
8.3.1.1	Format of Control Frames	SCMS	If interfacing to an SMCS over DSRC is implemented by the system, the DSRC radio subsystem shall comply with the control frame format with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-011]
8.3.1.4	ACK Frame Format	SCMS	If interfacing to an SMCS over DSRC is implemented, the DSRC radio subsystem shall comply with the ACK frame format with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-012]

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802.11 Clause	Title (802.11 Clause)	Required For	Requirement
8.3.2	Data Frames	V2V, SCMS	The DSRC radio subsystem shall comply with the data frame with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-013]
8.3.2.1	Data Frame Format	V2V	The DSRC radio subsystem shall set the data frame fields to the following values for V2V operation: Frame Control (bits) • Protocol version = 00 • Type = 10 (Data) • Subtype = 1000 (QoS Data) • ToDS = 0 • FromDS = 0 • More Fragments = 0 • Retry = 0 • Power Mgmt = 0 • More Data = 0 • Protected Frame = 0 • Order = 0 Duration ID = 0 Address 1 (destination) = ff ff ff ff ff ff Address 2 (source) = <random 6 octets, changed per rules defined in 6.5.1> Address 3 (BSS ID) = ff ff ff ff ff ff Sequence Control • Fragment Number = 0 • Sequence Number = <incrementing value for each transmitted frame> Address 4 field is omitted. QoS Control • TID (bits 0-3) = <User Priority (0-7)> • EOSP (bit 4) = 0 • Ack Policy: bit 5=1, bit 6=0 (No ACK) • A-MSDU Present (bit 7) = 0 • Tx Op Duration Req (bits 8-15) = 0 HT Control field is omitted. Frame body content is defined by the higher layers and shall not exceed 1500 octets. FCS contains the 32-bit CRC of the MAC header and frame body field. [6.1.1-V2V-STD-802.11-014]

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802.11 Clause	Title (802.11 Clause)	Required For	Requirement
8.4.2.31	EDCA Parameter Set Element	V2V	The DSRC Radio Subsystem shall comply with the EDCA parameter set element, with the default EDCA values set as specified in 6.3.4 of this standard. [6.1.1-V2V-STD-802.11-015]
9	MAC Sublayer Functional Description		
9.2	MAC Architecture		
9.2.4	Hybrid Coordination Function	V2V	The DSRC radio subsystem shall comply with the enhanced distributed channel access (EDCA) mechanism of the hybrid coordination function (HCF) with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-016]
9.7	Multirate Support	V2V	The DSRC radio subsystem shall comply with the multirate support feature with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-017]
9.19	HCF		
9.19.1	General	V2V	The DSRC radio subsystem shall comply with the HCF general features with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-018]
9.19.2	HCF Contention-Based Channel Access (EDCA)	V2V	The DSRC radio subsystem shall comply with the HCF contention-based channel access (EDCA) features with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-019]
10	MLME		
10.1	Synchronization	V2V	The DSRC radio subsystem shall comply with the MLME synchronization feature with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-020]
10.20	STAs Communicating Data Frames Outside the Context of a BSS	V2V	The DSRC radio subsystem shall comply with the protocol of STAs communicating data frames outside the context of a BSS, with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-021]
18	Orthogonal Frequency Division Multiplexing (OFDM) PHY Specification	V2V	The DSRC radio subsystem shall comply with the orthogonal frequency division multiplexing (OFDM) PHY specification with dot11OCBActivated equal to true, with 10 MHz channel spacing, as specified. [6.1.1-V2V-STD-802.11-022]
Annex C	ASN.1 Encoding of the MAC and PHY MIB	V2V	The DSRC radio subsystem shall include the MIB items used to support OFDM and the features with dot11OCBActivated equal to true, as specified. [6.1.1-V2V-STD-802.11-023]
Annex D	Regulatory References		
Annex D.1	External Regulatory References	V2V	The DSRC radio subsystem shall comply with the United States Federal Communications Commission (FCC) specifications, as specified. [6.1.1-V2V-STD-802.11-024]
Annex D.1	External Regulatory References	V2V	The DSRC radio subsystem shall comply with the behavior limits set for ITS_mobile_operations, as specified. [6.1.1-V2V-STD-802.11-025]
Annex D2.1	Transmit and Receive In-Band and Out-of-Band Spurious Emissions	V2V	The DSRC radio subsystem shall comply with transmit and receive in-band and out-of-band spurious emissions, as specified. [6.1.1-V2V-STD-802.11-026]
Annex D2.2	Transmit Power Levels	V2V	The DSRC radio subsystem shall comply with the transmit power level requirements for STA transmit power classification C, as specified. [6.1.1-V2V-STD-802.11-027]
Annex D2.3	Transmit Spectrum Mask	V2V	The DSRC radio subsystem shall comply with the transmit spectrum mask requirements for 10 MHz channel spacing for STA transmit power classification C, as specified. [6.1.1-V2V-STD-802.11-028]
Annex E	Country Elements and Operating Classes		

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802.11 Clause	Title (802.11 Clause)	Required For	Requirement
Annex E.1	Country Information and Operating Classes	V2V	The DSRC radio subsystem shall comply with the country element and operating classes, as specified to support operating class 17, channel *vChannelNumber*. [6.1.1-V2V-STD-802.11-029]
Annex E.1	Country Information and Operating Classes	SCMS	If interfacing to an SMCS over DSRC is implemented, the DSRC radio subsystem shall comply with the country element and operating classes, as specified to support operating class 17. [6.1.1-V2V-STD-802.11-030]
Annex E.2.3	5.9 GHz Band in the United States (5.850-5.925 GHz)	V2V	The DSRC radio subsystem shall comply with the rules to support the 5.9 GHz band in the United States (5.850-5.925 GHz), as specified. [6.1.1-V2V-STD-802.11-031]

6.1.2 IEEE 1609.2 (1609.2)

This section specifies the requirements from IEEE 1609.2 [3] to support V2V and SCMS operation, as described in 6.1.

6.1.2.1 PICS Proforma (Informative)

This section provides an example protocol implementation conformance statement (PICS) from IEEE 1609.2 [3]. Implementers typically use a PICS to indicate compliance with particular features in the standard. The “item” column contains a feature identifier; the “security configuration” column contains a feature description; the “reference” column contains the clause number for the 1609.2 standard; and the “status” column indicates if the feature is mandatory or optional. Items marked with “M” are mandatory, and items marked with “O” are optional. Multiple items marked with O followed by a number (e.g., O1) indicate that the implementer chooses at least one of the options. Finally, items marked C followed by a number (e.g., C1) indicate that the implementer chooses one of the two features.

In this section, items marked “Y” in the “support” column correspond to requirements in the security profile provided in 6.1.2.2, and items marked “N” do not correspond to the security profile. In some cases a value is included to provide further guidance to implementers. Items marked N/A do not apply to the security profile, and items marked O are optional within the security profile.

Table 6 - IEEE 1609.2 security services conformance statement

Item	Security Configuration (Top-Level)	Reference	Status	Support
S1.	**Support Secure Data Service**		O1	Y
S1.1.	**Secure Data Exchange Entity (SDEE) Identification**	4.2.2.1	S1:M	Y
S1.1.1.	Support only one SDEE	4.2.2.1	S1:C1	Choose one of these items
S1.1.2.	Distinguish between SDEEs	4.2.2.1	S1:C1	
S1.2.	**Generate SPDU**		S1:O2	Y
S1.2.1.	**Create Ieee1609Dot2Data containing Unsecured Data**	4.2.2.2.2	S1.2:O3	N
S1.2.2.	**Create Ieee1609Dot2Data containing valid SignedData**	4.2.2.2.3, 5.2, 5.3.1, 5.3.3, 5.3.7, 6.3.4, 6.3.9, 9.3.9.1	S1.2:O3	Y
S1.2.2.1.	Using a valid HashAlgorithm	6.3.5	S1.2.2:M	Y
S1.2.2.1.1.	Support signing with hash algorithm SHA-256	6.3.5	S1.2.2:O3a	Y
S1.2.2.1.2.	Support signing with hash algorithm SHA-384	6.3.5	S1.2.2:O3a	N
S1.2.2.1.3.	Support signing with other hash algorithm	6.3.5	S1.2.2:O	N
S1.2.2.2.	Containing a Signed Data payload	6.3.6	S1.2.2:M	Y
S1.2.2.2.1.	… with payload containing data	6.3.7	S1.2.2.2:O4	Y
S1.2.2.2.2.	… with payload containing extDataHash	6.3.7	S1.2.2.2: O4	N

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Item	Security Configuration (Top-Level)	Reference	Status	Support
S1.2.2.2.3.	… with generationTime in the security headers	6.3.9, 6.3.11	S1.2.2.2: O	Y
S1.2.2.2.4.	… with expiryTime in the security headers	6.3.9, 6.3.11	S1.2.2.2: O	N
S1.2.2.2.5.	… with generationLocation in the security headers	6.3.9, 6.3.12	S1.2.2.2: O	N
S1.2.2.2.6.	… with p2pcdLearningRequest in the security headers	6.3.9, 6.3.26	S1.2.2.2: O	O
S1.2.2.2.7.	… with missingCrlIdentifier in the security headers	6.3.9, 6.3.16	S1.2.2.2: O	O
S1.2.2.2.8.	… with encryptionKey in the security headers	6.3.9, 6.3.18	S1.2.2.2: O	N
S1.2.2.2.8.1.	… … With a PublicEncryptionKey	6.3.9, 6.3.18, 6.3.19	S1.2.2.2.8:O5	N
S1.2.2.2.8.2.	… … With a SymmetricEncryptionKey	6.3.9, 6.3.18, 6.3.20	S1.2.2.2.8:O5	N
S1.2.2.3.	Support a SignerIdentifier	6.3.25	S1.2.2:M	Y
S1.2.2.3.1.	… of type self	6.3.24	S1.2.2.3:O6	N
S1.2.2.3.2.	… of type digest	6.3.27	S1.2.2.3:O6	Y
S1.2.2.3.3.	… of type certificate	6.4.2	S1.2.2.3:O6	Y
S1.2.2.3.3.1.	… … Maximum number of certificates included in the SignerIdentifier	6.3.25	S1.2.2.3.2 1:M > 1:O	1
S1.2.2.4.	Support a Signature	6.3.28	S1.2.2:M	Y
S1.2.2.4.1.	… an ecdsa256Signature	6.3.29	S1.2.2.4:M	Y
S1.2.2.4.1.1.	… … using NIST p256	6.3.29	S1.2.2.4.1:O7	Y
S1.2.2.4.1.2.	… … using Brainpool p256r1	6.3.29	S1.2.2.4.1:O7	N
S1.2.2.4.1.3.	… … with a x-only r value	6.3.29	S1.2.2.4.1:O8	Choose at least one of these items
S1.2.2.4.1.4.	… … with a compressed r value	6.3.29	S1.2.2.4.1:O8	
S1.2.2.4.1.5.	… … with an uncompressed r value	6.3.29	S1.2.2.4.1:O8	N
S1.2.2.4.1.6.	… an ecdsa384Signature using Brainpool p384r1	6.3.29a	S1.2.2.4:O6a	N
S1.2.2.4.1.7.	… … with a x-only r value	6.3.29a	S1.2.2.4.2:O8	N/A
S1.2.2.4.1.8.	… … with a compressed r value	6.3.29a	S1.2.2.4.2:O8	N/A
S1.2.2.4.1.9.	… … with an uncompressed r value	6.3.29a	S1.2.2.4.2:O8	N/A
S1.2.2.5.	Determine that certificate used to sign data is valid (part of a consistent chain, valid at the current time and location, has not been revoked)	5.2	S1.2.2:M	Y
S1.2.2.5.1.	Determine that the generation location is consistent with the region in the certificate	5.2.3.2.2, 6.4.17	S1.2.2.5:M	Y
S1.2.2.5.1.1.	Support a circularRegion	6.4.17, 6.4.18	S1.2.2.5.1:O9	N
S1.2.2.5.1.2.	Support a rectangularRegion	6.4.17, 6.4.20	S1.2.2.5.1:O9	N
	Maximum number of rectangularRegions supported	6.4.17, 6.4.20	S1.2.2.5.1.2 8:M > 8:O	N/A
S1.2.2.5.1.3.	Support a polygonalRegion	6.4.17, 6.4.21	S1.2.2.5.1:O9	N
	Maximum number of points in a polygonalRegion	6.4.17, 6.4.21	S1.2.2.5.1.3 8:M > 8:O	N/A
S1.2.2.5.1.4.	Support identifiedRegion	6.4.17, 6.4.22	S1.2.2.5.1:O9	Y

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Item	Security Configuration (Top-Level)	Reference	Status	Support
	Maximum number of identifiedRegions supported	6.4.17, 6.4.22	S1.2.2.5.1.4: 8:M > 8:O	Minimum of 3
	Support IdentifiedRegion of type CountryOnly	6.4.22, 6.4.23	S1.2.2.5.1.4:O10	Y
	Support IdentifiedRegion of type CountryAndRegions	6.4.22, 6.4.24	S1.2.2.5.1.4:O10	N
	Support IdentifiedRegion of type CountryAndSubregions	6.4.22, 6.4.25	S1.2.2.5.1.4:O10	N
	List of supported IdentifiedRegions	5.2.3.4, 6.4.22	S1.2.2.5.1.4:M	At least US, Canada, and Mexico (see item 1.2.2.5.1.4.1 and the security profile)
S1.2.2.5.2.	Determine that the certificate has the proper appPermissions	6.4.8, 6.4.28	S1.2.2.5:M	Y
S1.2.2.5.2.1.	Maximum number of PsidSsp in the appPermissions sequence	6.4.8, 6.4.28	S1.2.2.5.2 8:M > 8:O	8 Note: This document only describes behavior associated with PSID 0x20.
S1.2.2.5.3	Maximum supported length of the full chain (sending)	5.1.2.2	S1.2.2.5: 2:M >2:O	2 Note: The length of the full chain may be up to 8 but is recommended to be as short as possible.
S1.2.2.6.	Determine that key and certificate used to sign are a valid pair	5.3.7	S1.2.2:M	Y
S1.2.2.7.	Support signing with explicit certificates	6.4.6	S1.2.2.5:O11	N
S1.2.2.8.	Support signing with implicit certificates	5.3.2, 6.4.5	S1.2.2.5:O11	Y
S1.2.2.9.	Generate ECDSA keypairs using a high-quality random number generator	5.3.6	S1.2.2.4.1: M	Y
S1.2.3.	**Create Ieee1609Dot2Data containing EncryptedData**	4.2.2.3.2, 5.3.4, 6.3.30	S1.2:O2	N
S1.3.	**Receive SPDU**		S1:O2	Y
S1.3.1.	Support preprocessing SPDUs	4.2.2.3.1	S1.3.2.3.1, S3.2 S3.3:M	Y
S1.3.2.	Verify Ieee1609Dot2Data containing SignedData	4.2.2.2.3, 5.2, 5.3.1, 5.3.3, 5.3.7, 6.3.4, 6.3.9	S1.3:O17	Y
S1.3.2.1.	Using a valid HashAlgorithm		S1.3.2:M	Y
S1.3.2.1.1.	Verify signed data using HashAlgorithm SHA-256	6.3.5	S1.3.2.1:O17a	Y
S1.3.2.1.2.	Verify signed data using a HashAlgorithm other than SHA-384	6.3.5	S1.3.2.1:O17a	N
S1.3.2.1.3.	Verify signed data using another HashAlgorithm	6.3.5	S1.3.2.1:O17a	N
S1.3.2.2.	Containing a Signed Data payload	6.3.6	S1.3.2:M	Y
S1.3.2.2.1.	… with payload containing data	6.3.7	S1.3.2.2:O18	Y
S1.3.2.2.2.	… with payload containing extDataHash	6.3.7	S1.3.2.2:O18	N
S1.3.2.2.3.	… with generationTime in the security headers	6.3.9, 6.3.11	S1.3.2.2:O	Y
S1.3.2.2.4.	… with expiryTime in the security headers	6.3.9, 6.3.11	S1.3.2.2:O	N
S1.3.2.2.5.	… with generationLocation in the security headers	6.3.9, 6.3.12	S1.3.2.2:O	N

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Item	Security Configuration (Top-Level)	Reference	Status	Support
S1.3.2.2.6.	… with missingCertIdentifier in the security headers	6.3.9, 6.3.24	S1.3.2.2:O	O
S1.3.2.2.7.	… with missingCrlIdentifier in the security headers	6.3.9, 6.3.16	S1.3.2.2:O	O
S1.3.2.2.8.	… with encryptionKey in the security headers	6.3.9, 6.3.18	S1.3.2.2:O	N
S1.3.2.2.8.1.	… … With a PublicEncryptionKey	6.3.9, 6.3.18, 6.3.19	S1.3.2.2.8:O19	N
S1.3.2.2.8.2.	… … With a SymmetricEncryptionKey	6.3.9, 6.3.18, 6.3.20	S1.3.2.2.8:O19	N
S1.3.2.3.	Support a SignerIdentifier	6.3.23	S1.3.2:M	Y
S1.3.2.3.1.	… of type self	6.3.24	S1.3.2.3:O20	N
S1.3.2.3.2.	… of type digest	6.3.25	S1.3.2.3:O20	Y
S1.3.2.3.3.	… of type certificate	6.4.2	S1.3.2.3:O20	Y
S1.3.2.3.3.1.	… … Maximum number of certificates included in the SignerIdentifier	6.3.25	S1.3.2.3.2 1:M > 1:O	1
S1.3.2.3.4.	… of type self		S1.3.2.3:O20	N
S1.3.2.4.	Support a Signature	6.3.28	S1.3.2:M	Y
S1.3.2.4.1.	… a ecdsa256Signature	6.3.30	S1.3.2.4:M	Y
S1.3.2.4.1.1.	… … using NIST p256	6.3.30	S1.3.2.4.1:O21	Y
S1.3.2.4.1.2.	… … using Brainpool p256r1	6.3.30	S1.3.2.4.1:O21	N
S1.3.2.4.1.3.	… … with a x-only r value	6.3.30	S1.3.2.4.1:O22	Y
S1.3.2.4.1.4.	… … with a compressed r value	6.3.30	S1.3.2.4.1:O22	Y
S1.3.2.4.1.5.	… … with a compressed r value and fast verification	6.3.30	S1.3.2.4.1:O22	O
S1.3.2.4.1.6.	… … with a uncompressed r value	6.3.30	S1.3.2.4.1:O22	N
S1.3.2.4.1.7.	… … with a uncompressed r value and fast verification	6.3.30	S1.3.2.4.1:O22	N
S1.3.2.4.2.	… a ecdsa384Signature using Brainpool p384r1	6.3.29	S1.3.2.4:M	N
S1.3.2.4.2.1.	… … with a x-only r value	6.3.29	S1.3.2.4.1:O22	N/A
S1.3.2.4.2.2.	… … with a compressed r value	6.3.29	S1.3.2.4.1:O22	N/A
S1.3.2.4.2.3.	… … with a compressed r value and fast verification	6.3.29	S1.3.2.4.1:O22	N/A
S1.3.2.4.2.4.	… … with a uncompressed r value	6.3.29	S1.3.2.4.1:O22	N/A
S1.3.2.4.2.5.	… … with a uncompressed r value and fast verification	6.3.29	S1.3.2.4.1:O22	N/A
S1.3.2.5.	SignedData verification fails if the certificate is not valid (part of a consistent chain, valid at the current time and location, has not been revoked)	5.2, 6.4.2	S1.3.2:M	Y
S1.3.2.5.1.	Reject data based on generation location being inconsistent with certificate	6.4.8, 6.4.17	S1.3.2.5:O	Y Note: the position information comes from the BSM payload.
S1.3.2.5.1.1.	… using a circularRegion	6.4.17, 6.4.18	S1.3.2.5.1:O23	N
S1.3.2.5.1.2.	Support a rectangularRegion	6.4.17, 6.4.20	S1.3.2.5.1:O23	N
S1.3.2.5.1.3.	Maximum number of rectangularRegions supported	6.4.17, 6.4.20	S1.3.2.5.1.2 8:M > 8:O	N/A
S1.3.2.5.1.4.	Support a polygonalRegion	6.4.17, 6.4.21	S1.3.2.5.1:O23	N
S1.3.2.5.1.5.	Maximum number of points in a polygonalRegion	6.4.17, 6.4.21	S1.3.2.5.1.4 8:M > 8:O	N

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Item	Security Configuration (Top-Level)	Reference	Status	Support
S1.3.2.5.1.6.	Support identifiedRegion	6.4.17, 6.4.22	S1.3.2.5.1 8:M > 8:O	Y
	Maximum number of identifiedRegions supported	6.4.17, 6.4.22	S1.3.2.5.1.6: 8:M > 8:O	8
	Support IdentifiedRegion of type CountryOnly	6.4.22, 6.4.23	S1.3.2.5.1.6:O24	Y
	Support IdentifiedRegion of type CountryAndRegions	6.4.22, 6.4.24	S1.3.2.5.1.6:O24	N
	Support IdentifiedRegion of type CountryAndSubregions	6.4.22, 6.4.25	S1.3.2.5.1.6:O24	N
1.3.2.5.1.6.5	List of supported IdentifiedRegions	6.4.17, 6.4.22	S1.2.2.5.1.4: M	At least US, Canada and Mexico (see security profile)
S1.3.2.5.2.	Reject data if the certificate does not have the proper appPermissions	6.4.8, 6.4.28	S1.3.2.5:M	Y
S1.3.2.5.3.	Maximum number of PsidSsp in the appPermissions sequence	6.4.8, 6.4.28	S1.3.2.5 8:O > 8:O	Minimum of 2
S1.3.2.5.4.	Determine that the assuranceLevel is an acceptable level	6.4.8, 6.4.27	S1.3.2.5:O	N
S1.3.2.5.5.	Maximum length of the full chain (receiving)	5.1.2.2	S1.2.2.5: 2:M >2:O	8
S1.3.2.6.	Support verifying SPDUs signed with explicit authorization certificates	6.4.5	S1.3.2:O25	N
S1.3.2.7.	Support verifying SPDUs signed with implicit authorization certificates	5.3.2, 6.4.5	S1.3.2:O25	Y
S1.3.2.8.	Support explicit CA certificates	6.4.2, 6.4.6	S1.3.2:M	Y
S1.3.2.9.	Support receiving implicit CA certificates	6.4.2, 6.4.5	S1.3.2:O	N
S1.3.2.10.	SignedData verification fails in the following circumstances:	6.3.4	S1.3.2:M	Y
S1.3.2.10.1.	… SPDU-Parsing: Invalid Input	6.3.4	S1.3.2.10:M	Y
S1.3.2.10.2.	… SPDU-Parsing: Unsupported critical information field	6	S1.3.2.10:M	Y
S1.3.2.10.3.	… SPDU-Parsing: Certificate not found	4.3, 6.3.13, 6.3.14, 6.3.15	S1.3.2.10:M	Y
S1.3.2.10.4.	… SPDU-Parsing:Generation time not available	4.3, 6.3.13, 6.3.14, 6.3.15	S1.3.2.10:M	Y
S1.3.2.10.5.	… SPDU-Parsing:Generation location not available	4.3, 6.3.13, 6.3.14, 6.3.15	S1.3.2.10:M	Y
S1.3.2.10.6.	… SPDU-Certificate-Chain: Not enough information to construct chain	5.1.2	S1.3.2.10:M	Y
S1.3.2.10.7.	… SPDU-Certificate-Chain: Chain ended at untrusted root	5.1.2	S1.3.2.10:M	Y
S1.3.2.10.8.	… SPDU-Certificate-Chain: Chain was too long for implementation	5.1.2	S1.3.2.10:M	Y
S1.3.2.10.9.	… SPDU-Certificate-Chain: Certificate revoked	5.1.2	S1.3.2.10:M	Y
S1.3.2.10.10.	… SPDU-Certificate-Chain: Overdue CRL	5.1.2	S1.3.2.10:M	Y
S1.3.2.10.11.	… SPDU-Certificate-Chain: Inconsistent expiry times	5.1.2	S1.3.2.10:M	Y
S1.3.2.10.12.	… SPDU-Certificate-Chain: Inconsistent start times	5.1.2	S1.3.2.10:M	Y
S1.3.2.10.13.	… SPDU-Certificate-Chain: Inconsistant chain permissions	5.1.2	S1.3.2.10:M	Y

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Item	Security Configuration (Top-Level)	Reference	Status	Support
S1.3.2.10.14.	… SPDU-Crypto: Verification failure	5.3.1	S1.3.2.10:M	Y
S1.3.2.10.15.	… SPDU-Consistency: Future certificate at generation time	5.2.3	S1.3.2.10:M	Y
S1.3.2.10.16.	… SPDU-Consistency: Expired certificate at generation time	5.2.3	S1.3.2.10:M	Y
S1.3.2.10.17.	… SPDU-Consistency: Expiry date too early	5.2.3	S1.3.2.10:M	Y
S1.3.2.10.18.	… SPDU-Consistency: Expiry date too late	5.2.3	S1.3.2.10:M	Y
S1.3.2.10.19.	… SPDU-Consistency: Generation location outside validity region	5.2.3	S1.3.2.10:M	Y
S1.3.2.10.20.	… SPDU-Consistency: Unauthorized PSID	5.2.3	S1.3.2.10:M	Y
S1.3.2.10.21.	… SPDU-Internal-Consistency: Expiry time before generation time	6.4.8, 6.4.14, 5.2.3	S1.3.2.10:M	N/A
S1.3.2.10.22.	… SPDU-Internal-Consistency: extDataHash doesn't match	5.2.3	S1.3.2.10:M	Y
S1.3.2.10.23.	… SPDU-Local-Consistency: PSIDs don't match	5.2.3	S1.3.2.10:O	Y
S1.3.2.10.24.	… SPDU-Local-Consistency: Chain was too long for SDEE	5.2.3	S1.3.2.10:M	Y
S1.3.2.10.25.	… SPDU-Relevance: SPDU Too Old	5.2.4	S1.3.2.10:O	Y
S1.3.2.10.26.	… SPDU-Relevance: Future SPDU	5.2.4	S1.3.2.10:O	Y
S1.3.2.10.27.	… SPDU-Relevance: Expired SPDU	5.2.4	S1.3.2.10:O	N
S1.3.2.10.28.	… SPDU-Relevance: SPDU Too Distant	5.2.4	S1.3.2.10:O	N
S1.3.2.10.29.	… SPDU-Relevance: Replayed SPDU	5.2.4	S1.3.2.10:O	N
S1.3.3.	Decrypt Ieee1609Dot2Data containing EncryptedData	4.2.2.3.3, 5.3.5, 6.3.32	S1.3:O17	N

Table 7 - IEEE 1609.2 CRL verification entity conformance statement

Item	Security Configuration (Top-Level)	Reference	Status	Support
S2.	Support CRL Validation Entity	7	O1	Y
S2.1.	Correctly verify received CRL	7.4	S2:M	Y
S2.1.1.	…using hash ID-based revocation	5.1.3.5	S2.1:O29	Y
S2.1.1.1.	… of type fullHashCrl	7.3.2	S2.1.1:M	Y
S2.1.1.2.	… of type deltaHashCrl	7.3.2	O	O
S2.1.2.	… using linkage-based revocation	5.1.3.4	S2.1:O29	Y
S2.1.2.1.	… of type fullLinkedCrl	7.3.2	S2.1.2:M	Y
S2.1.2.2.	… of type deltaLinkedCrl	7.3.2	O	O
S2.1.2.3.	… containing individual linkage values	7.3.6	S2.1.2:M	Y
S2.1.2.4.	… containing group linkage values	7.3.6	O	Y

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Table 8 - IEEE 1609.2 peer-to-peer certificate distribution conformance statement

Item	Security Configuration (Top-Level)	Reference	Status	Support
S3.	Support Peer to Peer Certificate Distribution (P2PCD)	8	O	O (following P2PCD features are conditional on this being implemented)
S3.1.	Number of supported SDEEs	8.2.6	S3.2: 1:O > 1:O	Minimum of 1
S3.2.	Support out-of-band P2PCD operations	8	S3:O30	O
S3.3.	Support SSME and SDS operations for out-of-band P2PCD in the requester role	8.2.4.1.1	S3:O30	Y
S3.3.1.	Under at least one condition, trigger request processing on receiving a trigger SPDU	8.2.4.1.1	S3.3:M	Always trigger learning based on the conditions in 1609.2-2016 clause 8.2.4.1 step a.
S3.3.2.	Do not trigger request processing on receiving a trigger SPDU for which a request is already active	8.2.4.1.1	S3.3:M	Y
S3.3.3.	Number of simultaneously active P2PCD learning requests	8.2.4.1.1	S3.3: 1:O > 1:O	Minimum of 1
S3.3.4.	When request processing is triggered, include a P2PCD learning request in the next SPDU for the trigger SDEE except in the following exception cases	8.2.4.1.1, 8.2.6	S3.3: M	Y
S3.3.4.1.	Do not include a P2PCD learning request if a learning request for the same certificate has been received within **p2pcd_observedRequestTimeout**	8.2.4.1.1	S3.3.4:O	Y
S3.3.4.2.	Only include one P2PCD learning request no matter how many learning requests have been triggered	8.2.4.1.1	S3.3.4: M	Y
S3.3.5.	Receive notifications from a P2PCDE that a P2PCD learning response has been received and use those to update the list of known certificates	8.2.4.1.1	S3.3: M	Y
S3.4.	**Support SSME and SDS operations for out-of-band P2PCD in the responder role**	8.2.4.2.2	S3:O30	Y
S3.4.1.	Trigger response processing on receiving a P2PCD learning request	8.2.4.2.2	S3.4:M	Y
S3.4.2.	Number of simultaneously active P2PCD learning responses	8.2.4.2.2, 8.2.6	S3.4: 1:O > 1:O	Minimum of 1
S3.4.3.	Do not trigger response processing if less than **p2pcd_responseActiveTimeout** has passed since last triggered	8.2.4.2.2	S3.4: M	Y
S3.4.4.	Trigger sending response after random back-off time unless threshold number of responses have been observed	8.2.4.2.2	S3.4: M	Y
S3.4.5.	Increment number of responses observed based on input from P2PCDE	8.2.4.2.2	S3.4: M	Y
S3.5.	Support P2PCDE operations for P2PCD	8.2.4.2.2	S3:O30	Y

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Item	Security Configuration (Top-Level)	Reference	Status	Support
S3.5.1.	Receive responses and provide to SSME	8.2.4.1.2, 8.2.4.2.2, 8.3.1	S3.5: M	Y
S3.5.2.	Send responses when triggered by SSME	8.2.4.2.2, 8.3.1	S3.5: O	Y
S3.5.3.	Send responses over WSMP	8.2.4.2.2	S3.5.2: M	Y
S3.6.	Support inline P2PCD operations	8	S3:O30	N

6.1.2.2 BSM Security Profile Proforma (Normative)

A security profile proforma is provided in IEEE 1609.2 [3]. This profile is used to further state requirements for BSM transmissions on channel *vChannelNumber.* An example hexadecimal representation of a message based on this security profile and the corresponding ASN.1 is provided in Appendix A.9.

6.1.2.2.1 IEEE 1609.2 Security Profile Identification

- The system shall use the security profile identified in Table 9 [6.1.2-V2V-STD-1609.2-001].

Table 9 - Security profile identification

Field	Value	Notes
Security Profile Identification	IEEE Std 1609.2a-2017	
Name	Security profile for BSMs transmitted on channel *vChannelNumber* for light vehicles	
PSIDs	0x20	
Other Considerations	This security profile is to be used for BSMs transmitted on channel *vChannelNumber*	

6.1.2.2.2 Sending

The system shall use the security profile for sending defined in Table 10 [6.1.2-V2V-STD-1609.2-002].

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Table 10 - Security profile for transmitting BSMs

Field	Value	Notes
Sign Data	True	All BSMs are signed
Signed Data in Payload	True	BSM data is encapsulated in the signed data
External Data	False	No additional data is signed
External Data Source	N/A	
External Data Hash Algorithm	N/A	
Set Generation Time in Security Headers	True	Prevents replay attacks
Set Generation Location in Security Headers	False	Not necessary, already included in BSM
Set Expiry Time in Security Headers	False	Not necessary, application discards old messages
Signed SPDU Lifetime	N/A	
Signer Type Self	Prohibited	Certificates must be issued by a CA (SCMS)
Signer Identifier Policy Type	Simple	Refer to IEEE 1609.2 [3]
Simple Signer Identifier Policy: Minimum Inter Cert Time	*vMaxCertDigestInterval*	Attach certificates at a rate of 1/*vMaxCertDigestInterval*
Simple Signer Identifier Policy: Exceptions	True	Use full certificate when an event flag in DE_VehicleEventFlags is set
Simple Signer Identifier Policy: Signer Identifier Cert Chain Length	1	
Text Signer Identifier Policy	N/A	When sending certificates, send only the BSM signer certificate and not the CA certificates
Sign with Fast Verification	Optional	Doesn't change signature
EC Point Format	Compressed	Reduces packet size; refer to IEEE 1609.2 [3]
p2pcd_flavor	Out of Band (Optional)	Items indented in the rows below only apply if this option is implemented
p2pcd_maxResponseBackoff	*vP2pcd_maxResponseBackoff*	Wait no more than *vP2pcd_maxResponseBackoff*seconds before deciding to send a response
p2pcd_responseActiveTimeout	*vP2pcd_responseActiveTimeout*	Send a response no more than 1/*vP2pcd_responseActiveTimeout* per second
p2pcd_requestActiveTimeout	*vP2pcd_requestActiveTimeout*	*vP2pcd_requestActiveTimeout*
p2pcd_observedRequestTimeout	*vP2pcd_observedRequestTimeout*	*vP2pcd_observedRequestTimeout*
p2pcd_currentlyUsedTriggerCertificateTime	*vP2pcd_currentlyUsedTriggerCertificateTime*	Response only to requests for certificates that have been used within the *vP2pcd_currentlyUsedTriggerCertificateTime*
p2pcd_responseCountThreshold	*vP2pcd_responseCountThreshold*	Respond only if fewer than *vP2pcd_responseCountThreshold* responses were seen during the backoff time
Repeat Signed SPDUs	False	Each BSM is uniquely signed before transmission
Time Between Signing	N/A	
Encrypt Data	False	Encryption is not used for BSMs

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6.1.2.2.3 Receiving

When the system chooses to verify a received BSM, it shall use the security profile for receiving defined in Table 11 [6.1.2-V2V-STD-1609.2-003].

Table 11 - Security profile for receiving BSMs

Field	Value	Notes
Use Preprocessing	True	Store certificates even for messages that aren't being verified; used to verify a digest
Verify Data	True	
Maximum Certificate Chain Length	8	
Relevance: Replay	False	Application handles duplication within the validity period
Relevance: Generation Time in Past	True	Guards against replay attacks
Validity Period	30 seconds	Corresponds to ± the maximum value of DE_DSecond/2
Relevance: Generation Time in Future	Yes	Refer to IEEE 1609.2 [3]; applies to messages newer than the validity period allows; applications can filter messages inside the validity period
Acceptable Future Data Period	30 seconds	Corresponds to ± the maximum value of DE_DSecond/2
Generation Time Source	Security headers	Refer to IEEE 1609.2 [3]
Relevance: Expiry Time	False	
Expiry Time Source	N/A	
Consistency: Generation Location	True	Use the position data from the BSM to compare to validity region
Relevance: Generation Location Distance	False	
Validity Distance	N/A	
Generation Location Source	Payload	BSM
Additional Geographic Consistency Conditions	False	
Identified Region Representation Accuracy	Not Specified	Generation location consistency check is implementation specific (i.e., left to the implementer).
Overdue CRL Tolerance	3 years	Setting that corresponds to the number of certificates with which a system is initially equipped
Relevance: Certificate Expiry	True	Default recommended in IEEE 1609.2 [3]
Encrypted Data	False	Encryption is not used for BSMs

6.1.2.2.4 Security Management

The system shall comply with the security management profile defined in Table 12 [6.1.2-V2V-STD-1609.2-004].

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Table 12 - Security management profile

Field	Value	Notes
Signing Key Algorithm	ECDSA-256	This is the only supported signing algorithm in this standard
Encryption Algorithm	N/A	Encryption is not used for BSMs
Implicit or Explicit Certificates	Implicit	Reduces packet size; reduces verification time for verification on demand (refer to IEEE 1609.2 [3])
EC Point Format	Compressed	Reduces packet size
Supported Geographic Regions	Identified country only	U.S., Canada, Mexico Note: Any other geographic regions that choose to apply this standard will have to modify this item.
Maximum Certificate Chain Length	8	
Use Individual Linkage ID	True	Support privacy preserving revocation
Use Group Linkage ID	True	Support privacy preserving revocation
Signature Algorithms in Chain or CRL	ECDSA-256	This is the only supported signing algorithm in this (SAE J2945/1) standard

6.1.2.2.5 Other

Table 13 identifies security fields that may be subject to future policy updates.

Table 13 - Fields subject to policy updates

Field	Value	Notes
Fields that May be Subject to Policy Update	Overdue CRL tolerance, supported geographic regions, use peer-to-peer certificate distribution, signing key algorithm, signature algorithms in chain or CRL	These fields may be updated by a SCMS in the future

6.1.3 IEEE 1609.3 (1609.3)

This section specifies the requirements from IEEE 1609.3 [4] to support V2V and SCMS operation, as described in 6.1. Using the PICS from IEEE 1609.3 [4], the profile for BSM transmissions on channel *vChannelNumber* is provided in this section. Items left blank in the Support column are optional. Items marked V2V are required for transmitting and receiving signed BSMs, and items marked V2V may also be used for peer-to-peer certificate learning options in 6.1.2.2.2. Items marked SCMS are required only if interfacing to an SCMS over DSRC is supported. See 6.1.2.1 for further description regarding how to interpret an IEEE PICS.

- The DSRC radio subsystem shall comply with the V2V items identified in Table 14. If values are specified in the table, the items are set as stated [6.1.3-V2V-STD-1609.3-001].

- If the system supports interfacing to an SCMS over DSRC, the DSRC radio subsystem shall comply with the SCMS items identified in Table 14 [6.1.3-V2V-STD-1609.3-002].

Table 14 - IEEE 1609.3 requirements (PICS proforma)

Item	Feature	Value	Reference	Status	Support
N1	**DATA PLANE**		—	—	
N1.1	**LLC**		5.2	M	V2V
N1.1.1.	LLC extensions for WSMP		7.5	N1.3:M	V2V
N1.2	**IPv6**		5.3, 6.4	O1	SCMS
N1.2.1	Use stateless configuration		6.4	O	SCMS
N1.2.2	IP readdressing		6.4.2	M	SCMS
N1.2.3.	Send IP datagrams		5.3	O2	SCMS
N1.2.4.	Receive IP datagrams		5.3	O2	SCMS
N1.2.4.1.	Receive by link-local address		6.4	M	SCMS
N1.2.4.2.	Receive by global address		6.4	M	SCMS

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Item	Feature	Value	Reference	Status	Support
N1.2.4.3.	Receive by host multicast addresses		6.4	O3	
N1.2.4.4.	Receive by router multicast addresses		6.4	O3	
N1.2.5.	UDP		5.4	O	
N1.2.6.	TCP		5.4	O	SCMS
N1.2.7.	Other IETF protocols	( )[a]	5.4	O	
N1.3.	**WSMP**		5.5	O1	V2V
N1.3.1.	**WSM reception**		5.5.3	O4	V2V
N1.3.1.1.	Check WSMP version number	( )[b]	5.5.3, 8.3.2	M	V2V (Version = 3)
N1.3.1.2.	Check subtype field	( )[r]	5.5.3, 8.3.2	M	V2V (Subtype = 0 or 1)
N1.3.1.3.	Check TPID field	( )[s]	5.5.3, 8.3.2	M	V2V (TPID = 0)
N1.3.1.4.	WAVE info elem extension field		8.1.1	M	V2V
N1.3.1.5.	Deliver message based on address info (PSID)		5.5.3	M	V2V
N1.3.2.	**WSM transmission**		5.5.2	O4	V2V
N1.3.2.1.	Insert WSMP version number		8.3.2	M	V2V (Version = 3)
N1.3.2.2.	Insert address info (PSID)		8.3.3	M	V2V
N1.3.2.3.	Outbound message size	( )[c]	5.5.2	M	V2V (at least 1400 bytes)
N1.3.2.4.	Transmit channel number		8.3.4.2	O	
N1.3.2.5.	Transmit data rate		8.3.4.3	O	
N1.3.2.6.	Transmit power used Transmit power used is optionally included in the WSMP-N-Header for use by the WSM recipient. This standard uses this header to represent the radiated power (calculated) in units of dBm. As described in IEEE Std 802.11, transmit power used is represented by a signed integer number in the range -128 to 127. Note, however, that the field transmit power used is encoded for over the air transport using UPER, therefore, the transmit power used field in the WSM header has a range of 0 to 255 (refer to 8.1 of IEEE 1609.3). It is one octet in length.		8.3.4.4	O	V2V (set to calculated radiated power (RP) in dBm, see calculated RP per 6.3.8.7). Note: This can be used to support congestion control testing and verification.
N1.3.2.7.	Channel load		8.3.4.5	O	
N1.3.2.8.	Insert subtype features	( )[r]	8.3.2	M	V2V (Subtype = 0)
N1.3.2.9.	Insert TPID features	( )[s]	8.3.2	M	V2V (TPID = 0)
N2.	**MANAGEMENT PLANE**		—	—	
N2.1.	**User role**		6.2.1	O	SCMS
N2.1.1.	Receive WSAs over WSMP		6.3.2	O5	SCMS
N2.1.2.	Verify and accept secured WSA		6.3.3, 8.2.1	O5	SCMS
N2.1.3.	Accept unsecured WSA		6.3.3, 8.2.1	O5	
N2.1.4.	WAVE info elem extension fields		8.1.1	M	SCMS
N2.1.5.	Calculate avail service link quality		6.3.4	O	
N2.1.6.	**WSA header**		8.2.2	M	SCMS
N2.1.6.1.	Check WSA version number	( )[d]	8.2.2.2	M	SCMS (version = 3)
N2.1.6.2.	Check WSA identifier		8.2.2.4	O	
N2.1.6.3.	Check content count		8.2.2.5	O	
N2.1.6.4.	WSA header info element ext field		8.2.2.6	M	SCMS
N2.1.6.4.1.	Repeat rate		8.2.2.6.1	O	
N2.1.6.4.2.	2DLocation		8.2.2.6.2	O	
N2.1.6.4.3.	3DLocation		8.2.2.6.3	O	
N2.1.6.4.4.	Advertiser identifier		8.2.2.6.4	O	
N2.1.6.4.5.	Other info elements	( )[e]	8.2.2.6	O	

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Item	Feature	Value	Reference	Status	Support
N2.1.7.	**Service info segment**		8.2.3	M	SCMS
N2.1.7.1.	Number of service info instances	( )[f]	8.2.3	M	SCMS (at least 1)
N2.1.7.2.	WAVE information element extension		8.2.3.5	M	SCMS
N2.1.7.2.1.	PSC		8.2.3.5.1	O	SCMS
N2.1.7.2.2.	IPv6 address		8.2.3.5.2	O	SCMS
N2.1.7.2.3.	Service port		8.2.3.5.3	O	
N2.1.7.2.4.	Provider MAC address		8.2.3.5.4	O	SCMS
N2.1.7.2.5.	RCPI threshold		8.2.3.5.5	O	
N2.1.7.2.6.	WSA count threshold		8.2.3.5.6	O	
N2.1.7.2.6.1.	WSA count threshold interval		8.2.3.5.7	O	
N2.1.7.2.7.	Other info elements	( )[g]	8.2.3.5	O	
N2.1.8.	**Channel info segment**		8.2.4	M	SCMS
N2.1.8.1.	Number of channel info instances	( )[h]	8.2.4	M	SCMS (at least 1)
N2.1.8.2.	WAVE info elem extension field		8.2.4.8	M	SCMS
N2.1.8.2.1.	EDCA parameter set		8.2.4.8.1	O	SCMS
N2.1.8.2.2.	Channel access		8.2.4.8.2	O	SCMS
N2.1.8.2.3.	Other info elements	( )[i]	8.2.4.8	O	
N2.1.9.	**WAVE router advertisement**		8.2.5.1	O	SCMS
N2.1.9.1.	WAVE info elem extension field		8.2.5.7	M	SCMS
N2.1.9.1.1.	Secondary DNS		8.2.5.7.1	O	SCMS
N2.1.9.1.2.	Gateway MAC address		8.2.5.7.2	O	SCMS
N2.1.9.1.3.	Other info elements	( )[j]	8.2.5.7	O	
N2.2.	**Provider role**		6.2.1	O	Note: The provider role is not required for the system. Thus, the provider section of the PICS is not included.
N2.2.1.	**Timing advertisement**		—		
N2.3.1	Timing advertisement generation		6.2.4.3	O	
N2.4.	**MIB maintenance**		6.5	—	
N2.4.1	Managed WAVE device		6.5	O	
N2.4.2	MIB per standard		6.5	N2.4.1:M	
N2.4.3	Other MIB	( )[q]	6.5	O	

[a] List protocols supported.
[b] List version numbers supported.
[c] Enter maximum WAVE short message length.
[d] List version numbers supported.
[e] List any other WSA header WAVE information elements processed on reception.
[f] Enter maximum number of service info instances processed on reception.
[g] List any other service info segment WAVE information elements processed on reception.
[h] Enter maximum number of channel info instances processed on reception.
[i] List any other channel info segment WAVE information elements processed on reception.
[j] List any other WAVE routing advertisement WAVE information elements processed on reception.
[k] List any other WSA header WAVE information elements supported on transmission.
[l] Enter maximum number of service info instances supported on transmission.
[m] List any other service info segment WAVE information elements supported on transmission.
[n] Enter maximum number of channel info instances supported on transmission.
[o] List any other channel info segment WAVE information elements supported on transmission.
[p] List any other WAVE routing advertisement WAVE information elements supported on transmission.
[q] List any other MIBs supported.
[r] List subtype values supported.
[s] List TPID values supported.

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6.1.4 IEEE 1609.4 (1609.4)

This section specifies the requirements from IEEE 1609.4 [5] to support V2V and SCMS operation, as described in 6.1. Using the protocol implementation conformance statement from IEEE 1609.4 [5], the profile for BSM transmissions on channel *vChannelNumber* is provided in this section. Items left blank in the Support column are not identified for use by this standard. Items marked V2V are required for transmitting and receiving BSMs, and items marked SCMS are required only if interfacing to an SCMS over DSRC is supported. See 6.1.2.1 for further description regarding how to interpret an IEEE PICS.

- The DSRC radio subsystem shall comply with the V2V items identified in Table 15. If values are specified in the table, the items are set as stated [6.1.4-V2V-STD-1609.4-001].

- If the system supports interfacing to an SCMS over DSRC, the DSRC radio subsystem shall comply with the SCMS items identified in Table 15 [6.1.4-V2V-STD-1609.4-002].

Table 15 - IEEE 1609.4 requirements (PICS Proforma)

Item	Feature	Value	Reference	Status	Support
M1.	OCBActivated communication		5.1	M	V2V
M2.	Operation on CCH	( )[a]	5.2	O4	
M2.1.	Continuous CCH access		6.3.1	O	
M3.	Operation on SCH	( )[b]	5.2.1, 5.2.3	O4	V2V: USA, channel 172, Class C SCMS: USA, channels 172, 174, 176, 180, 182, 184, Class C
M3.1.	Continuous SCH access		6.3.1	O	V2V
M4.	**Mixed operation**		5.2	O	SCMS
M4.1.	Immediate access		6.3.3	O	SCMS
M4.2.	Alternating access		6.3.2	O	SCMS: Optional
M4.2.1.	Use common time reference		5.2.2, 6.2.2	M	
M4.2.1.1.	Derive timing from GPS		6.2.3	O5	
M4.2.1.2.	Derive timing from timing advertisement frame		6.2.3	O5	
M4.2.1.3.	Derive timing from other timing source	( )[c]	6.2.3	O5	
M4.2.2.	Guard interval on transmit		6.2.5	M	SCMS
M4.2.3.	Medium busy at end of guard interval		6.2.5	M	SCMS
M5.	**Transmit**		5.3.2	O2	V2V
M5.1.	EDCA and user priority		5.4	M	V2V
M5.2.	Cancel transmissions		5.3.2	O	
M5.3.	Send TA		6.2.6	O	
M5.4.	Send other IEEE 802.11 frames	( )[d]	6.4	O	
M5.5.	Send WSM		5.3.3	O3	V2V
M5.5.1.	Expiry time		5.3.3	O	
M5.6.	Send IPv6		5.3.4	O3	SCMS
M5.6.1.	Send IPv6 on SCH only		5.2.3	M	SCMS
M6.	**Receive**		5.3.5	O2	V2V
M6.1.	Receive TA		6.2.7	O	
M6.2.	Receive WSM		5.3.3	O3	V2V
M6.3.	Receive IPv6		5.3.4	O3	SCMS
M7.	**Device readdressing**		6.6	O	V2V

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Item	Feature	Value	Reference	Status	Support
M8.	**MIB maintenance**		6.5	—	
M8.1.	Managed WAVE device		3.1, 6.5	O	
M8.2.	IEEE 1609.4 MIB per Annex E		6.5	M8.1: M	
M8.3.	Other MIB	( )[e]	6.5	O	

[a] List supported control channel(s), including country and operating class.
[b] List supported service channel(s), including country and operating class.
[c] Indicate device's timing source(s).
[d] Enter IEEE 802.11 management frames/service request primitives supported.
[e] Enter references to other management information bases supported.

6.1.5 IEEE 1609.12 (1609.12)

This section specifies the features from IEEE 1609.12 [6] required for V2V and SCMS operation, as described in 6.1.

Table 16 - IEEE 1609.12 requirements

1609.12 Clause	Title (1609.12 Clause)	Required For	Requirement
4	WAVE Identifiers		
4.1.x	Provider Service Identifier (PSID)	V2V	The System shall set the PSID value to the value assigned to "vehicle-to-vehicle safety and awareness" (hex 20). [6.1.5-V2V-STD-1609.12-001]
4.1.x	Provider Service Identifier (PSID)	SCMS	The System shall set the PSID value to the value assigned to "WAVE security management" (hex 23). [6.1.5-V2V-STD-1609.12-002]
4.3	Ethertype	V2V	The System shall set the Ethertype value to the value assigned to WSMP (hex 88DC). [6.1.5-V2V-STD-1609.12-003]
4.3	Ethertype	SCMS	The System shall set the Ethertype value to the value assigned to IPv6 (hex 86DD). [6.1.5-V2V-STD-1609.12-004]

6.1.6 SAE J2735 (J2735)

This section identifies the BSM data frames and data elements used in this standard. All requirements in Table 17 refer to SAE J2735 unless otherwise noted. The functional and performance requirements corresponding to these data frames and data elements are specified in 6.2 through 6.6 of this standard.

Table 17 - SAE J2735 requirements

Title (SAE J2735 Clause)	Requirement
Message Encoding	The system shall conform to the ASN.1 representation of the basic safety message as specified. [6.1.6-V2V-STD-J2735-001]
MSG_MessageFrame	
Data Element: DE_DSRC_MessageID	The system shall conform to the data element DE_DSRC_MessageID, as specified. [6.1.6-V2V-STD-J2735-002]
Message: MSG_BasicSafetyMessage (BSM)	The system shall conform to Part I of the basic safety message, as specified. [6.1.6-V2V-STD-J2735-003]
Message: MSG_BasicSafetyMessage (BSM)	The system shall conform to Part II of the basic safety message as specified in SAE J2735 [1], this table, and 6.3 of this standard. [6.1.6-V2V-STD-J2735-004]
Data Frames	Note: The system includes the data frames required for Part I and Part II in the basic safety message as specified in SAE J2735 [1], this table, and 6.3 of this standard.
Data Frame: DF_AccelerationSet4Way	The system shall conform to DF_AccelerationSet4Way, as specified. [6.1.6-V2V-STD-J2735-005]
Data Frame: DF_BrakeSystemStatus	The system shall conform to DF_BrakeSystemStatus, as specified. [6.1.6-V2V-STD-J2735-006]

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Title (SAE J2735 Clause)	Requirement
Data Frame: DF_BSMcoreData	The system shall conform to DF_BSMcoreData, as specified. [6.1.6-V2V-STD-J2735-007]
Data Frame: DF_PathHistory	The system shall conform to DF_PathHistory, as specified. [6.1.6-V2V-STD-J2735-008]
Data Frame: DF_PathHistoryPointList	The system shall conform to DF_PathHistoryPointList, as specified. [6.1.6-V2V-STD-J2735-009]
Data Frame: DF_PathHistoryPoint	The system shall conform to DF_PathHistoryPoint, as specified. [6.1.6-V2V-STD-J2735-010]
Data Frame: DF_PathPrediction	The system shall conform to DF_PathPrediction, as specified. [6.1.6-V2V-STD-J2735-011]
Data Frame: DF_PositionalAccuracy	The system shall conform to DF_Positional Accuracy, as specified. [6.1.6-V2V-STD-J2735-012]
Data Frame: DF_VehicleSafetyExtensions	The system shall conform to DF_VehicleSafetyExtensions, as specified. [6.1.6-V2V-STD-J2735-013]
Data Frame: DF_VehicleSize	The system shall conform to DF_VehicleSize. [6.1.6-V2V-STD-J2735-014]
Data Elements	Note: The system includes the data elements from Part I and Part II of the basic safety message, as specified in SAE J2735 [1], this table, and 6.3 of this standard.
Data Element: DE_Acceleration	The system shall conform to DE_Acceleration, as specified. [6.1.6-V2V-STD-J2735-015]
Data Element: DE_AntiLockBrakeStatus	The system shall conform to DE_AntiLockBrakeStatus, as specified. [6.1.6-V2V-STD-J2735-016]
Data Element: DE_AuxiliaryBrakeStatus	The system shall conform to DE_AuxiliaryBrakeStatus, as specified. [6.1.6-V2V-STD-J2735-017]
Data Element: DE_BrakeAppliedStatus	The system shall conform to DE_BrakeAppliedStatus, as specified. [6.1.6-V2V-STD-J2735-018]
Data Element: DE_BrakeBoostApplied	The system shall conform to DE_BrakeBoostApplied, as specified. [6.1.6-V2V-STD-J2735-019]
Data Element: DE_Confidence	The system shall conform to DE_Confidence, as specified. [6.1.6-V2V-STD-J2735-020]
Data Element: DE_DSecond	The system shall conform to DE_DSecond, as specified. [6.1.6-V2V-STD-J2735-021]
Data Element: DE_Elevation	The system shall conform to DE_Elevation, as specified. [6.1.6-V2V-STD-J2735-022]
Data Element: DE_ExteriorLights	When DE_ExteriorLights is included in the BSM, the system shall conform to DE_ExteriorLights, as specified. [6.1.6-V2V-STD-J2735-023]
Data Element: DE_Heading	The system shall conform to DE_Heading, as specified. [6.1.6-V2V-STD-J2735-024]
Data Element: DE_Latitude	The system shall conform to DE_Latitude, as specified. [6.1.6-V2V-STD-J2735-025]
Data Element: DE_Longitude	The system shall conform to DE_Longitude, as specified. [6.1.6-V2V-STD-J2735-026]
Data Element: DE_MsgCount	The system shall conform to DE_MsgCount, as specified. [6.1.6-V2V-STD-J2735-027]
Data Element: DE_OffsetLL-B18	The system shall conform to DE_OffsetLL-B18, as specified. [6.1.6-V2V-STD-J2735-028]
Data Element: DE_RadiusOfCurvature	The system shall conform to DE_RadiusOfCurvature, as specified. [6.1.6-V2V-STD-J2735-029]
Data Element: DE_SemiMajorAxisAccuracy	The system shall conform to DE_SemiMajorAxisAccuracy, as specified. [6.1.6-V2V-STD-J2735-030]
Data Element: DE_SemiMajorAxisOrientation	The system shall conform to DE_SemiMajorAxisOrientation, as specified. [6.1.6-V2V-STD-J2735-031]
Data Element: DE_SemiMinorAxisAccuracy	The system shall conform to DE_SemiMinorAxisAccuracy, as specified. [6.1.6-V2V-STD-J2735-032]
Data Element: DE_Speed	The system shall conform to DE_Speed, as specified. [6.1.6-V2V-STD-J2735-033]

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Title (SAE J2735 Clause)	Requirement
Data Element: DE_StabilityControlStatus	The system shall conform to DE_StabilityControlStatus, as specified. [6.1.6-V2V-STD-J2735-034]
Data Element: DE_SteeringWheelAngle	The system shall conform to DE_SteeringWheelAngle, as specified. [6.1.6-V2V-STD-J2735-035]
Data Element: DE_TemporaryID	The system shall conform to DE_TemporaryID, as specified. [6.1.6-V2V-STD-J2735-036]
Data Element: DE_TimeOffset	The system shall conform to DE_TimeOffset, as specified. [6.1.6-V2V-STD-J2735-037]
Data Element: DE_TractionControlStatus	The system shall conform to DE_TractionControlStatus, as specified. [6.1.6-V2V-STD-J2735-038]
Data Element: DE_TransmissionState	The system shall conform to DE_TransmissionState, as specified. [6.1.6-V2V-STD-J2735-039]
Data Element: DE_VehicleEventFlags	When DE_VehicleEventFlags is included in the BSM, the system shall conform to DE_VehicleEventFlags as specified in SAE J2735 [1] and this standard. [6.1.6-V2V-STD-J2735-040]
Data Element: DE_VehicleLength	The system shall conform to DE_VehicleLength, as specified. [6.1.6-V2V-STD-J2735-041]
Data Element: DE_VehicleWidth	The system shall conform to DE_VehicleWidth, as specified. [6.1.6-V2V-STD-J2735-042]
Data Element: DE_VerticalAcceleration	The system shall conform to DE_VerticalAcceleration, as specified. [6.1.6-V2V-STD-J2735-043]
Data Element: DE_VertOffset-B12	The system shall conform to DE_VertOffset-B12, as specified. [6.1.6-V2V-STD-J2735-044]
Data Element: DE_YawRate	The system shall conform to DE_YawRate, as specified. [6.1.6-V2V-STD-J2735-045]

6.1.7 FCC 47 CFR, Parts 0, 1, 2, and 95 (Informative)

Regulatory requirements for the DSRC radio subsystem within the United States are available from the Federal Communications Commission (FCC), Title 47 of the Code of Federal Regulations (CFR), Parts 0, 1, 2, and 95.

NOTE: Regulatory domains outside the United States may be subject to different regulatory requirements. FCC type certifications are required in the United States as defined in the CFR.

6.2 Positioning and Timing Requirements (POSTIM)

6.2.1 Position Determination (POSDETER)

- The positioning subsystem shall include a GNSS receiver. In the United States, this requires that the GNSS receiver includes GPS [6.2.1-V2V-POSTIM-POSDETER-001].

NOTE: Additional positioning and augmentation capabilities can be used.

- The system shall determine the position[5] of the vehicle as defined in 6.2.3 at a nominal rate of *vPosDetRate* and the Coordinated Universal Time (UTC) when at that position [6.2.1-V2V-POSTIM-POSDETER-002].

NOTE: In order to send a position estimate that is more recent than that provided by the positioning subsystem, the system can estimate vehicle position at the current time, for example using the method described in Appendix A.3. DE_DSecond represents the time at which position and positional accuracy were determined (see 6.3.6.4).

NOTE: Position and UTC are estimates and are subject to the accuracy requirements in 6.3.6.4, 6.3.6.5, and 6.3.6.6.

[5] Position is based on the position estimate provided by the positioning subsystem and any additional sensors. The method of computation of position estimates is outside the scope of this standard.

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6.2.2 Wide Area Augmentation System (WAAS)

- The positioning subsystem shall use WAAS corrections, when the WAAS signal is available to the subsystem, in order to improve the position accuracy. If the WAAS signal is not available to the positioning subsystem, the GNSS and other sensors can still be used [6.2.2-V2V-POSTIM-WAAS-001].

NOTE: This requirement establishes a minimum requirement for common source of positional corrections used by the GNSS receivers in V2V systems and provides better accuracy of the resulting relative position between vehicles. Additional augmentation and correction systems may be used in addition to WAAS.

6.2.3 Coordinate System and Reference (COORDSYSREF)

The vehicle position (position reference) reported in a BSM shall be a point (latitude, longitude, and elevation) projected onto the surface of the roadway (road plane) with reference to the WGS-84 coordinate system and its reference ellipsoid. This point is the center of the rectangle on the road plane, oriented about the vehicle that encompasses the farthest forward, rearward, and side-to-side points on the vehicle, including original equipment such as outside side view mirrors (see Figure 21) [6.2.3-V2V-POSTIM-COORDSYSREF-001].

NOTE: The GNSS antenna position is not the same as the position reference. See Appendix A.4 for an example of how to translate from the position of the GNSS antenna to the position reference. Incline of the road results in negligible change in position reference, but testing can be done under flat road test conditions (grade < 0.2% and cross-slope < 2%) to avoid introducing related biases during the test.

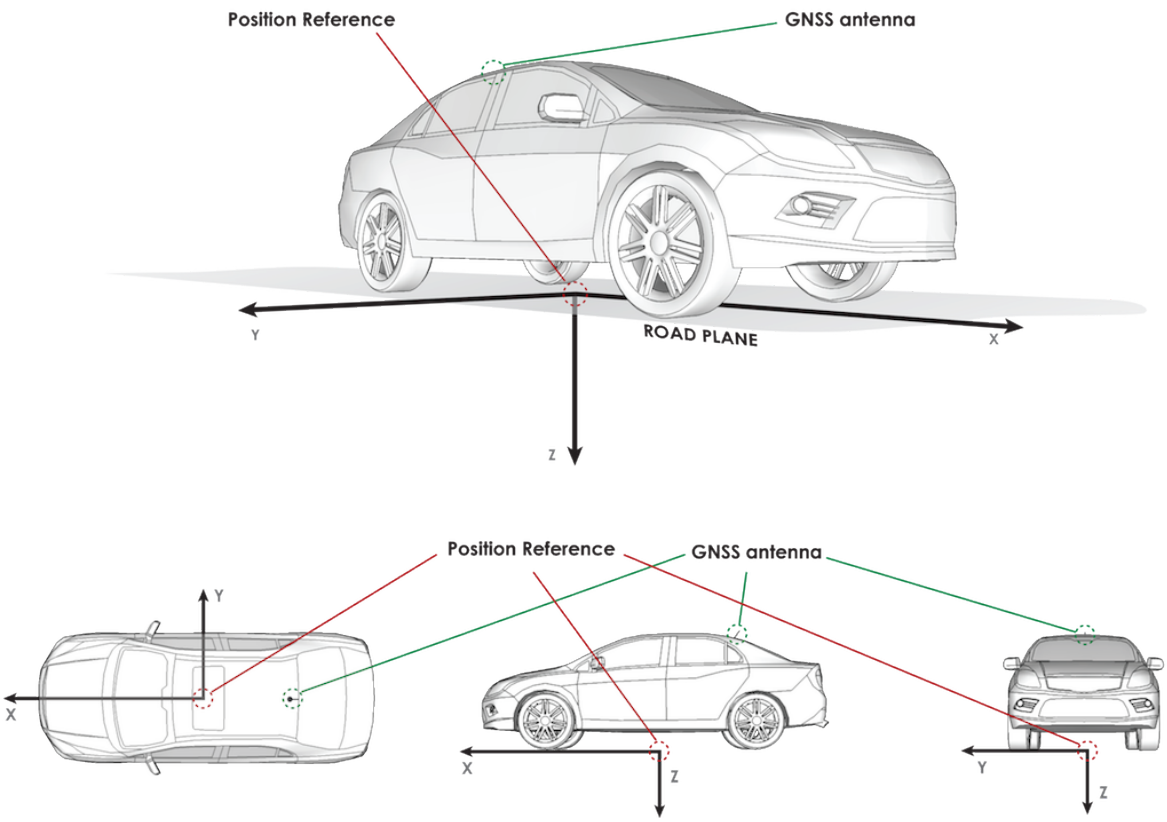

Figure 21 - BSM position reference

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6.2.4 System Time Coordination (SYSTIMCOORD)

- The system shall include a reference clock with an output that conforms to UTC [15]. [6.2.4-V2V-POSTIM-SYSTIMCOORD-001]

- The UTC-conformant output of the reference clock shall be accurate to within *vTimeAccuracy* of the UTC reference. [6.2.4-V2V-POSTIM-SYSTIMCOORD-002]

- The System shall determine the time at which position is determined, using the UTC-conformant reference. [6.2.4-V2V-POSTIM-SYSTIMCOORD-003]

NOTE: The system has time (DE_DSecond) synchronized to UTC in order to support position and time extrapolation, and security requirements. The system may implement the reference clock using the GNSS receiver and the corresponding one pulse per second (1PPS).

6.3 BSM Transmission Requirements on Channel vChannelNumber (BSMTX)

6.3.1 BSM Contents (BSMCONT)

- If the BSM meets the minimum criteria for BSM transmission specified in Tables 19 and 20, the BSM shall be transmitted using a WAVE short message as defined in IEEE 1609.3 [4] and 6.1.3, and according to the timing requirements in 6.3.3 and 6.3.8 [6.3.1-V2V-BSMTX-BSMCONT-001].

- When transmitting a BSM, the system shall generate the corresponding MSG_MessageFrame containing MSG_BasicSafetyMessage and the data frames and data elements as specifed in this standard and SAE J2735 [1] [6.3.1-V2V-BSMTX-BSMCONT-002].

- All BSMs contain the BSM Part I content as specified in SAE J2735 [1]. The system shall include additional BSM content as follows:

  - The System transmitting a BSM shall include the DF_VehicleSafetyExtensions data frame in Part II [6.3.1-V2V-BSMTX-BSMCONT-003].

  - The DF_VehicleSafetyExtensions data frame shall include:

    - DF_PathHistory

    - DF_PathPrediction

    [6.3.1-V2V-BSMTX-BSMCONT-004]

- The DF_VehicleSafetyExtensions data frame shall include DE_ExteriorLights only if one or more bits in the data element are set [6.3.1-V2V-BSMTX-BSMCONT-005].

- If one or more event conditions corresponding to an event flag is met, the BSM Part II shall include the DF_VehicleSafetyExtensions data element, DE_VehicleEventFlags. DE_VehicleEventFlags is not included if no event flags are set [6.3.1-V2V-BSMTX-BSMCONT-006].

NOTE: Vehicle type (DE_VehicleType) is omitted from BSM Part II for light vehicles. However, it can be inferred for light vehicles as long as all other classes of vehicles include the field in their BSMs as defined in future revisions of this standard, or in other standards in the SAE J2945 family of standards.

NOTE: SAE J2735 [1] specifies that each BSM is is encoded using unaligned packed encoding rules (UPER)

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6.3.2 Channel and Data Rate (CHDATARATE)

- The system shall transmit BSMs on channel *vChannelNumber* with 10 MHz channel spacing (refer to 802.11 [2]) [6.3.2-V2V-BSMTX-CHDATARATE-001].

- The system shall transmit BSMs using an 802.11 [2] data rate of *vDataRate* [6.3.2-V2V-BSMTX-CHDATARATE-002].

6.3.3 Generation of the First BSM after System Device Startup and Generation Timing (GENTIM)

The timing requirements in this section apply only if the BSM meets the minimum criteria for BSM transmission specified in Tables 19 and 20. See 6.3.5 for behavior when the BSM does not meet the minimum transmission criteria.

- After system device startup, generation of the first BSM shall have a random start time. The start time is the basis for the computational intervals used in 6.3.8 and shown in Figure 22 [6.3.3-V2V-BSMTX-GENTIM-001].

NOTE: The intent of the requirement is to avoid synchronization to a common BSM generation epoch, such as a 1-PPS signal from the GNSS. This reduces packet collisions of BSMs generated by multiple systems.

- After the first BSM generated after system device startup, the system shall generate each subsequent BSM within -*vBSMRateTolerance* and +*vBSMRateTolerance* of its scheduled generation time [6.3.3-V2V-BSMTX-GENTIM-002].

NOTE: In 802.11-based systems, there may be a relatively small non-determistic queuing delay between the time at which a packet is generated, and the time at which it is transmitted. Hence, this standard uses “generate” in place of “transmit” where MAC queuing can introduce small latencies between the time the BSM is formed and its transmission. Queuing performance within the 802.11 MAC is outside the scope of this standard, but if system tests related to BSM generation requirements are performed by testing a system’s BSM transmissions without the presence of BSM transmissions from other systems, 802.11 MAC performance has a negligible impact on the test results.

6.3.4 User Priority and EDCA Settings (UPEDCA)

- The system shall set the user priority field defined in 802.11 [2] to “5” for BSMs with no critical event flags [6.3.4-V2V-BSMTX-UPEDCA-001].

- The system shall set the user priority field defined in 802.11 [2] to “7” for BSMs that include one or more critical event flags [6.3.4-V2V-BSMTX-UPEDCA-002].

- The system shall set the EDCA parameters defined in 802.11 [2] as shown in Table 18 for BSM transmissions on channel *vChannelNumber* [6.3.4-V2V-BSMTX-UPEDCA- 003].

Table 18 - EDCA parameter set

User Priority	AC	CWmin	CWmax	AIFSN	TXOP Limit OFDM/CCKOFDM PHY
1, 2	AC_BK	15	1023	9	0
0, 3	AC_BE	15	1023	6	0
4, 5	AC_VI	15	1023	4	0
6, 7	AC_VO	3	7	2	0

NOTE: The CWmin and AIFSN values for AC_VI are different than the default EDCA parameter set when dot11OCBActivated is equal to true (refer to 802.11 [2]). AIFSN was changed from a default of 3 to 4, CWmin was changed from a default of 7 to 15, and CWmax was changed from a default of 15 to 1023. These changes improve throughput for BSMs and prioritize BSMs over other potential background and best-effort messages on the same channel.

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6.3.5 Minimum Transmission Criteria (MINTX)

- The system shall transmit a BSM only if the BSM meets the minimum criteria for BSM transmission specified in Tables 19 and 20. If at any time the system cannot formulate a BSM that meets the minimum transmission criteria, the system ceases transmitting BSMs until the criteria are met [6.3.5-V2V-BSMTX-MINTX-001].

Table 19 - BSM Part I: Minimum criteria for BSM transmission

Data Element/Frame	Can be set to unavailable, or represent an unknown value, as specified in SAE J2735 [1]?	Section Reference (this standard)
DE_DSRC_MessageID	No	6.3.6.1
DE_MsgCount	No	6.3.6.2
DE_TemporaryID	No	6.3.6.3
DE_DSecond	No	6.3.6.4
DE_Latitude	No	6.3.6.5
DE_Longitude	No	6.3.6.5
DE_Elevation	No	6.3.6.6
DF_PositionalAccuracy	No	6.3.6.7
DE_SemiMajorAxisAccuracy	No	6.3.6.7
DE_SemiMinorAxisAccuracy	No	6.3.6.7
DE_SemiMajorAxisOrientation	No	6.3.6.7
DE_Speed	No	6.3.6.8
DE_TransmissionState	Yes	6.3.6.9
DE_Heading	No	6.3.6.10
DE_SteeringWheelAngle	Yes	6.3.6.11
DF_AccelerationSet4Way		
DE_Acceleration (Longitudinal)	No	6.3.6.12
DE_Acceleration (Lateral)	Yes	6.3.6.12
DE_VerticalAcceleration	Yes	6.3.6.12
DE_YawRate	No	6.3.6.12
DF_BrakeSystemStatus	Yes Note: All parameters in DF_BrakeSystemStatus may be set to unavailable.	6.3.6.13
DF_VehicleSize		
DE_VehicleWidth	No	6.3.6.14
DE_VehicleLength	No	6.3.6.14

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Table 20 - BSM Part II: Minimum criteria for BSM transmission

Data Element/Frame	Can a BSM be transmitted without the data frame/element?	Notes	Section Reference (this standard)
DE_VehicleEventFlags	Yes	DE_VehicleEventFlags is not included in the BSM if no event conditions are met	6.3.6.15
DF_PathHistory	No	A BSM cannot be transmitted without DF_PathHistory	6.3.6.16
DF_PathPrediction	No	A BSM cannot be transmitted without DF_PathPrediction	6.3.6.17
DE_ExteriorLights	Yes	DE_ExteriorLights is not included in the BSM if the corresponding status is unavailable or the exterior lights are turned off	6.3.6.18

6.3.6 Data Element Accuracy (DATAACC)

6.3.6.1 DE_DSRC_MessageID

- The system shall set the DE_DSRC_MessageID to the value assigned to basicSafetyMessage (refer to SAE J2735 [1]) [6.3.6-V2V-BSMTX-DATAACC-001].

6.3.6.2 DE_MsgCount

- The system shall initialize the DE_MsgCount to a random value within the range defined by SAE J2735 [1] when sending the first BSM after system device startup [6.3.6-V2V-BSMTX-DATAACC-002].

- If the certificate used to sign the BSM has changed since transmitting the most recent BSM, the system shall re-initialize the DE_MsgCount field to a new random value within the range defined by SAE J2735 [1] before transmitting the next BSM [6.3.6-V2V-BSMTX-DATAACC-003].

- The system shall set DE_MsgCount equal to one greater than the value used in the previously transmitted BSM, according to SAE J2735 [1] if the certificate used to sign the BSM has not changed since sending the most recent BSM. For this element, the value after 127 is zero per SAE J2735 [1] [6.3.6-V2V-BSMTX-DATAACC-004].

NOTE: DE_MsgCount, DE_TemporaryID, and the layer 2 address are each randomly reinitialized when the security certificate changes.

6.3.6.3 DE_TemporaryID

- The system shall initialize the DE_TemporaryID to a random value expressable within the range defined by SAE J2735 [1] for the first BSM generated after system device startup [6.3.6-V2V-BSMTX-DATAACC-005].

- If the certificate used to sign the BSM has changed since transmitting the most recent BSM, the system shall re-initialize the DE_TemporaryID to a new random value within the range defined by SAE J2735 [1] before transmitting the next BSM [6.3.6-V2V-BSMTX-DATAACC-006].

- The system shall not change DE_TemporaryID if the certificate has not changed [6.3.6-V2V-BSMTX-DATAACC-007].

NOTE: DE_MsgCount, DE_TemporaryID and the layer 2 address are each randomly reinitialized when the security certificate changes.

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6.3.6.4 DE_DSecond

- The system shall set the DE_DSecond as specified in SAE J2735 [1], using UTC as the time of reference [6.3.6-V2V-BSMTX-DATAACC-008].

- The time represented by the value of DE_DSecond shall be the time from the reference clock (6.2.3) at which the vehicle position data contained in the BSM Part I (6.3.6.5 and 6.3.6.6) was determined by the system [6.3.6-V2V-BSMTX-DATAACC-009].

- In order to ensure the transmitted information is current, the difference between UTC at which the BSM is generated and the time represented by the value of DE_DSecond shall be greater than or equal to zero and less than *vMaxPosAge* [6.3.6-V2V-BSMTX-DATAACC-010].

NOTE: The result of this requirement is that BSMs do not contain information that is older than UTC minus *vMaxPosAge*.

NOTE: SAE J2735 [1] states in a remark that other measurements present in the BSM are aligned to DE_DSecond insofar as possible in the implementation. Practical implementations to date have used the most recent measurement updates known to the transmitter at the time when the BSM is composed.

6.3.6.5 DE_Latitude and DE_Longitude

- The system shall set the DE_Latitude and DE_Longitude data elements to its corresponding two-dimensional (2D) horizontal position reference in the WGS-84 coordinate system [6.3.6-V2V-BSMTX-DATAACC-011].

- The position of the system transmitting a BSM shall be accurate to within *vPosAccuracy* of the vehicle's actual 2D horizontal position reference over 68% of measurements under open sky test conditions. See Appendix A.7 for the definition of open sky test conditions [6.3.6-V2V-BSMTX-DATAACC-012].

NOTE: The intent of the requirements is for the position to be accurate enough to support safety applications requiring lane-level accuracy. The accuracy requirement is based on the minimum Federal Highway Administration (FHWA) recommended width (3.0 m) of any roadway equal to or wider than a collector roadway [8]. The 68% accuracy requirement under open sky test conditions results in 95% confidence for relative positioning with lane-level granularity, based on testing done to support the use cases described in Section 4.

6.3.6.6 DE_Elevation

- The system shall set the DE_Elevation data element to its corresponding elevation ("height above reference ellipsoid") of the position reference, above or below the WGS-84 reference ellipsoid [6.3.6-V2V-BSMTX-DATAACC-013].

- The DE_Elevation data element shall be accurate to within *vElevAccuracy* of the actual elevation over 68% of measurements, under open sky test conditions. See Appendix A.7 or the definition of open sky test conditions [6.3.6-V2V-BSMTX-DATAACC-014].

NOTE: The 68% accuracy requirement under open sky test conditions results in 95% confidence for relative positioning with lane-level granularity, based on testing done to support the use cases described in Section 4.

6.3.6.7 DF_PositionalAccuracy

- The system shall set the values in the DF_PositionalAccuracy data frame of the BSM with values corresponding to its accuracy estimate for the vehicle position data included in the corresponding BSM. The methodology for estimating positional accuracy (confidence) is outside the scope of this standard [6.3.6-V2V-BSMTX-DATAACC-015].

- DF_PositionalAccuracy shall provide the DE_SemiMajorAxisAccuracy and DE_SemiMinorAxisAccuracy of the error ellipsoid at one standard deviation, as well as the DE_SemiMajorAxisOrientation for the semi-major axis [6.3.6-V2V-BSMTX-DATAACC-016].

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6.3.6.8 DE_Speed

- The DE_Speed data element in this data frame shall be accurate to within *vSpeedAccuracy* of the actual vehicle speed over 68% of measurements, under open sky test conditions [6.3.6-V2V-BSMTX-DATAACC-017].

NOTE: See Appendix A.7 for the definition of open sky test conditions.

6.3.6.9 DE_TransmissionState

- The DE_TransmissionState data element shall correctly reflect the state of the vehicle's transmission [6.3.6-V2V-BSMTX-DATAACC-018].

6.3.6.10 DE_Heading

- DE_Heading shall describe the direction of the vehicle reference point, and its value increases clockwise from north [6.3.6-V2V-BSMTX-DATAACC-019].

- DE_Heading shall be accurate to within *vHeadAccuracyB* of the actual vehicle heading over 68% of measurements when the vehicle speed is less than or equal to *vHeadingSpeedThresh* under open sky test conditions [6.3.6-V2V-BSMTX-DATAACC-020].

- DE_Heading shall be accurate to within *vHeadAccuracyA* of the actual vehicle heading over 68% of measurements when the vehicle speed is greater than *vHeadingSpeedThresh* under open sky test conditions [6.3.6-V2V-BSMTX-DATAACC-021].

- If the system cannot meet the accuracy specified in V2V-BSMTX-DATAACC-020 when the speed is below *vHeadLatchThresh*, the system shall latch the value of DE_Heading to the last known heading value when the speed was greater or equal to *vHeadLatchThresh* and populate DE_Heading with the latched value when the vehicle speed drops below *vHeadLatchThresh*. (The accuracy of DE_Heading is not subject to *vHeadAccuracyB* under this condition.) [6.3.6-V2V-BSMTX-DATAACC-022].

- If the heading is latched, the system shall unlatch the value of DE_Heading when the vehicle speed exceeds *vHeadUnlatchThresh* [6.3.6-V2V-BSMTX-DATAACC-023].

NOTE: The ability to meet the accuracy in V2V-BSMTX-DATAACC-020 at speeds below *vHeadLatchThresh* is based on system design and testing, not a real-time determination. See Appendix A.7 for the definition of open sky test conditions.

6.3.6.11 DE_SteeringWheelAngle

- If the DE_SteeringWheelAngle is used, it shall be accurate to within *vStWhAnAccuracy* of the actual vehicle steering wheel angle over 95% of measurements, or DE_SteeringWheelAngle is set to unavailable [6.3.6-V2V-BSMTX-DATAACC-024].

NOTE: This information is available from the vehicle CAN bus or other vehicle interface.

6.3.6.12 DF_AccelerationSet4Way

- The DE_Acceleration (longitudinal) and DE_Acceleration (lateral) data elements in this data frame shall be accurate to within *vAccelAccuracy* of the actual vehicle longitudinal and lateral accelerations, respectively, over 68% of measurements under open sky test conditions and flat road test conditions (grade < 0.2% and cross-slope < 2%) [6.3.6-V2V-BSMTX-DATAACC-025].

- The DE_VerticalAcceleration data element in this data frame shall be accurate to within *vVertAccelAccuracy* of the actual vehicle vertical acceleration over 68% of measurements under open sky test conditions and flat road test conditions (grade < 0.2% and cross-slope < 2%) [6.3.6-V2V-BSMTX-DATAACC-026].

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- The DE_YawRate data element in this data frame shall be accurate to within *vYawRateAccuracy* of the actual vehicle yaw rate over 68% of measurements under open sky test conditions and flat road test conditions (grade < 0.2% and cross-slope < 2%) [6.3.6-V2V-BSMTX-DATAACC-027].

NOTE: See Appendix A.7 for the definition of open sky test conditions.

6.3.6.13 DF_BrakeSystemStatus

- If available, the system shall use the vehicle bus as the data source for DF_BrakeSystemStatus [6.3.6-V2V-BSMTX-DATAACC-028].

- When braking status for each wheel is available, the system shall set each bit in the wheelBrakes field to 1 (= true) or 0 (= false) based on the brake status for the corresponding wheel, and set the wheelBrakesUnavailable field to 0 (= false) [6.3.6-V2V-BSMTX-DATAACC-029].

- If only one braking status indication is available (individual wheel status not available), the system shall set the bits for all wheels in the wheelBrakes field on or off depending on the braking status and set the wheelBrakesUnavailable field to 0 (= false) [6.3.6-V2V-BSMTX-DATAACC-030].

- When no braking status is available, the system shall set the wheelBrakesUnavailable field to 1 (= true) [6.3.6-V2V-BSMTX-DATAACC-031].

- The system shall set the traction, ABS, scs, brakeBoost, and auxBrakes fields in accordance with SAE J2735 [1] [6.3.6-V2V-BSMTX-DATAACC-032].

6.3.6.14 DF_VehicleSize

- The DE_VehicleLength and DE_VehicleWidth data elements in this data frame shall be accurate to within *vSizeAccuracy* of the actual vehicle length and vehicle width, respectively [6.3.6-V2V-BSMTX-DATAACC-033].

6.3.6.15 DE_VehicleEventFlags

- The difference between the time at which a critical event condition is initially met (refer to SAE J2735 [1]) and the time of generation of the initial BSM with the corresponding DE_VehicleEventFlags bit set shall be less than *vEventDetectLatency*. This requirement still holds if a different critical event condition was already ongoing [6.3.6-V2V-BSMTX-DATAACC-034].

- The system shall set the Hard Braking event flag when the corresponding event condition is met (refer to SAE J2735 [1]). If the information is available, the system sets the ABS, traction control, and stability control critical event flags when the corresponding critical event conditions occur, and the system may support other event flags [6.3.6-V2V-BSMTX-DATAACC-035].

6.3.6.16 DF_PathHistory

- The system shall populate DF_PathHistory in the BSM Part II DF_VehicleSafetyExtensions data frame as follows:
  - crumbData: DF_PathHistoryPointList [6.3.6-V2V-BSMTX-DATAACC-036].
  - Within the DF_PathHistoryPointList, the system shall populate the DF_PathHistoryPoint data frame with the following data elements:
    - latOffset: DE_OffsetLL-B18
    - lonOffset: DE_OffsetLL-B18

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  - elevationOffset: DE_VertOffset-B12
  - timeOffset: DE_TimeOffset

    [6.3.6-V2V-BSMTX-DATAACC-037]

  - DF_PathHistory and DF_PathHistoryPoint shall not include any additional data elements or frames within the BSMs transmitted on channel *vChannelNumber* [6.3.6-V2V-BSMTX-DATAACC-038].

- The system shall populate DF_PathHistory with path history (PH) points so that the represented PH distance (i.e., the distance between the first and last PH point along the vehicle path) is at least *vMinPHistDistance* unless initially, or due to the unavailability of position, there is less than *vMinPHistDistance* of PH [6.3.6-V2V-BSMTX-DATAACC-039].

NOTE: DF_PathHistory contains five points or fewer (≤40 bytes prior to UPER encoding) 91.3% of the time based on extensive testing.

The system shall maintain a vehicle path comprised of time, latitude, longitude, and elevation derived from the positioning subsystem, representing the vehicle's recent movement over a corresponding distance [6.3.6-V2V-BSMTX-DATAACC-040].

- The system shall populate DF_PathHistory with PH points such that the perpendicular distance between any point on the vehicle path and the straight line connecting two adjacent PH points is less than *vPathPerpendicularDist* over at least 95% of measurements under open sky conditions [6.3.6-V2V-BSMTX-DATAACC-041].

NOTE: Testing conformance to path history requirements should be performed in an uncongested environment (i.e., BSMs are being transmitted approximately every 100 ms), because the vehicle path will be derived from Part I data elements (DE_Second, DE_Latitude, DE_Longitude, and DE_Elevation) transmitted in previous BSMs.

- The system shall populate DF_PathHistory with PH points such that the maximum distance between two consecutive points is *vMaxChordLength* [6.3.6-V2V-BSMTX-DATAACC-042].

- The system shall populate DF_PathHistory with the minimum number of PH points, selected as a subset of the available vehicle path position data, necessary to satisfy *vPathPerpendicularDist, vMaxChordLength,* and *vMinPHistDistance* [6.3.6-V2V-BSMTX-DATAACC-043].

NOTE: Requirements [6.3.6-V2V-BSMTX-DATAACC-039], [6.3.6-V2V-BSMTX-DATAACC-042] and [6.3.6-V2V-BSMTX-DATAACC-043] ensure that the maximum PH distance is limited to the sum of *vMinPHistDistance* and *vMaxChordLength.*

- The system shall populate DF_PathHistory with time-ordered PH points, with the first PH point being the closest in time to the current UTC time [6.3.6-V2V-BSMTX-DATAACC-044].

NOTE: Time-ordered PH points are not required to be spaced equally in time.

- If the number of PH points needed to meet the requirements previously stated in this section exceeds *vMaxPHistPoints*, the system shall populate DF_PathHistory with no more than *vMaxPHistPoints* points from the computed set of points (effectively the distance requirement is relaxed) [6.3.6-V2V-BSMTX-DATAACC-045].

NOTE: An example PH algorithm is provided in Appendix A.5.

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6.3.6.17 DF_PathPrediction

- The system shall populate the DF_PathPrediction data frame in the BSM Part II DF_VehicleSafetyExtensions data frame as follows:
  - DE_RadiusOfCurvature
  - DE_Confidence

  [6.3.6-V2V-BSMTX-DATAACC-046]

- The system shall populate DF_PathPrediction with a calculated radius, and when the actual radius is greater than or equal to *vMinCurveRadius* and less than or equal to *vMaxCurveRadius* in magnitude, the calculated radius must be less than *vPPredRadiusError* from the actual radius over at least 68% of measurements when the vehicle is in both steady state and open sky conditions [6.3.6-V2V-BSMTX-DATAACC-047].

NOTE: The actual radius is the ratio of the average speed (in meters per second) to the average yaw rate (in radians per second) when the vehicle is in steady state condition while driving on a curve with a constant radius. For the purposes of path prediction (PP), steady state conditions occur when the vehicle is driving on a curve with a constant radius. In steady state the average of the absolute value of the change of yaw rate over time is smaller than 1 degree/s^2.

- When transitioning from a curve with constant radius of curvature onto a curve with a new constant radius of curvature, the System shall populate at least one DE_RadiusOfCurvature within *vPPredTransitionTime*, with a value that is within *vPPredRadiusError* of the curve's actual radius. [6.3.6-V2V-BSMTX-DATAACC-048].

NOTE: Requirement 6.3.6-V2V-BSMTX-DATAACC-047 combined with requirement 6.3.6-V2V-BSMTX-DATAACC-048 are intended to result in the system converging on the new actual radius of curvature after *vPPredTransitionTime*.

- The system shall report a "straight path" radius of value 32,767 and confidence of value 100% (corresponds to a value of 200 for the data element) when the transmitting vehicle speed is less than *vStationarySpeedThresh* [6.3.6-V2V-BSMTX-DATAACC-049].

NOTE: An example PP algorithm is provided in Appendix A.6. Using constant radii provides sufficiently high confidence to support the target use cases because the PP is updated for every BSM transmission.

6.3.6.18 DE_ExteriorLights

- If the DF_VehicleSafetyExtensions data frame includes DE_ExteriorLights, the system shall set the individual light indications in the DE_ExteriorLights according to the available vehicle status data [6.3.6-V2V-BSMTX-DATAACC-050].

6.3.6.19 Additional Data Elements

- The system shall not include any additional data elements or data frames in transmitted BSMs beyond those required in this standard [6.3.6-V2V-BSMTX-DATAACC-051].

- On reception, the system shall be capable of ignoring unused data frames/elements [6.3.6-V2V-BSMTX-DATAACC-052].

NOTE: This requirement is needed to ensure that vehicle safety communications are not subject to undesired channel congestion caused by excessive message size.

NOTE: Other standards within the SAE J2945 family of standards that apply to vehicle types other than light vehicles may have additional requirements that permit use of additional data frames and elements.

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6.3.7 Data Persistency (DATAPERSIST)

6.3.7.1 Heading

- If the the system is unable to meet the accuracy specified in V2V-BSMTX-DATAACC-020 when the speed is below *vHeadLatchThresh*, the system shall store the last known (latched) heading value per V2V-BSMTX-DATAACC-022 in persistent memory upon system device shutdown [6.3.7-V2V-BSMTX-DATAPERSIST-001].

NOTE: This is to enable the use of the last known heading to be retrieved upon system device startup. The vehicle speed is expected to be below *vHeadLatchThresh* at the time of system device shutdown (i.e., the vehicle ignition is being turned off).

- Upon system device startup, if the the system is unable to meet the accuracy specified in V2V-BSMTX-DATAACC-020 when the speed is below *vHeadLatchThresh*, the system shall populate DE_Heading with the heading stored upon last system device shutdown [6.3.7-V2V-BSMTX-DATAPERSIST-002].

NOTE: This is to enable the use of a last known heading upon system device startup. If the speed is below *vHeadLatchThresh, and* the system is unable to maintain the heading accuracy specified in V2V-BSMTX-DATAACC-020, the system continues to use the stored heading until requirements in V2V-BSMTX-DATAACC-020 can be met.

NOTE: These data persistency requirements ensure robustness of the system upon device startup and address crash scenarios involving stationary and stopped vehicles.

6.3.7.2 Path History

- The system shall store the last known PH in persistent memory upon device shutdown [6.3.7-V2V-BSMTX-DATAPERSIST-003].

NOTE: This is to enable the PH to be retrieved upon system device startup.

- Upon system device startup, the system shall initialize DF_PathHistory by populating it with points based on the current position and the path history prior to the last system device shut down [6.3.7-V2V-BSMTX-DATAPERSIST-004].

NOTE: This is to enable the use of a last-known PH upon system device startup.

6.3.8 BSM Scheduling and Congestion Control (CONGCTRL)

This section specifies the requirements for scheduling BSMs and congestion control, based on the research and development documented in [18], [19], and [20]. See Appendix A.8 for additional implementation details of the congestion control algorithm.

- The system shall generate BSMs using the congestion control algorithm defined in 6.3.8.1 through 6.3.8.8. Those sections use the following computational intervals: *vCBPMeasInt*, *vPERSubInterval*, *vPERInterval,* and *vTxRateCntrlInt.* The relationship between those intervals is shown in Figure 22 [6.3.8-V2V-BSMTX-CONGCTRL-001].

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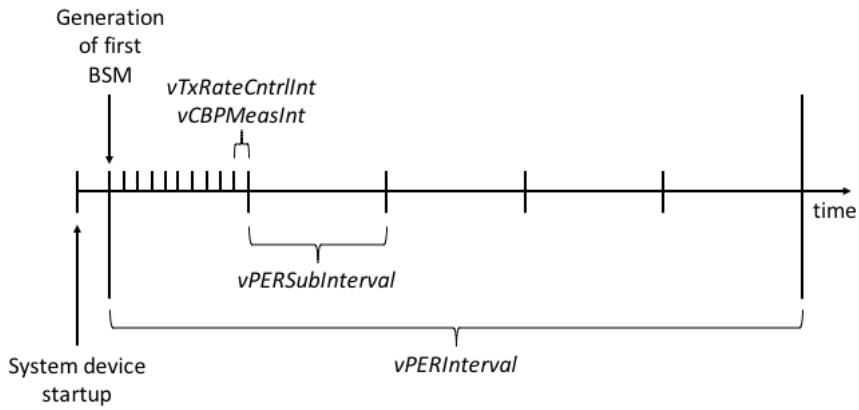

Figure 22 - Relationship between computational intervals: vCBPMeasInt, vPERSubInterval, vPERInterval, and vTxRateCntrlInt

The calculations corresponding to a given computational interval are always performed at the end of that computational interval, and *k* denotes a given instance of that interval. The calculations use inputs collected during the given interval and are written as a function of *k*. For example, if F represents a specific function, F(*k*) is a computation using inputs collected during the kth instance of the interval (also referred to as the kth interval).

6.3.8.1 Inputs

This section defines the inputs that are used by the congestion control algorithm. Raw channel busy percentage (RawCBP) and smooth channel busy percentage (CBP) are calculated at the end of the kth instance of *vCBPMeasInt* as follows:

- RawCBP: RawCBP is the percentage of time the channel was busy during a given instance of *vCBPMeasInt*. The channel is busy when the DSRC radio subsystem is either transmitting or its clear channel assessment function indicates the channel is busy, as defined in 802.11 [2]. RawCBP includes channel busy time due to both carrier sensing and energy detection, but it does not include virtual channel busy time based on the network allocation vector [2]. RawCBP(*k*) is calculated as follows:

$$RawCBP(k) = \frac{(100 \times Duration\ Channel\ Indicated\ as\ Busy)}{vCBPMeasInt} \qquad \text{(Eq. 1)}$$

- Smooth CBP: The system calculates the smooth CBP, CBP(*k*), based on RawCBP(*k*) and CBP(*k*-1), to reduce the impact of measurement noise, as follows:

$$CBP(k) = vCBPWeightFactor \times RawCBP(k) + (1 - vCBPWeightFactor) \times CBP(k-1) \qquad \text{(Eq. 2)}$$

Packet error ratio (PER), channel quality indicator (Π), and vehicle density in range (N) are calculated at the end of the kth instance of *vPERSubInterval* as follows:

- PER: The PER is calculated between a pair of vehicles, the host vehicle (HV) and a given remote vehicle, RV_i. The system calculates the sliding-window PER over the interval, *vPERInterval*, at the end of the kth instance of *vPERSubinterval* as follows:

  Let δ_k be the kth instance of *vPERInterval*, and w_k be the kth instance of *vPERSubInterval*. In Figure 23 the duration of δ_k is *n* times the duration of w_k, where *n* denotes the number of subintervals within δ_k. At the end of w_k, the number of missed BSMs and the total number of expected BSMs during the corresponding interval δ_k are calculated for RV_i. The PER, $PER_i(k)$, for RV_i, during δ_k is computed as follows:

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$$PER_i(k) = \frac{number\ of\ missed\ BSMs\ from\ RV_i\ during\ [w_{k-n+1}, w_k]}{total\ expected\ BSMs\ from\ RV_i\ during\ [w_{k-n+1}, w_k]} \quad \text{(Eq. 3)}$$

where $k \geq n$.

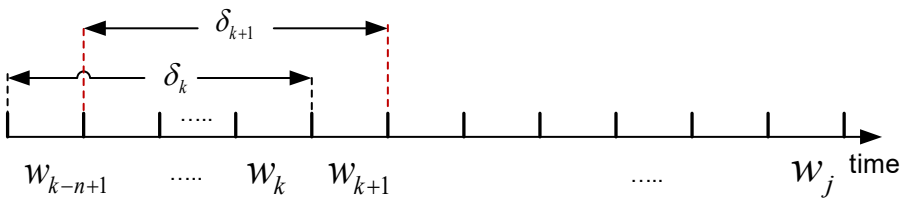

Figure 23 - Sliding window

PER is calculated using the DE_MsgCount data elements contained in the BSMs for each RV from which two or more BSMs were received during δ_k. The number of expected BSMs for a given RV is one plus the difference between the values of DE_MsgCount in the last and first BSMs received from that RV within δ_k. The number of missed BSMs for the RV is the difference between the number of expected BSMs and the number of received BSMs from the RV within δ_k. When calculating the number of expected and missed BSMs from a given RV, the modulus of the DE_MsgCount data element is accounted for based on the range of DE_MsgCount as defined in SAE J2735 [1].

If only one packet is received from a given RV during δ_k, the PER corresponding to that RV is undefined and not used in PER calculations for δ_k. See Appendix A.8.3 for examples of special PER calculation cases.

An RV is within *vPERRange* if the last BSM received from that RV during w_k contains a 2D position within *vPERRange* of the HV's most recent 2D position reference as of the time when PER is calculated. If a BSM is not received from that RV during w_k, the RV is not within *vPERRange*.

- Channel Quality Indicator (Π): $\Pi(k)$ is calculated at the end of w_k as the average of PER(k) for all RVs within *vPERRange* and for which PER(k) is calculated, with the following constraint:

$$\text{If } (\Pi(k) > vPERMax)$$
$$\Pi(k) = vPERMax \quad \text{(Eq. 4)}$$

where PER(k) for a given RV, RV_i, is $PER_i(k)$ from Equation 3.

- Vehicle Density in Range (N): The HV (system) calculates N(k) at the end of w_k as the number of unique RVs within *vPERRange* (an RV is determined to be unique if it has a unique DE_TemporaryID included in its BSM).

6.3.8.2 Calculate Tracking Error

This section defines the steps used to calculate the tracking error used in the congestion control algorithm. Note that the tracking error is a result of the delay with which RVs receive the HV's kinematic state data due to transmission latencies and packet losses, and it is distinct from accuracy-related errors of the positioning subsystem.

The system performs the following operations in order at the end of the kth instance of *vTxRateCntrlInt*:

- The system estimates the position of the HV at the current time (the HV local estimate), per Appendix A.3, using the configuration parameters *vHVLocalPosEstIntMin* and *vHVLocalPosEstIntMax* in place of HVPosEstIntMin and HVPosEstIntMax, respectively. If per Appendix A.3 Delta_time_ms > *vHVLocalPosEstIntMax*, e(k) = 0, and the remaining steps in this section (6.3.8.2) are not performed.

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- Using the latest HV state information assumed received at RVs (see 6.3.8.8), the HV (system) calculates its position at the current time as it perceives the RVs have calculated its position (the HV remote estimate), per Appendix A.3. The HV remote estimate uses the configuration parameters *vHVRemotePosEstIntMin* and *vHVRemotePosEstIntMax in place of* HVPosEstIntMin and HVPosEstIntMax, respectively. If per Appendix A.3 Delta_time_ms > *vHVRemotePosEstIntMax*, e(*k*) = 0, and the remaining steps in this section (6.3.8.2) are not performed.

- The system calculates the tracking error, e(*k*), as the 2D distance between the HV local estimate and the HV remote estimate.

NOTE: A reference implementation of the tracking error calculation is shown in Appendix A.8.2.

6.3.8.3 Calculate Transmission Probability

The system performs the following operations at the end of the kth instance of *vTxRateCntrlInt*:

- The system calculates a transmission probability, p(*k*), using the calculated tracking error as follows:

$$p(k) = \begin{cases} 1 - \exp(-\alpha \times |e(k) - T|^2) & \text{if } T \le e(k) < S \\ 1 & \text{if } e(k) \ge S \\ 0 & \text{otherwise} \end{cases} \qquad \text{(Eq. 5)}$$

where *T* is the minimum communications-induced tracking error threshold, *vTrackingErrMin*, α is the error sensitivity *vErrSensitivity,* and *S* is the communications-induced tracking error saturation value *vTrackingErrMax*.

NOTE: The design rationale of Equation 5 is that, when the tracking error is below the pre-defined threshold *T*, an HV does not broadcast a BSM due to tracking error. When the tracking error exceeds this threshold, the larger magnitude of tracking error would result in a higher transmission probability, unless the tracking error exceeds the threshold *S* in which case a BSM will always be sent due to tracking error. Since not all HVs have the same tracking error, they will use different levels of transmission probabilities to broadcast BSMs.

6.3.8.4 Calculate Maximum Inter-Transmit Time (Maximum BSM Generation Interval)

This section defines the steps used to calculate the maximum BSM generation interval, Max_ITT. *vMinITT* is the minumum BSM generation interval. The system performs the following operations at the end of the kth instance of *vTxRateCntrlInt*:

- The system smooths the calculated current vehicle density in range, N(*k*), as follows:

$$N_s(k) = \lambda \times N(k) + (1 - \lambda) \times N_s(k-1) \qquad \text{(Eq. 6)}$$

where λ is the weight factor *vDensityWeightFactor*, and $N_s(k)$ is the smoothed current vehicle density in range.

The system calculates Max_ITT(*k*) as follows:

$$Max_ITT(k) = \begin{cases} vMin_ITT & N_s(k) \le B \\ vMin_ITT \times \dfrac{N_s(k)}{B} & B < N_s(k) < \dfrac{vMax_ITT}{vMin_ITT} \times B \\ vMax_ITT & \dfrac{vMax_ITT}{vMin_ITT} \times B \le N_s(k) \end{cases}$$

(Eq. 7)

where Max_ITT(*k*) is the message generation interval in milliseconds, *B* is the density co-efficient *vDensityCoefficient*, *vMax_ITT* is the maximum threshold used in Equation 7.

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6.3.8.5 Transmission Decision

This section defines the steps for making the transmission decision.

The system performs the following operations at the end of the kth instance of *vTxRateCntrlInt*, and when a critical event condition initially occurs:

- The following logic is used to make the transmission decision (New_Tx_Decision_Critical_Event, TxDecision_Critical_Event, TxDecision_Dynamics, and TxDecision_Max_ITT are initialized to zero prior to each transmission decision):
  - Transmission based on critical event condition: If a critical event condition is met (see 3.1), then
    - Set TxDecision_Critical_Event = 1
  - If TxDecision_Critical_Event == 1 and previous_TxDecision_Critical_Event == 0
    - NEW_Tx_Decision_Critical_Event = 1
  - Else
    - NEW_Tx_Decision_Critical_Event = 0
  - Next, set previous_TxDecision_Critical_Event to TxDecision_Critical_Event.
  - Transmission based on vehicle dynamics (i.e., based on tracking error in 6.3.8.2): Use the transmission probability from Equation 5 and draw a uniform random number between zero and one for a Bernoulli trial, *rand().* If the outcome of the Bernoulli trial is true, and the time until the next scheduled BSM is greater than or equal to *vRescheduleTh,* i.e.,
  - If (rand()<=p(*k*) && (NextScheduledMsgTime - CurrentTime)>=*vRescheduleTh*), then
  - Set TxDecision_Dynamics = 1
  - Transmission based on change in Max_ITT: Changes in the value of Max_ITT affect the transmission decision as follows:
  - If (NextScheduledMsgTime - (LastTxTime + Max_ITT(*k*)) >= *vRescheduleTh*), then
  - Set TxDecision_Max_ITT = 1
  - Where NextScheduledMsgTime is the UTC in milliseconds at which the next BSM is scheduled for transmission, *CurrentTime* is the current UTC in milliseconds, LastTxTime is the UTC in milliseconds when the previous BSM was generated by the system, and *vRescheduleTh* is the threshold in milliseconds used in the transmission decision. The resolution of time is defined by *vTimeAccuracy*.

6.3.8.6 Schedule Transmission

This section defines the steps used to schedule BSM transmissions. The system performs the following operations immediately after the transmission decision (6.3.8.5):

- The system schedules BSMs for transmission using the following logic:
  - If (((TxDecision_Critical_Event == 1 or TxDecision_Dynamics == 1) and ((NextScheduledMsgTime - LastTxTime) > (100 + vTxRand))) or NEW_TxDecision_Critical_Event == 1)
    - Cancel scheduled transmission
    - Schedule transmission now, i.e., NextScheduledMsgTime = CurrentTime

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- Else if TxDecision_Max_ITT == 1
  - Cancel scheduled transmission
  - Schedule next transmission at NextScheduledMsgTime = max(CurrentTime,LastTxTime + Max_ITT(*k*))
- Else, BSM is generated as previously scheduled, i.e., if TxDecision_Critical_Event == 0 and TxDecision_Dynamics == 0 and TxDecision_Max_ITT == 0
  - No change to scheduled BSM transmission

6.3.8.7 Calculate Radiated Power

This section specifies the steps used to calculate the radiated power (RP) for the transmitted BSM. The corresponding transmit power setting may be derived from RP and calculated as follows (see 6.4.1):

- Transmit power = RP - MinSectorAntGain + CLoss

The system performs the following operations when CurrentTime = NextScheduledMsgTime.

If TxDecision_Critical_Event == 1 or TxDecision_Dynamics == 1,

RP = *vRPMax*

Otherwise, the system calculates the radiated power as follows:

$$f(CBP) = \begin{cases} vRPMax & CBP \le vMinChanUtil \\ vRPMax - \left(\dfrac{vRPMax - vRPMin}{vMaxChanUtil - vMinChanUtil}\right) \times (CBP - vMinChanUtil) & vMinChanUtil < CBP < vMaxChanUtil \\ vRPMin & vMaxChanUtil \le CBP \end{cases}$$

(Eq. 8)

$$\text{Base_RP} = \text{Previous_ RP} + vSUPRAGain \times (\text{f(CBP)} - \text{Previous_RP}) \quad \text{(Eq. 9)}$$

$$\text{RP} = \text{Base_RP} \quad \text{(Eq. 10)}$$

$$\text{Previous_RP} = \text{Base_RP} \quad \text{(Eq. 11)}$$

- where CBP is calculated at the end of the most recent instance of *vCBPMeasInt* (Equation 2), *vRPMax* is the higher threshold for radiated power, *vRPMin* is the lower threshold for radiated power, *vMinChanUtil* is the lower threshold for channel utilization, *vMaxChanUtil* is the higher threshold for channel utilization, and *vSUPRAGain* is the stateful utilization-based power adaptation (SUPRA) gain. The initial value of RP (for the first BSM transmitted after system startup) is *vRP*.
- If derived transmit power is higher than the system supports, the system uses the highest supported transmit power (MaxTxPowerCap). Transmit power granuality is defined by *vTxPwrCtrlStep* (see 6.4.1.1).

6.3.8.8 Generate BSM and Schedule Next BSM Generation

This section defines the final steps in generating and transmitting the BSM.

The system performs the following operations in order when CurrentTime is NextScheduledMsgTime.

- The system generates the BSM for transmission at the transmit power necessary to achieve the calculated RP from 6.3.8.7.
- The system makes an assumption of the latest HV state information received by the RVs based on the channel quality indicator per Appendix A.8.1.

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- The system assigns LastTxTime = CurrentTime.
- The system schedules the next BSM to be generated at NextScheduledMsgTime = LastTxTime + Max_ITT + RandOffset.
  - RandOffset and the value of Max_ITT are defined as follows:
    - RandOffset is a pseudorandom number uniformly distributed between -*vTxRand* and +*vTxRand*.

NOTE: The randomization reduces repeated packet collisions of transmitted BSMs from multiple systems.

    - The value of Max_ITT is calculated at the end of the most recent instance of *vTxRateCntrlInt* using Equation 7.

6.4 RF Performance Requirements (RFPERF)

6.4.1 DSRC Radiated Power and Transmit Power Accuracy (DSRCTX)

A spherical coordinate system is shown in Figure 24. Measurements of antenna gain or radiated power are made on the surface of the sphere, the center of which is the DSRC antenna. As shown in Figure 24, the elevation angle, θ, is measured from the z-axis, and the azimuth angle, $\emptyset$, is measured from the x-axis.

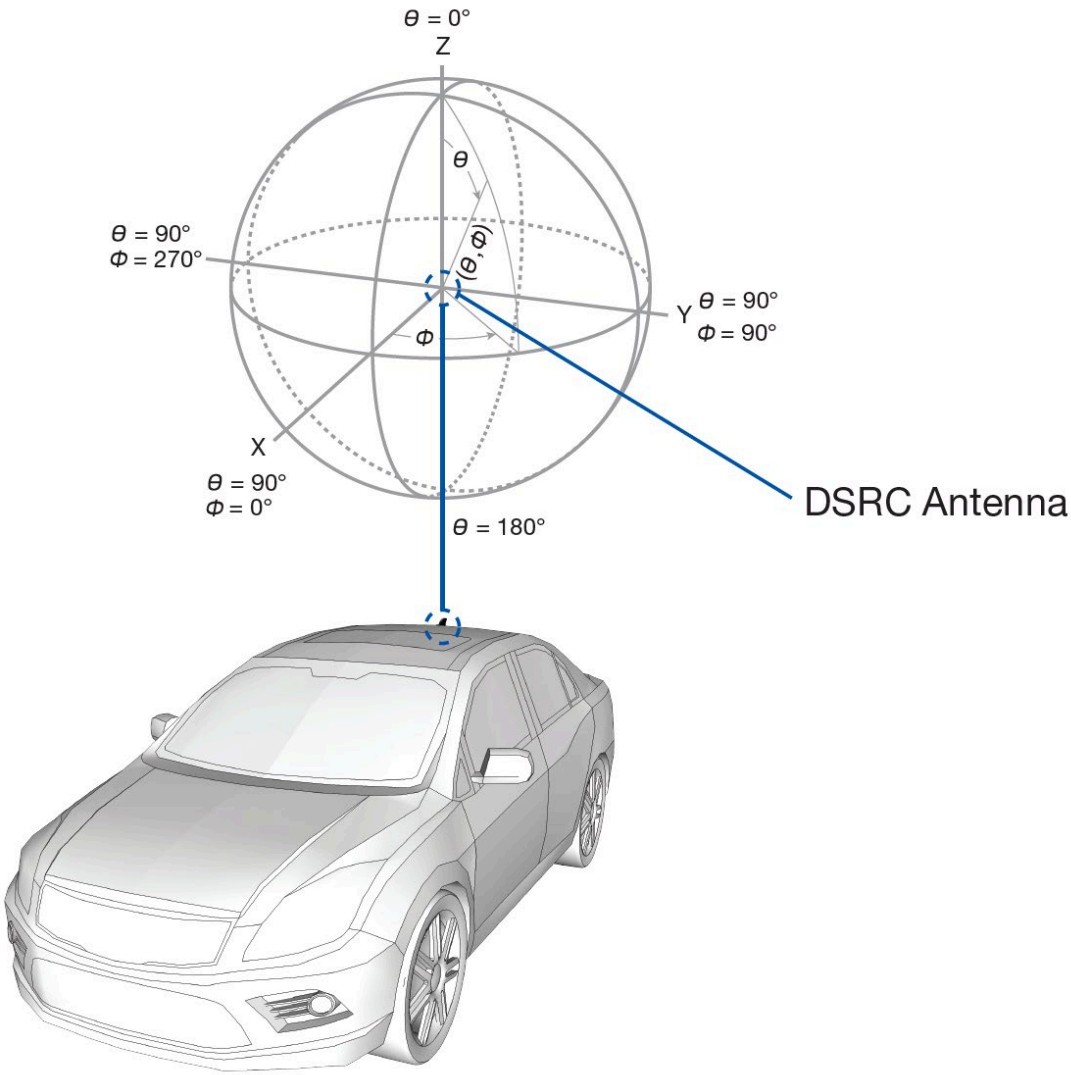

Figure 24 - Spherical coordinate system

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Right and left sectors are defined as shown in Figure 25; the line between $\emptyset = 0$ degree and $\emptyset = 180$ degrees separates the two sectors. *vUpperEl* and *vLowerEl* define the measurement range of elevation angle, θ.

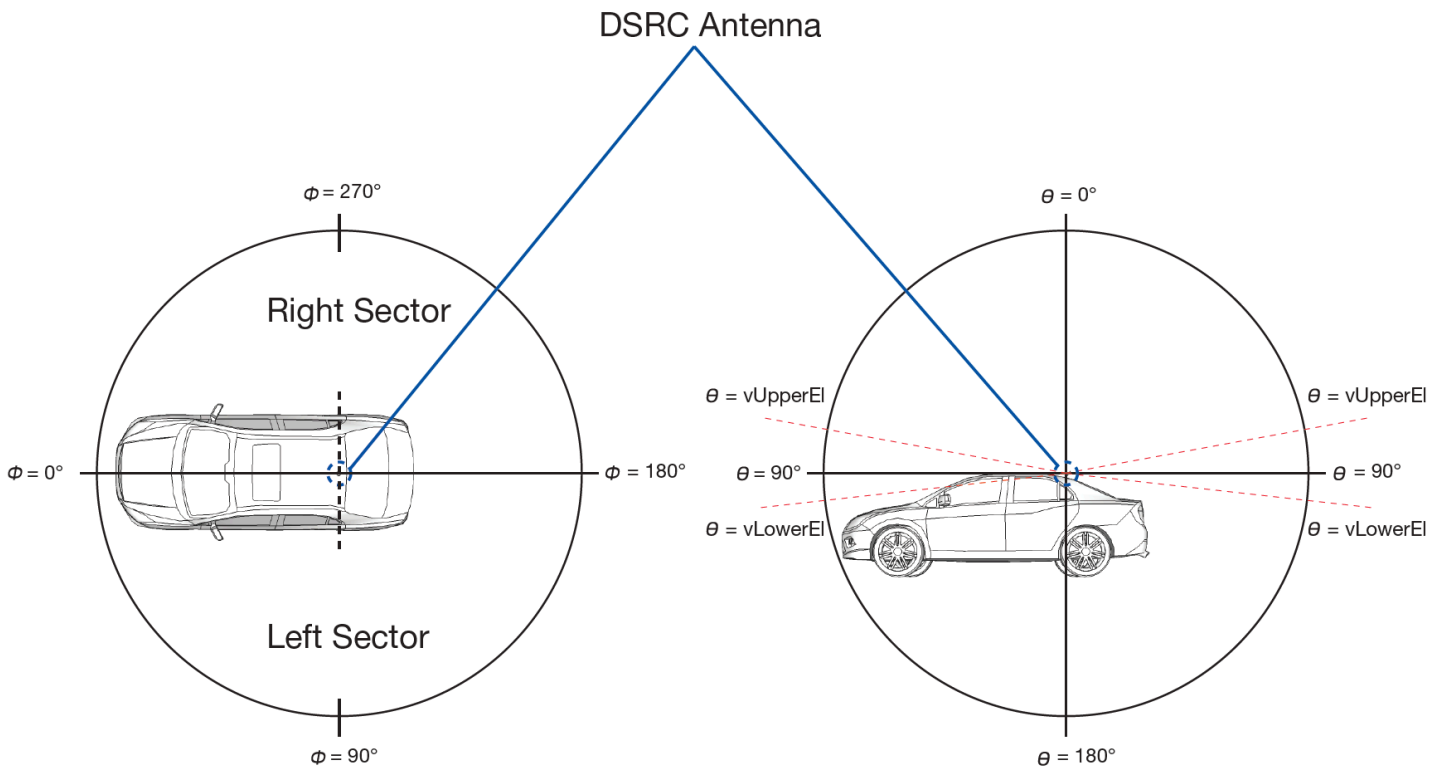

Figure 25 - Left and right sectors and elevation angle range

- The DSRC radio subsystem shall comply with the following:
  - MinSectorAntGain + MaxTxPowerCap - CLoss ≥ *vRP*, where:
    - CLoss is the total cable and connector losses in dB.
    - MaxTxPowerCap is the measured maximum conducted transmit power in dBm of the DSRC radio subsystem at which 802.11 [2] transmitter requirements are met.
    - MinSectorAntGain is calculated in dBi as follows:
      - Divide the area around the vehicle into right and left sectors as shown in Figure 25.
      - For azimuth angle ranges 0 ≤ azimuth <180 degrees (the left sector), in a maximum of 2-degree azimuth increments, the vertically polarized antenna gain in dBi is measured from *vUpperEl* to *vLowerEl* in maximum 2-degree elevation increments with the antenna(s) mounted on or in a vehicle (i.e., in situ).
      - For azimuth angle ranges 180 ≤ azimuth <360 degrees (the right sector), in a maximum of 2-degree azimuth increments, the vertically polarized antenna gain in dBi is measured from *vUpperEl* to *vLowerEl* in maximum 2-degree elevation increments with the antenna(s) mounted on or in a vehicle (i.e., in situ).

        NOTE: As shown in Figures 24 and 25, measurements are referenced to the location of the antenna as installed on the vehicle, which may or may not be installed along the vehicle's center line. The horizon plane is a plane that gives an elevation reference of θ = 90 degrees, as shown in Figure 25. The horizon plane cuts through the midpoint between the base and top of the antenna.

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- SectorAntGainAvg is calculated in dBi as the linear average of all of the measurements over one sector. It is calculated individually for the right and left sectors (see following note).

  NOTE: The linear average over one sector is calculated by converting each measurement in dBi, G_i, to $g_i = 10^{(Gi/10)}$, then averaging g_i over all the measurements (g_{iave}), and $10 \times \log_{10}(g_{iave})$ is the result in dBi of the linear average.

- MinSectorAntGain is the lesser of the right and left sector SectorAntGainAvg values in dBi.

[6.4.1-V2V-RFPERF-DSRCTX-001]

NOTE: Compliance with *vRP* can be verified using a vehicle-level radiated power test, without separately measuring antenna gain and cable and connector losses. In implementation multiple (diversity) antennas can be used to meet the requirement; the methodology by which antenna diversity is implemented and tested is outside the scope of this standard.

6.4.1.1 Transmit Power Accuracy

- The DSRC radio subsystem shall meet the 802.11 transmitter requirements for 10 MHz channel spacing with QPSK and 1/2 rate coding (e.g., frequency accuracy and relative constellation error specifications [2]) over the range of MaxTxPowerCap - PwrRange to MaxTxPowerCap [6.4.1-V2V-RFPERF-DSRCTX-002].

  - PwrRange is calculated in dB as follows:

    PwrRange = MinSectorAntGain + MaxTxPowerCap - CLoss - *vRPMax* + *vTxPwrRange*

NOTE: The intent is to support the range of “radiated power” (6.3.8.7) from *vRPMin* to the minimum of the maximum supported radiated power and *vRPMax*.

- The transmit power out of the DSRC radio subsystem as measured at the antenna connector of the subsystem housing shall be within *vTxPwrAcc* of its setting over 95% of measurements at each setting within the range MaxTxPowerCap - PwrRange to MaxTxPowerCap [6.4.1-V2V-RFPERF-DSRCTX-003].

- The transmit power out of the DSRC radio subsystem measured at the antenna connector of the subsystem housing shall be a monotonically increasing function of the transmit power setting in step sizes of *vTxPwrCtrlStep* over the range MaxTxPowerCap - PwrRange to MaxTxPowerCap [6.4.1-V2V-RFPERF-DSRCTX-004].

NOTE: Finer power control steps than *vTxPwrCtrlStep* are permitted. These requirements support the congestion control requirements in 6.3.8.

6.4.2 DSRC Receiver Sensitivity (DSRCRXSENS)

- The packet error rate of the DSRC radio subsystem shall be 10% or less when the PSDU length is 400 octets and the input level is *vRxSens* dBm at 6 Mbps (QPSK with 1/2 rate coding), at room temperature (21 °C ± 5 °C). The minimum input levels are measured at the antenna connector of the system housing [6.4.2-V2V-RFPERF-DSRCRXSENS-001].

- The DSRC radio subsystem shall comply with the optional enhanced (dot11ACRType = 2) adjacent and non-adjacent channel rejection requirements for 6 Mbps (QPSK with 1/2 rate coding), as specified in 802.11 [2], with the following measurement modifications:

  - The receiver sensitivity is *vRxSens.*

  - The PSDU length is 400 octets.

    [6.4.2-V2V-RFPERF-DSRCRXSENS-002]

NOTE: Receiver tests can be done in a controlled environment (“bench test”), and the input to the system can be via a cable.

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6.5 Security and Privacy Requirements (SECPRIV)

6.5.1 Identification Randomization (IDRAND)

- The system shall randomize its layer 2 address upon power-up [6.5.1-V2V-SECPRIV-IDRAND-001].

- If the certificate used to sign the BSM has changed since transmitting the most recent BSM, the system shall re-randomize its layer 2 address [6.5.1-V2V-SECPRIV-IDRAND-002].

NOTE: DE_MsgCount, DE_TemporaryID, and the layer 2 address are each randomly reinitialized when the security certificate changes. Refer to 6.3.6.2 and 6.3.6.3 for randomization requirements for the BSM DE_MsgCount and DE_Temporary ID fields.

6.5.2 BSM Signing (BSMSIGN)

- The system shall sign every BSM using the security credentials defined by 1609.2 [3], as profiled in 6.1.2 [6.5.2-V2V-SECPRIV-BSMSIGN-001].

- The system shall attach a certificate or certificate digest to every BSM as defined by 1609.2 [3] [6.5.2-V2V-SECPRIV-BSMSIGN-002].

- The system shall attach a certificate to a BSM when the time interval between the current BSM and the generation of a previous BSM with an attached certificate (not certificate digest) is greater than or equal to *vMaxCertDigestInterval* [6.5.2-V2V-SECPRIV-BSMSIGN-003].

- The system shall attach the entire certificate (not a certificate digest) to the BSM when a critical event flag is set [6.5.2-V2V-SECPRIV-BSMSIGN-004].

- The system shall not transmit BSMs when it has no valid certificates [6.5.2-V2V-SECPRIV-BSMSIGN-005].

- The system shall not transmit BSMs with an expired certificate [6.5.2-V2V-SECPRIV-BSMSIGN-006].

- The system shall attach the entire certificate (not a certificate digest) to the first BSM generated after system device startup [6.5.2-V2V-SECPRIV-BSMSIGN-007].

- The system shall attach the entire certificate (not a certificate digest) to the first BSM generated after the certificate has been changed [6.5.2-V2V-SECPRIV-BSMSIGN-008].

- The System shall attach a certificate digest (not a full certificate) if a full certificate is not required (according to the other BSM signing requirements in 6.5.2) [6.5.2-V2V-SECPRIV-BSMSIGN-009].

6.5.3 Certificate Change (CERTCHG)

- To preserve privacy, the system shall not use the same certificate for more than *vCertChangeInterval* consecutive minutes, unless one or more of the following exceptions holds.

  - Exceptions:

    - The absolute distance between the current position and the location at which the last certificate change occurred is less than *vCertChangeDistance* (with a tolerance of *vCertChangeDistanceTolerance*).

    - One or more critical event flags are set.

[6.5.3-V2V-SECPRIV-CERTCHG-001]

NOTE: For example, if a vehicle changes its certificate when more than *vCertChangeDistance* minus *vCertChangeDistanceTolerance* from the location at which the last certificate change occurred, and no critical event flags are set, it complies with the requirement.

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NOTE: Non-consumer vehicles used for commercial or government agency purposes may use alternate certificate change/privacy protection mechanisms.

- The system shall not change its certificate as long as one or more critical event flags are set, unless the certificate expires [6.5.3-V2V-SECPRIV-CERTCHG-002].

- The system shall not change its certificate if it is less than *vCertChangeDistance* (with a tolerance of *vCertChangeDistanceTolerance*) in absolute distance from the location at which the last certificate change occurred, unless the certificate expires, or unless system device shutdown and startup have occurred since the last certificate change [6.5.3-V2V-SECPRIV-CERTCHG-003].

NOTE: Absolute distance is the shortest distance between two points, not total distance driven between two points. The intent of the requirements is that the certificate does not change if the corresponding vehicle is within a circle *vCertChangeDistance* in radius, centered at the location of the last certificate change. A system may change its certificate any time before *vCertChangeInterval* expires if the corresponding vehicle is outside of that circle and no Critical Event Flags are set.

NOTE: DE_MsgCount, DE_Temporary ID and the layer 2 address are each randomly reinitialized when the security certificate changes. While DE_MsgCount, DE_Temporary ID, and the layer 2 address are required to be randomized at system device startup, certificates are not required to be changed at System device startup.

6.5.4 BSM Cryptographic Verification (BSMVERIFY)

- If a system chooses to verify a BSM, the system shall verify the BSM using the signature of the BSM and the attached certificate. If a certificate digest instead of a certificate is received with the BSM, the system verifies the BSM using the certificate corresponding to the digest if the certificate was received with an earlier BSM from the same transmitter [6.5.4-V2V-SECPRIV-BSMVERIFY-001].

6.5.5 Certificate Revocation (CERTREV)

- The system shall not transmit any messages with a certificate that is included in a certificate revocation list (CRL) that has been received and verified by the system [6.5.5-V2V-SECPRIV-CERTREV-001].

6.6 Security Management (SECMGMT)

The system interfaces to a device configuration manager (DCM) function for bootstrap processing and to the SCMS to request and download pseudonym certificates. Refer to [9] and [10] for a description of the DCM and SCMS.

The DCM function is expected to be a proprietary capability developed by device manufacturers to support the functions described in 6.6.1. The system to DCM interface is expected to be trusted and proprietary.

The system to SCMS protocol will be described in an interface specification published by the SCMS entity. The system uses the interface specification protocol to communicate with the SCMS, or any other secure protocol supported by the SCMS. This standard describes the system functions based on the informative text in 6.6.1.

The interface between the system and the device configuration manager (DCM) may be via a direct, wired, or wireless communication. The system can interface with the SCMS via DSRC, cellular, or other wireless network. The connection types are outside of the scope of this specification.

For the purpose of the security management requirements in this standard, the system is assumed to have access to a DCM connection during the bootstrap process and to an SCMS connection for subsequent security management functions. The SCMS connection is not always available on a continuous basis (e.g., if the system is using DSRC and is outside the range of an RSE with connectivity to the SCMS).

The system needs to be capable of resolving IP addresses using a secure domain name system (DNS). The methodology for doing this is outside the scope of this specification.

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6.6.1 Bootstrap: Initialization and Enrollment Processing (Informative)

The bootstrap process takes place in a secure environment. The exact implementation of the bootstrap process is proprietary and outside the scope of this standard.

6.6.1.1 Initialization Processing

- The system interfaces to a DCM to acquire the root certificate authority (CA) certificate.
- The system interfaces to a DCM to acquire the uniform resource locator (URL) of the registration authority (RA), pseudonym certificate authority (PCA) certificates, intermediate certificate authority (ICA) certificates, misbehavior authority (MA) certificate, certificate revocation list generator (CRLG) certificate, security policies, and the current global certificate revocation list (CRL).
- The system is expected to obtain the RA certificate from the DCM. If the system does not receive an RA certificate from the DCM, it requests the certificate from the RA.
- The system resolves the IP address of the RA using a secure domain name system (DNS).

6.6.1.2 Enrollment Processing

- After a system completes initialization, it interfaces to a DCM to acquire an enrollment certificate.
- The system generates an enrollment public/private key pair, and provides the public key to the DCM for use in generating the enrollment certificate. Alternatively, an external-to-the-device and trusted entity may generate a public/private key pair and provide them to the system.

6.6.2 Certificate Loading (CERTLOAD)

- The system securely generates encryption and signing key pairs.
- The system obtains pseudonym certificates from the SCMS over a secure interface.
- The system obtains new batches of pseudonym certificates when necessary and connectivity to the SCMS is available.

NOTE: The three bulleted items above are related to SCMS protocol and interface requirements, which will be defined by the SCMS entity.

- The system shall be capable of securely updating root CA certificates [6.6.2-V2V-SECMGMT-CERTLOAD-001].

6.6.3 Certificate Storage (CERTSTORE)

- The system shall have at least *vCertNvMemSize* of non-volatile memory for storage of pseudonym certificates [6.6.3-V2V-SECMGMT-CERTSTORE-001].
- The system shall have at least *vSecMemSize* of secure memory available for data requiring secure storage [6.6.3-V2V-SECMGMT-CERTSTORE-002].
- The system shall store the individual, pseudonym certificates, the RA address, RA, intermediate CA, and PCA certificates, system configurations, and security policies in non-volatile memory [6.6.3-V2V-SECMGMT-CERTSTORE-003].
- The system shall store the root CA certificate, enrollment certificate, and system private keys in secure, tamper evident, non-volatile memory [6.6.3-V2V-SECMGMT-CERTSTORE-004].

NOTE: Tamper evidence requirements are defined by FIPS 140-2 [16].

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6.6.4 Certificate Revocation List Loading (CRLLOAD)

- The system shall store the most up-to-date CRL information it has received in non-volatile memory, subject to the minimum storage requirements below [6.6.4-V2V-SECMGMT-CRLLOAD-001].

- The system shall have at least *vCrlStoreSize* of non-volatile memory for storing the CRL [6.6.4-V2V-SECMGMT-CRLLOAD-002].

NOTE: The minimum CRL storage requirement (see Section 7) enables the storage of 10000 CRL entries of 40 bytes each.

6.6.5 Secure Hardware (SECHW)

- The system shall incorporate secure hardware that complies with FIPS 140-2 [16] requirements as specified for security level 2:
  - Operator authentication
  - Physical security
  - Operating system requirements
  - Cryptographic key management
  - Design assurance

  [6.6.5-V2V-SECMGMT-SECHW-001]

- The system shall perform all private key operations within the secure hardware [6.6.5-V2V-SECMGMT-SECHW-002].

7. PARAMETER SETTINGS

This section contains the values assigned to the parameters identified in Section 6 of this standard (Table 21).

Table 21 - Parameter settings for this standard

Section Reference(s)	Parameter	Value	Rationale(s)
6.1.1, 6.1.2.2.2, 6.1.3, 6.1.4, 6.3, 6.3.2, 6.3.4, 6.3.6.16	vChannelNumber	172	1
6.1.2.2.2	vP2pcd_maxResponseBackoff	0.25 second	7
6.1.2.2.2	vP2pcd_responseActiveTimeout	0.25 second	7
6.1.2.2.2	vP2pcd_requestActiveTimeout	0.25 second	7
6.1.2.2.2	vP2pcd_observedRequestTimeout	0.25 second	7
6.1.2.2.2	vP2pcd_currentlyUsedTriggerCertificateTime	1 minute	7
6.1.2.2.2	vP2pcd_responseCountThreshold	3	7
6.2.1	vPosDetRate	10 Hz	2, 3
6.2.4, 6.3.6.5	vTimeAccuracy	1 ms	2, 3
6.3.2	vDataRate	6 Mbps	3, 4, 5
6.3.3	vBSMRateTolerance	10 ms	3
6.3.6.4	vMaxPosAge	150 ms	3
6.3.6.5	vPosAccuracy	1.5 m	2, 3
6.3.6.6	vElevAccuracy	3 m	2, 3
6.3.6.8	vSpeedAccuracy	0.28 m/s	2, 3
6.3.6.10	vHeadAccuracyA	2 degrees	2, 3
6.3.6.10	vHeadAccuracyB	3 degrees	2, 3
6.3.6.10	vHeadingSpeedThresh	12.5 m/s	2, 3
6.3.6.10	vHeadLatchThresh	1.10 m/s	2, 3

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Section Reference(s)	Parameter	Value	Rationale(s)
6.3.6.10	vHeadUnlatchThresh	1.38 m/s	2, 3
6.3.6.11	vStWhAnAccuracy	5 degrees	2, 3
6.3.6.12	vAccelAccuracy	0.3 m/s^2	2, 3
6.3.6.12	vVertAccelAccuracy	1 m/s^2	2, 3
6.3.6.12	vYawRateAccuracy	0.5 degrees/s	2, 3
6.3.6.14	vSizeAccuracy	0.2 m	2, 3
6.3.6.15	vEventDetectLatency	250 ms	2
6.3.6.16	vMinPHistDistance	200 m	2
6.3.6.16	vMaxChordLength	210 m	2
6.3.6.16	vPathPerpendicularDist	1 m	2
6.3.6.16	vMaxPHistPoints	15	SAE J2735 [1]
6.3.6.17	vPPredRadiusError	10%	2, 3
6.3.6.17	vMinCurveRadius	80 m	2, 3
6.3.6.17	vMaxCurveRadius	2500 m	2, 3
6.3.6.17	vPPredTransitionTime	4 seconds	2, 3
6.3.6.17	vStationarySpeedThresh	1 m/s	2, 3
6.3.8, 6.3.8.1, 6.3.8.7	vCBPMeasInt	100 ms	4
6.3.8, 6.3.8.1	vPERInterval	5000 ms	4
6.3.8, 6.3.8.1	vPERSubInterval	1000 ms	4
6.3.8, 6.3.8.2, 6.3.8.3, 6.3.8.4, 6.3.8.5, 6.3.8.6, 6.3.8.8	vTxRateCntrlInt	100 ms	4
6.3.8.1	vCBPWeightFactor	0.5	4
6.3.8.1	vPERRange	100 m	4
6.3.8.1	vPERMax	0.3	4
6.3.8.2	vHVLocalPosEstIntMin	0 ms	4
6.3.8.2	vHVLocalPosEstIntMax	200 ms	4
6.3.8.2	vHVRemotePosEstIntMin	0 ms	4
6.3.8.2	vHVRemotePosEstIntMax	3000 ms	4
6.3.8.3	vTrackingErrMin	0.2 m	4
6.3.8.3	vTrackingErrMax	0.5 m	4
6.3.8.3	vErrSensitivity	75	4
6.3.8.4	vDensityWeightFactor	0.05	4
6.3.8.4	vDensityCoefficient	25	4
6.3.8.4	vMax_ITT	600 ms	4
6.3.8.4	vMin_ITT	100 ms	4
6.3.8.5	vRescheduleTh	25 ms	4
6.3.8.7, 6.4.1.1	vRPMax	20 dBm	4, 5
6.3.8.7, 6.4.1.1	vRPMin	10 dBm	4
6.3.8.7	vSUPRAGain	0.5	4
6.3.8.7	vMinChanUtil	50%	4
6.3.8.7	vMaxChanUtil	80%	4
6.3.8.8	vTxRand	5 ms	4
6.4.1, 6.3.8.7	vRP	15 dBm	4
6.4.1	vLowerEl	96 degrees	4
6.4.1	vUpperEl	80 degrees	4
6.4.1.1	vTxPwrRange	10 dB	4
6.4.1.1	vTxPwrAcc	2 dB	4
6.4.1.1, 6.3.8.7	vTxPwrCtrlStep	1 dB	4
6.4.2	vRxSens	-92 dBm	4, 5
6.5.2	vMaxCertDigestInterval	450 ms	6, 7
6.5.3	vCertChangeInterval	5 minutes	7
6.5.3	vCertChangeDistance	2100 m	7
6.5.3	vCertChangeDistanceTolerance	100 m	3

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Section Reference(s)	Parameter	Value	Rationale(s)
6.6.3	vCertNvMemSize	500 kilobytes	6
6.6.3	vSecMemSize	100 kilobytes	6
6.6.4	vCrlStoreSize	400 kilobytes	6
A.8.1	vMaxSuccessiveFail	3	4

Rationales:

1. The setting is the RF channel designated for public safety applications involving safety of life and property according to FCC rules (refer to [7]).

2. The setting is based on the need to provide accurate and timely safety alerts for the use cases described in Section 4 (refer to [12]).

3. The setting was obtained by extensively testing commercially available equipment and automotive sensors in a wide variety of driving environments, and the numbers were proven to be reasonable based on the equipment and sensor capabilities, while also supporting the use cases described in Section 4 (refer to [8] and [12] through [14]).

4. The parameter setting improves RF performance (reduces packet collisions) and/or is based on extensive congestion control research (refer to [12] and [18] through [20]).

5. The setting is based on the necessary range for V2V safety communications, which for the purposes of this document is 300 m.

6. The settings are derived from calculations based on the size of certificates and/or best engineering judgment (refer to [9], [10]).

7. The setting is based on recommendations from IEEE 1609.

8. NOTES

8.1 Revision Indicator

A change bar (I) located in the left margin is for the convenience of the user in locating areas where technical revisions, not editorial changes, have been made to the previous issue of this document. An (R) symbol to the left of the document title indicates a complete revision of the document, including technical revisions. Change bars and (R) are not used in original publications, nor in documents that contain editorial changes only.

PREPARED BY THE SAE DSRC (DEDICATED SHORT RANGE COMMUNICATION) TECHNICAL COMMITTEE

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APPENDIX A

A.1 IMPLEMENTATION CONFORMANCE STATEMENT (NORMATIVE)

A conformance table is provided in Table A1. The "Conformant?" column is intended to be filled out by an implementer or tester to indicate compliance with mandatory and optional features.

Table A1 - Implementation conformance table

Section (This Standard)	Requirement (Req) Category	Req Number	Mandatory/ Optional (M/O)	Conformant? (Y/N)	Requirement or Section Title (This Standard)
6.1	**Standards compliance**				
6.1.1	*IEEE 802.11: General description*				
	V2V-STD-802.11	001	M		STA transmission of data frames outside the context of a BSS
6.1.1	*IEEE 802.11: MAC service definition*				
	V2V-STD-802.11	002	M		Overview of MAC services
		003	M		MAC data service specification
6.1.1	*IEEE 802.11: Layer management*				
	V2V-STD-802.11	004	M		Reset
		005	O		Get TSF timer
6.1.1	*IEEE 802.11: PHY service specification*				
	V2V-STD-802.11	006	M		Scope
		007	M		PHY functions
		008	M		Detailed PHY service specification
6.1.1	*IEEE 802.11: Frame Formats*				
	V2V-STD-802.11	009	M		General requirements
		010	M		MAC frame formats
6.1.1	*IEEE 802.11: Format of individual frame types*				
	V2V-STD-802.11	011	O		Format of control frames
		012	O		ACK frame format
		013	M		Data frames
		014	M		Data frame format
		015	M		EDCA parameter set element
6.1.1	*IEEE 802.11: MAC sublayer functional description*				
	V2V-STD-802.11	016	M		Hybrid coordination function
		017	M		Multirate support
6.1.1	*IEEE 802.11: HCF*				
	V2V-STD-802.11	018	M		General
		019	M		HCF contention-based channel access (EDCA)
6.1.1	*IEEE 802.11: MLME*				
	V2V-STD-802.11	020	M		Synchronization
		021	M		STAs communicating data frames outside the context of a BSS
		022	M		Orthogonal frequency division multiplexing (OFDM) PHY specification
		023	M		ASN.1 encoding of the MAC and PHY MIB
6.1.1	*IEEE 802.11: Regulatory References*				
	V2V-STD-802.11	024	M		External regulatory references
		025	M		External regulatory references
		026	M		Transmit and receive in-band and out-of-band spurious emissions
		027	M		Transmit power levels
		028	M		Transmit spectrum mask

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Section (This Standard)	Requirement (Req) Category	Req Number	Mandatory/ Optional (M/O)	Conformant? (Y/N)	Requirement or Section Title (This Standard)
6.1.1	*IEEE 802.11: Country elements and operating classes*				
	V2V-STD-802.11	029	M		Country information and operating classes
		030	O		Country information and operating classes
		031	M		5.9 GHz band in the United States (5.850-5.925 GHz)
6.1.2	*IEEE 1609.2*				
	V2V-STD-1609.2	001	M		BSM Security Profile - ID
		002	M		BSM Security Profile - Sending
		003	M		BSM Security Profile - Receiving
		004	M		BSM Security Profile - Management
6.1.3	*IEEE 1609.3*				
		001	M		V2V-required features
		002	O		SCMS-required features
6.1.4	*IEEE 1609.4*				
		001	M		V2V-required features
		002	O		SCMS-required features
6.1.5	*IEEE 1609.12: WAVE identifiers*				
	V2V-STD-1609.12	001	M		Provider service identifier (PSID)
		002	O (SCMS only)		Provider service identifier (PSID)
		003	M		Ethertype
		004	O (SCMS only)		Ethertype
6.1.6	*SAE J2735: Message encoding*				
	V2V-STD-J2735	001	M		Message encoding
6.1.6	*SAE J2735: Message_MessageFrame*				
	V2V-STD-J2735	002	M		DE_DSRC_MessageID
		003	M		Message: MSG_BasicSafetyMessage (BSM)
		004	M		Message: MSG_BasicSafetyMessage (BSM)
6.1.6	*SAE J2735: Data frames*				
	V2V-STD-J2735	005	M		Data Frame: DF_AccelerationSet4Way
		006	M		Data Frame: DF_BrakeSystemStatus
		007	M		Data Frame: DF_BSMcoreData
		008	M		Data Frame: DF_PathHistory
		009	M		Data Frame: DF_PathHistoryPointList
		010	M		Data Frame: DF_PathHistoryPoint
		011	M		Data Frame: DF_PathPrediction
		012	M		Data Frame: DF_PositionalAccuracy
		013	M		Data Frame: DF_VehicleSafetyExtensions
		014	M		Data Frame: DF_VehicleSize
6.1.6	*SAE J2735: Data elements*				
	V2V-STD-J2735	015	M		Data Element: DE_Acceleration
		016	M		Data Element: DE_AntiLockBrakeStatus
		017	M		Data Element: DE_AuxiliaryBrakeStatus

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Section (This Standard)	Requirement (Req) Category	Req Number	Mandatory/ Optional (M/O)	Conformant? (Y/N)	Requirement or Section Title (This Standard)
		018	M		Data Element: DE_BrakeAppliedStatus
		019	M		Data Element: DE_BrakeBoostApplied
		020	M		Data Element: DE_Confidence
		021	M		Data Element: DE_DSecond
		022	M		Data Element: DE_Elevation
		023	O		Data Element: DE_ExteriorLights
		024	M		Data Element: DE_Heading
		025	M		Data Element: DE_Latitude
		026	M		Data Element: DE_Longitude
		027	M		Data Element: DE_MsgCount
		028	M		Data Element: DE_OffsetLL-B18
		029	M		Data Element: DE_RadiusOfCurvature
		030	M		Data Element: DE_SemiMajorAxisAccuracy
		031	M		Data Element: DE_SemiMajorAxisOrientation
		032	M		Data Element: DE_SemiMinorAxisAccuracy
		033	M		Data Element: DE_Speed
		034	M		Data Element: DE_StabilityControlStatus
		035	M		Data Element: DE_SteeringWheelAngle
		036	M		Data Element: DE_TemporaryID
		037			DE_TimeOffset
		038	M		Data Element: DE_TractionControlStatus
		039	M		Data Element: DE_TransmissionStatus
		040	M		Data Element: DE_VehicleEventFlags
		041	M		Data Element: DE_VehicleLength
		042	M		Data Element: DE_VehicleWidth
		043	M		Data Element: DE_VerticalAcceleration
		044	M		DE_VertOffset-B12
		045	M		Data Element: DE_YawRate
6.2	**Position and timing requirements**				
6.2.1	V2V-POSTIM-POSDETER	001	M		Position determination
		002	M		
6.2.2	V2V-POSTIM-WAAS	001	M		WAAS
6.2.3	V2V-POSTIM-COORDSYSREF	001	M		Coordinate system reference
6.2.4	V2V-POSTIM-SYSTIMCOORD	001	M		System time coordination
		002	M		
		003	M		
6.3	**BSM transmission requirements**				
6.3.1	BSMTX-BSMCONT	001	M		BSM contents
		002	M		

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Section (This Standard)	Requirement (Req) Category	Req Number	Mandatory/ Optional (M/O)	Conformant? (Y/N)	Requirement or Section Title (This Standard)
		003	M		
		004	M		
		005	M		
		006	M		
6.3.2	V2V-BSMTX-CHDATARATE	001	M		Channel and data rate
		002	M		
6.3.3	V2V-BSMTX-GENTIM	001	M		Generation of the first BSM after startup and generation timing
		002	M		
6.3.4	V2V-BSMTX-UPEDCA	001	M		User priority and EDCA settings
		002	M		
		003	M		
6.3.5	V2V-BSMTX-MINTX	001	M		Minimum transmission criteria
6.3.6	*Data element accuracy*				
6.3.6.1	V2V-BSMTX-DATAACC	001	M		DE_DSRC_MessageID
6.3.6.2		002	M		DE_MsgCount
		003	M		
		004	M		
6.3.6.3		005	M		DE_TemporaryID
		006	M		
		007	M		
6.3.6.4		008	M		DE_DSecond
		009	M		
		010	M		
6.3.6.5		011	M		DE_Latitude & DE_Longitude
		012	M		
6.3.6.6		013	M		DE_Elevation
		014	M		
6.3.6.7		015	M		DF_PositionalAccuracy
		016	M		
6.3.6.8		017	M		DE_Speed
6.3.6.9		018	M		DE_Transmission
6.3.6.10		019	M		DE_Heading
		020	M		
		021	M		
		022	M		
		023	M		
6.3.6.11		024	M		DE_SteeringWheelAngle
6.3.6.12		025	M		DF_AccelerationSet4Way
		026	M		
		027	M		
6.3.6.13		028	M		DF_BrakeSystemStatus
		029	M		
		030	M		
		031	M		
		032	M		
6.3.6.14		033	M		DF_VehicleSize
6.3.6.15		034	M		DE_VehicleEventFlags
		035	M		
6.3.6.16		036	M		DF_PathHistory
		037	M		

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Section (This Standard)	Requirement (Req) Category	Req Number	Mandatory/ Optional (M/O)	Conformant? (Y/N)	Requirement or Section Title (This Standard)
		038	M		
		039	M		
		040	M		
		041	M		
		042	M		
		043	M		
		044	M		
6.3.6.17		045	M		DF_PathPrediction
		046	M		
		047	M		
		048	M		
6.3.6.18		049	M		DE_ExteriorLights
6.3.6.19		050	M		Additional data elements
6.3.7	V2V-BSMTX-DATAPERSIST	001	M		Data persistency
		002	M		
		003	M		
		004	M		
6.3.8	V2V-BSMTX-CONGCTRL	001	M		BSM scheduling and congestion control
6.4	**RF Performance requirements**				
6.4.1	V2V-RFPERF-DSRCTX	001	M		Radiated power
6.4.1.1		002	M		Transmit power accuracy
		003			
		004	M		
6.4.2	V2V-RFPERF-DSRCRXSENS	001	M		Receive sensitivity
6.5	**Security and Privacy Tx Requirements**				
6.5.1	V2V-SECPRIV-IDRAND	001	M		Identification randomization
		002	M		
6.5.2	V2V-SECPRIV-BSMSIGN	001	M		BSM signing
		002	M		
		003	M		
		004	M		
		005	M		
		006	M		
		007	M		
		008	M		
6.5.3	V2V-SECPRIV-CERTCHG	001	M		Certificate change
		002	M		
		003	M		
6.5.4	V2V-SECPRIV-BSMVERIFY	001	O		BSM verification
6.5.5	V2V-SECPRIV-CERTREV	001	M		Certificate revocation
6.6	**Security management**				
6.6.2	V2V-SECMGMT-CERTLOAD	001	M		Certificate loading
6.6.3	V2V-SECMGMT-CERTSTORE	001	M		Certificate storage
		002	M		

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Section (This Standard)	Requirement (Req) Category	Req Number	Mandatory/ Optional (M/O)	Conformant? (Y/N)	Requirement or Section Title (This Standard)
		003	M		
		004	M		
6.6.4	V2V-SECMGMT-CRLLOAD	001	M		CRL loading
		002	M		
6.6.5	V2V-SECMGMT-SECHW	001	M		Secure hardware
		002	M		

A.2 COORDINATE TRANSFORMATION (NORMATIVE)

FUNCTION

ConvertXYtoLatLon(…)

INPUT

RefLat = e.g., REF_LATITUDE (rad)

RefLon = e.g., REF_LONGITUDE (rad)

RefHeading = e.g., REF_HEADING (rad)

Y = ACROSS_DISTANCE (m w.r.t. REF LATLON)

X = AHEAD_DISTANCE (m w.r.t. REF LATLON)

```
a = 6378137;                                    # semi-major axis of earth
f = 0.003353;                                   # flattening
f1 = (f*(2-f))^0.5;                             # eccentricity
f2 = a*(1-f1^2)/(1-f1^2*(sin(RefLat))^2)^(3/2); # radius of earth in meridian
f3 = a/(1-f1^2*(sin(RefLat))^2)^(1/2);          # radius of earth in prime vertical

E = (cos(RefHeading)*Y + sin(RefHeading)*X;
N = (cos(RefHeading)*X - sin(RefHeading)*Y;
```

OUTPUT

NEW_LATITUDE (rad) = (1/f2)*N + RefLat;

NEW_LONGITUDE (rad) = (1/(f3*cos(RefLat)))*E + RefLon;

NOTE: The vehicle's local coordinate frame (SAE) is represented in Figure A1.

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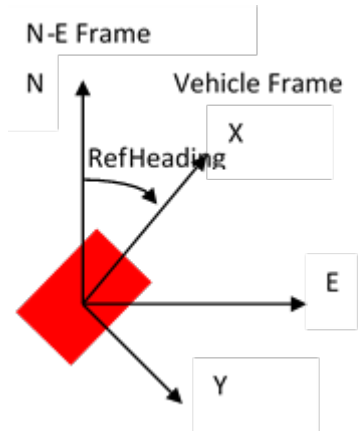

Figure A1 - Vehicle coordinate frame

A.3 2D POSITION EXTRAPOLATION (NORMATIVE)

Position extrapolation for the HV local estimate and HV remote estimate is performed up to the current time based on the extrapolation details presented here. Position extrapolation estimates the vehicle's current position at time T' (current time), based on the vehicle's last known position, heading, and speed at time T (older time). The estimation assumes that the vehicle is moving at a constant speed and constant heading. It uses two variables to determine when to extrapolate, HVPosEstIntMin and HVPosEstIntMax. These values are variables based on the HV local estimate and HV remote estimate.

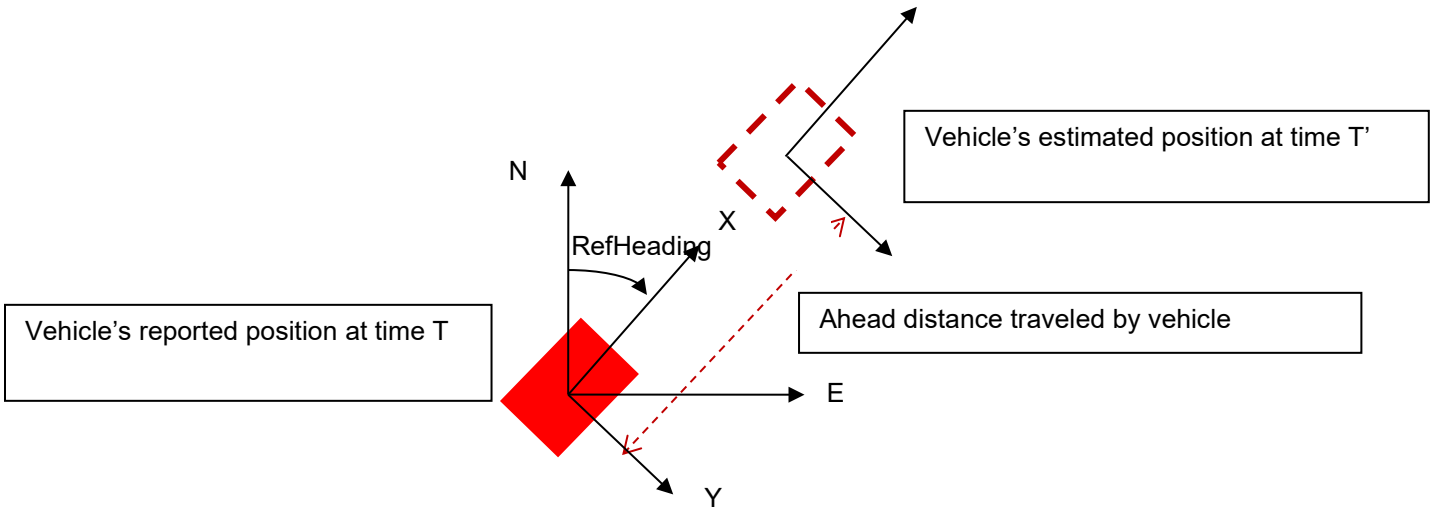

Figure A2 - Position extrapolation

1. First find Delta_time, the time since vehicle's last known position.
   - Delta_time_ms = T' - T
2. Do not perform position extrapolation in the following cases:
   - If Delta_time_ms < HVPosEstIntMin, (New_Latitude, New_Longitude) is the last known position
   - If Delta_time_ms > HVPosEstIntMax, then the vehicle has not received a position update for a very long time and its position is outdated

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3. If HVPosEstIntMin ≤ Delta_time_ms ≤ HVPosEstIntMax, then perform position extrapolation:

   - Calculate the estimated distance traveled by the vehicle in Delta_time_ms
   - Ahead_distance_m = Speed_mps * Delta_time_ms / 1000
   - Across_distance_m = 0

4. Use ConvertXYtoLatLon function (provided in Appendix A.2) to find the vehicle's new position at time T'.

5. The extrapolated vehicle position at the current time is (New_Latitude, New_Longitude).

NOTE: For 3D position extrapolation, the System may assume the elevation remains constant based on the last 3D position update from the Positioning Subsystem.

A.4 CALCULATIONS INTO VEHICLE'S POSITION REFERENCE POINT

Consider Figure 21 where the BSM vehicle position reference point with respect to GNSS antenna location on the vehicle is represented as follows:

- antOffsetX = Distance, in meters, to GNSS antenna location from vehicle position reference along the X axis (signed value).
- antOffsetY = Distance, in meters, to GNSS antenna location from vehicle position reference along the Y axis (signed value).
- antOffsetZ = Height, in meters, to the GNSS antenna from vehicle position reference along the Z-axis for an unloaded stationary vehicle on a planar surface. This value will always be negative. For example, for an antenna at the highest point on the roof, this value will be: - height of the antenna above ground (negative value) in meters. If the antenna is 1.05 m above the ground, then antOffsetZ = -1.05.

  Let:

  RefLat = e.g., GNSS Measured LATITUDE (rad)

  RefLon = e.g., GNSS Measured LONGITUDE (rad)

  RefHeading = e.g., GNSS Measured HEADING (rad)

  Y = -antOffsetY

  X = -antOffsetX

  Use ConvertXYtoLatLon function (provided in Appendix A.2) to find the vehicle's 2D position at position reference point.

  For transmitting V2V safety messages, use the calculated New_Latitude and New_Longitude as vehicle's 2D position at position reference point. These values are calculated for every V2V safety message transmitted.

  Finally,

  New_Elevation = antOffsetZ + GNSS measured elevation (in meters).

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A.5 PATH HISTORY REFERENCE DESIGN (INFORMATIVE)

A.5.1 Introduction

This example Path History (PH) module for the V2V communications system uses a history of the past Global Navigation Satellite System (GNSS) locations traversed by the host vehicle (HV) and computes an adaptable, concise PH representation of recent vehicle movement over a certain distance. The PH communicated by a vehicle provides other vehicles with information needed for predicting the roadway geometry. It plays an important role in target vehicle classification, relative to the HV, with reference to the roadway. There are different methods for design and implementation of the PH module. An example method is described here.

The PH module in the HV carries out these basic operations:

- Maintains a buffer of its recent GNSS positions (updated at 100 ms) over a certain travel distance.
- Computes concise representation(s) of the actual PH of the vehicle based on allowable position error tolerance between the vehicle path and its concise path history representation.
- Updates the PH concise representation as an output periodically for use by other V2V systems.
- Besides having the capability to represent its PH adequately and use it internally, the HV transmits the concise representation of the path history data wirelessly over-the-air (OTA) to other vehicles in the vicinity. Other vehicles use this information for predicting the roadway geometry and for target vehicle classification.

A.5.2 Path History Requirements

The PH module requirements are as follows:

- PH represents the HV vehicle path with a set of concise PH data elements. The concise PH data elements are a sampled subset of the actual vehicle path data elements. As shown in Figure A3, the orange circles represent the PH concise points, and the chord connecting two consecutive concise PH data elements represents an approximation of the actual vehicle path segment.
- The concise PH data elements are selected such that the perpendicular distance between any point on the actual vehicle path and the chord connecting two concise PH points (the concise representation of the actual vehicle path) is less than PH_ActualError, as shown in Figure A3.
- The size of the buffer containing the concise PH data elements is adaptable so that the represented PH distance computed using the elements of the buffer is at least a certain minimum length defined by the calibration parameter, K_PHDISTANCE_M (in meters). Referring to Figure A3, the total distance of all the chords connecting the orange concise data elements is a minimum distance of, K_PHDISTANCE_M meters. Note that the current vehicle position is not included in the PH and PH length (it is already in Part I of the BSM).

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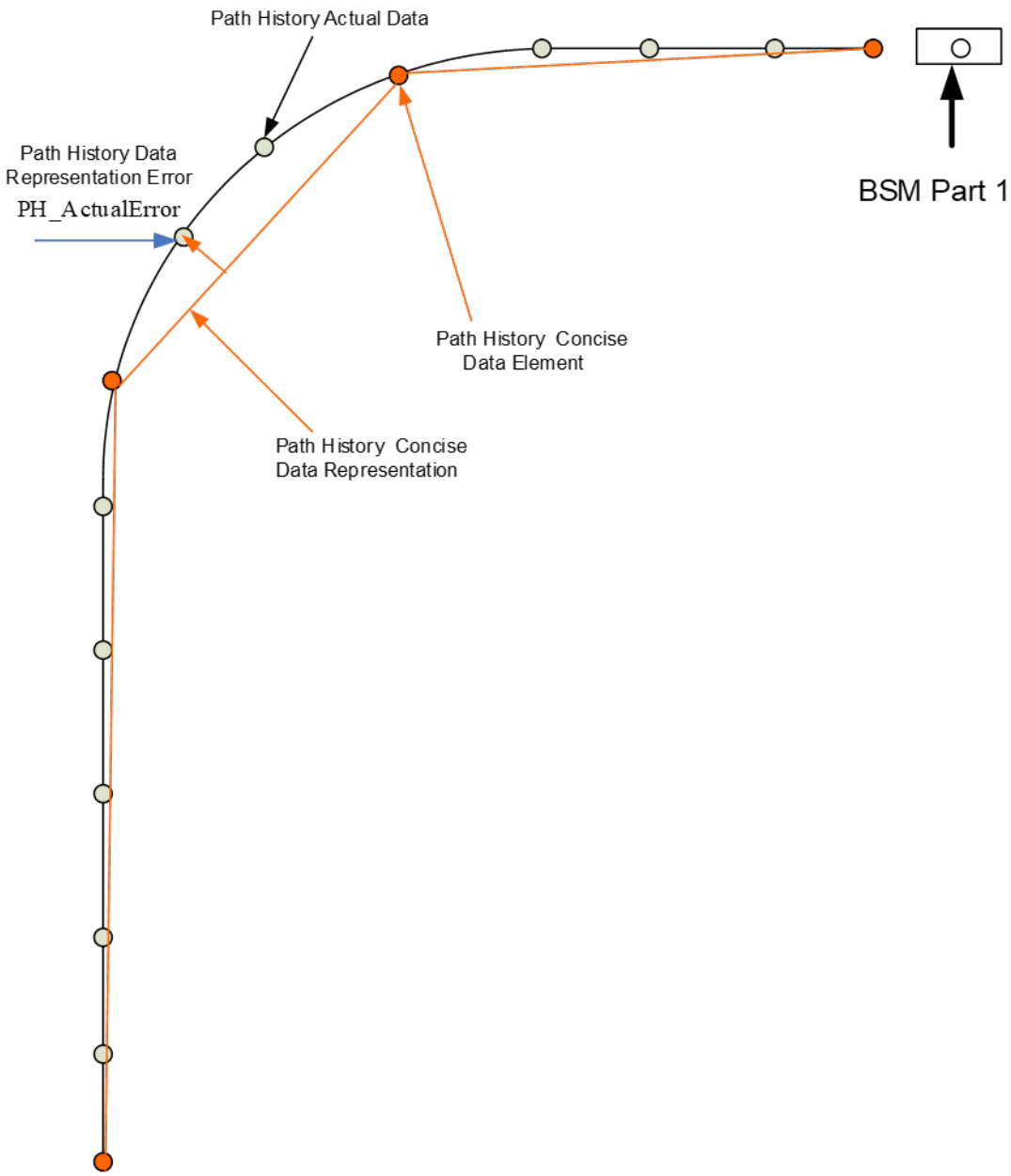

Figure A3 - Concise and actual path history representation

A.5.3 Path History Design

Design Preliminaries

An example design method for PH is presented below. This section defines some basic design preliminaries used by PH design.

a. It is assumed that the vehicle path is composed of the vehicle's recent GNSS positions

b. PH_ActualError is defined as the perpendicular distance between any point on the actual vehicle path and the chord connecting two concise PH points on the concise representation of the vehicle path. Some of the sampled points on the actual vehicle path may become part of the concise PH representation data elements according to the algorithm used. Please refer to Figure A4 for an illustration of PH_ActualError and actual and concise PH data elements.

c. Figure A4 illustrates points P_1, P_2, P_3, etc., that lie on a vehicle path. As illustrated, PH_ActualError varies based on the location of the points selected on the circular path.

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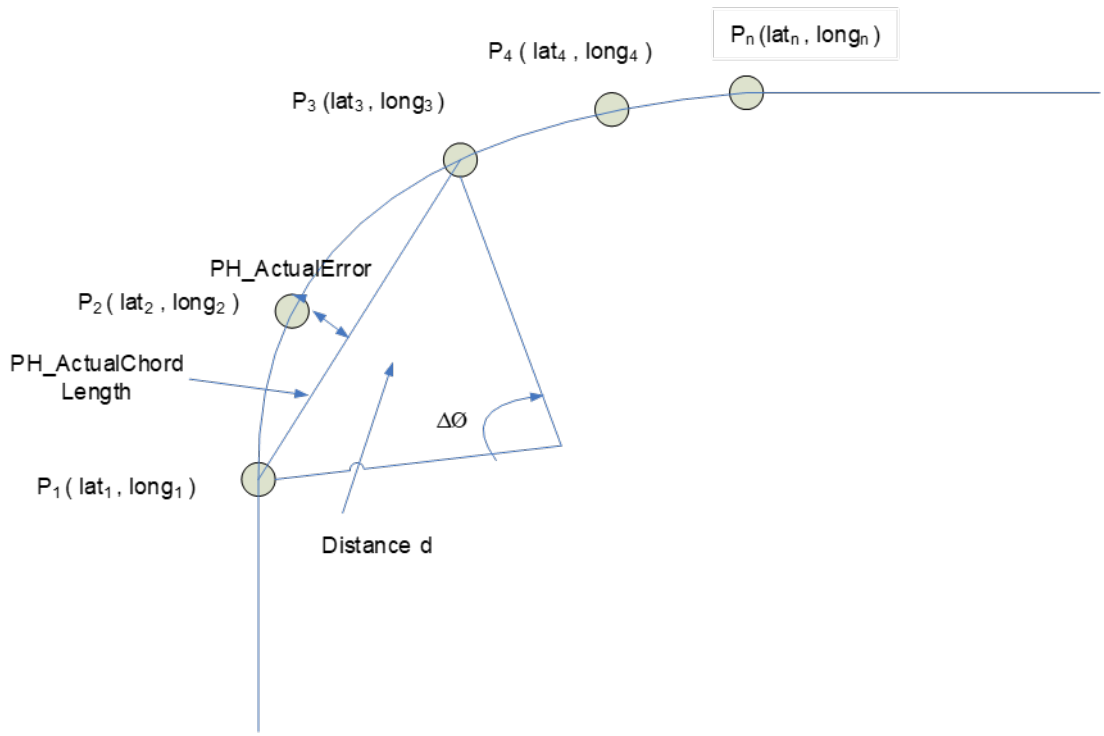

Figure A4 - Representation of Error

d. Referring to Figure A4, the actual chord length is defined as between two PH GNSS points on the circular vehicle path as PH_ActualChordLength. PH_ActualChordLength is the distance between two GNSS data points each defined by its latitude and longitude.

e. Let P_1 be defined by latitude, lat_1, and longitude, $long_1$. Similarly, let P_2 be defined by latitude, lat_2, and its longitude, $long_2$. These values are in radians. Define the radius of the earth (in meters) at the meridian as REarthMeridian. Then the actual distance of the chord is given by:

$$PH_ActualChordLength = REarthMeridian * \cos^{-1}[\cos(lat_1)\cos(lat_2)\cos(long_1 - long_2) + \sin(lat_1)\sin(lat_2)] \quad \text{(Eq. A1)}$$

NOTE: REarthMeridian is the radius of earth in meridian (in meters) and is computed as f2 defined in A.2.

A.5.3.1 Example PH Design Method

In this method, PH_ActualError is the maximum perpendicular distance between the actual vehicle path data elements and the chord connecting the concise PH representation data elements.

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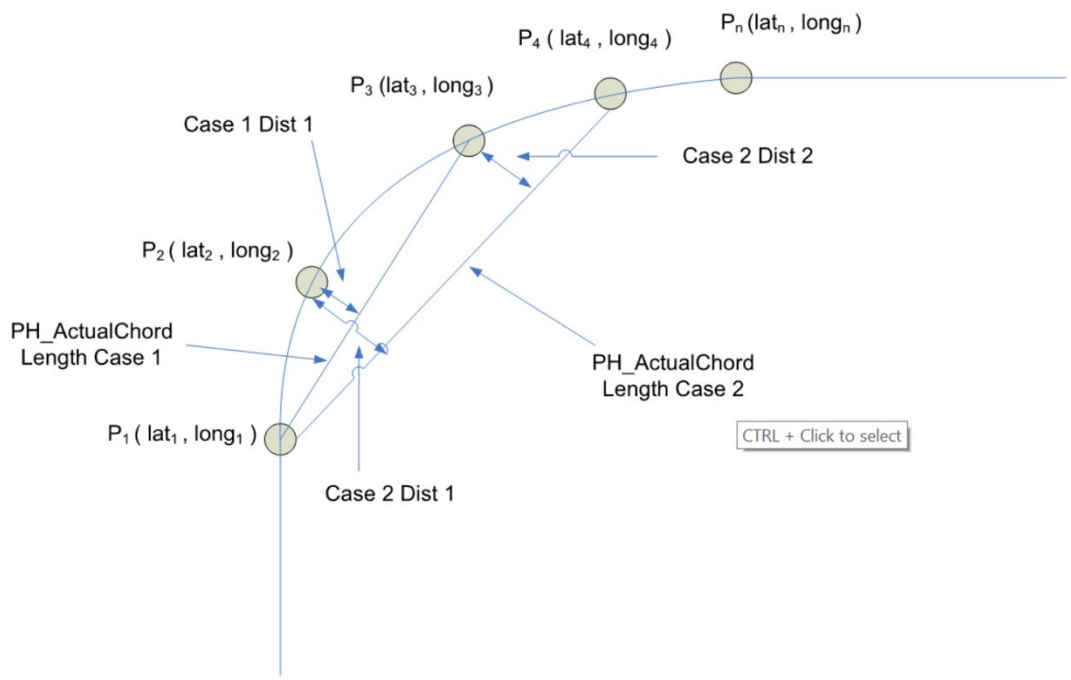

Figure A5 - Representation of PH Error for the Example PH Design Method

The steps involved in the design of the concise PH representation of a vehicle path using the example design method are described in pseudo code next.

Step One: Assume that a number of actual vehicle path GNSS data points that follow the vehicle path are sampled. The minimum number of points required is three. Initial conditions of these points are (see Figures A4 and A5):

i = 3

Starting Point, $P_{starting} = P_{i-2}$

Previous Point, $P_{previous} = P_{i-1}$

Next Point, $P_{next} = P_i$

elementPos = 0

totalDist = 0

incrementDist = 0

Include the GNSS point, $P_{starting,}$ as part of the concise PH representation data buffer and increment the elementPos by one as follows:

PH_ConciseDataBuffer[elementPos] = $P_{starting}$

elementPos++

Step Two: Calculate PH_ActualChordLength (i.e., chord length in meters) between two points, the Starting Point, $P_{starting}$, and the Next Point P_{next}, as shown in Figures A4 and A5, and Equation A1.

If PH_ActualChordLength > K_PH_CHORDLENGTHTHRESHOLD,

Set PH_ActualError to K_PHALLOWABLEERROR_M + 1

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Go to Step Four.

Otherwise, continue.

Step Three: Calculate PH_ActualError as follows:

Define PH data elements, such that P_1 is the Starting Point, P_n is the Next Point, and the Intermediate Points are P_2 through P_{n-1} as shown in Figure A5. Define the perpendicular distance between the Intermediate Points and the chord connecting $P_{starting}$ and P_{next} as D_i, where i = 2, …, n-1. Define PH_ActualError as

PH_ActualError = MAX(D_i); i = 2, …, n-1. (2)

The procedure for calculating the distances D_i is described next. Before performing the following calculations, the GNSS coordinates of the points must be represented into the north-east coordinate frame. The following provides a solution to finding the shortest distance from a point to a line or line segment.

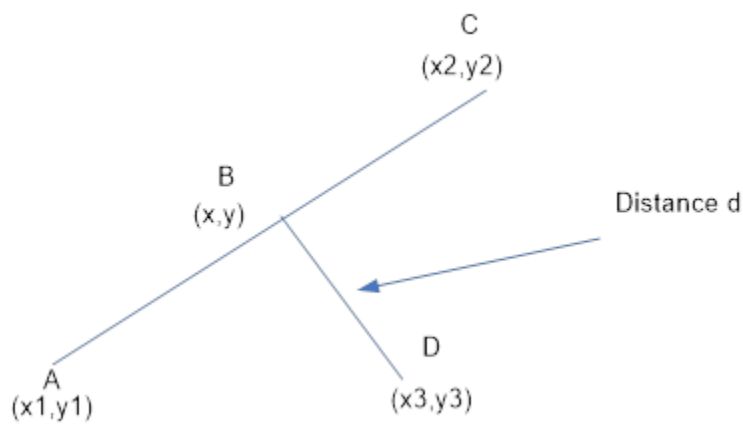

Figure A6 - Shortest Distance From a Point to a Line Segment

Consider Figure A6. A solution is provided to the shortest distance from point **D** to the line segment AC. The equation of a line segment defined through two points **A** (x1,y1) and **C** (x2,y2) is given by

B = **A** + u (**C** - **A**),

where u is a value between 0 and 1. The point **B** (x,y) on the line segment AC that is closest to **D**, satisfies

(**D** - **B**) dot (**C** - **A**) = 0,

where “dot” indicates the dot product of the vectors. Substituting for **B** in the above equation gives

[**D** - **A** - u(**C** - **A**)] dot (**C** - **A**) = 0.

Solving this gives the value of u as

$u = ((x3-x1)(x2-x1) + (y3-y1)(y2-y1)) / || C - A||^2$.

Substituting this into the equation of the line gives the point of intersection **B** (x,y) as

x = x1+ u(x2 - x1),

y = y1 + u(y2 - y1).

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The distance therefore between the point **D** and the line is the Euclidean distance between (x,y) and **D**:

$d = sqrt((x3-x)^2 + (y3-y)^2)$.

Note: Before computing the distance of the point to a line segment, it is necessary to first test that u lies between 0 and 1.

Step Four: If PH_ActualError is greater than the allowable PH error, K_PHALLOWABLEERROR_M, then add the previous point $P_{previous}$ to the concise data buffer as follows:

If PH_ActualError > K_PHALLOWABLEERROR_M

PH_ConciseDataBuffer[elementPos] = $P_{previous}$

elementPos++

Redefine three GNSS data points for further processing. The new points are set to the starting point, previous point, and next point as follows:

$P_{starting} = P_{i-1}$

$P_{next} = P_{i+1}$

$P_{previous} = P_i$

i = i + 1

Go to Step Six.

Step Five: If PH_ActualError ≤ K_PHALLOWABLEERROR_M, redefine the previous point and next point as:

$P_{next} = P_{i+1}$

$P_{previous} = P_i$

i = i + 1

Go to Step Two.

The algorithm repeats itself with the assigned values of starting point, previous point, and next point. This procedure repeats until the error violation occurs.

Step Six: Calculate the sum of the actual distances between the consecutive PH GNSS data points in the concise buffer PH_ConciseDataBuffer as follows:

totalDist = totalDist + incrementDist

totalDist is the sum of distances between PH GNSS points in the concise data buffer PH_ConciseDataBuffer.

incrementDist is the distance between the last two PH GNSS data points added to the concise data buffer. Hence, if the total distance is greater or equal to K_PHDISTANCE_M, then keep deleting elements from the bottom of the concise buffer (i.e., the oldest points) until the total distance becomes just enough to maintain a minimum value of K_PHDISTANCE_M. If the number of elements remaining in the concise buffer exceeds the maximum allowed (15), then keep deleting the oldest points in the buffer until only 15 points remain.

Go to Step Two.

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A.5.4 PH Module Signal Interface Description

In this subsection, the inputs, outputs, and calibration parameters used in the PH module are provided.

Inputs to the PH module are:

Coordinated Universal Time (UTC) time; latitude; longitude; altitude (elevation); speed; heading; yaw rate

Calibration parameters for the PH Module are:

K_PHDISTANCE_M200 (meters)

K_PHDATAPOINTSSAMPLETIME_S100 (milliseconds)

K_PHALLOWABLEERROR_M1 (meters)

K_PH_CHORDLENGTHTHRESHOLD210 (meters)

K_PH_MAXESTIMATEDRADIUS7FFFFF (meters)

The outputs are available in the concise PH data structure buffer and are the PH data elements. Outputs of the PH module are:

N; // number of PH concise representation data elements

PH_CONCISE_DATA_ELEMENT_1,

….

….

PH_CONCISE_DATA_ELEMENT_N,

where, PH_CONCISE_DATA_ELEMENT consists of,

PH_UTCTime; PH_Latitude; PH_Longitude; PH_Altitude; PH_Speed; PH_Heading.

A.5.5 Test Results

A.5.5.1 Concise PH Representation of a Vehicle Path

Data were collected over a certain vehicle path as shown in Figure A7. GNSS data were collected using the u-blox® 8 receiver with the data sampling interval of the actual vehicle position data being 100 ms. The actual vehicle path is represented by 2666 GNSS data points.

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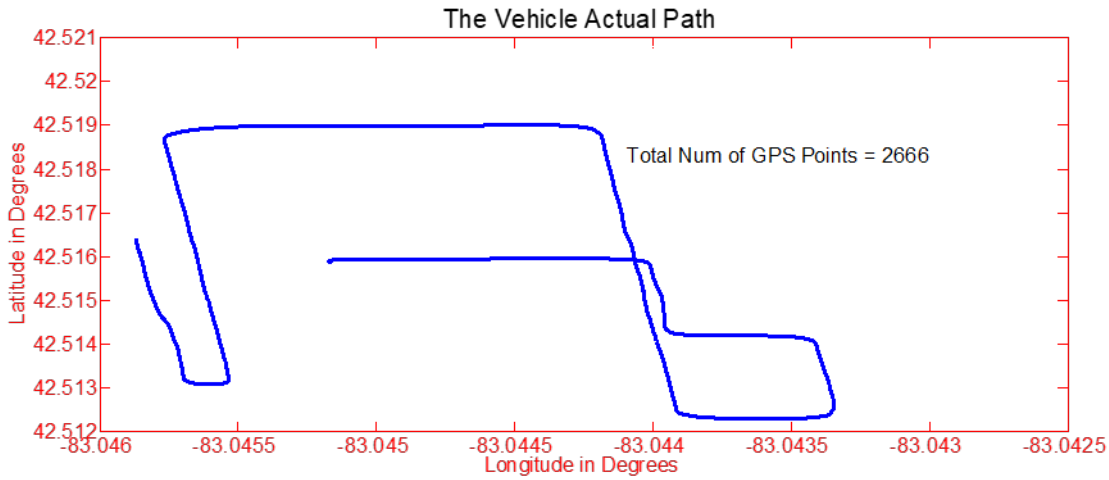

Figure A7 - Vehicle actual path

The test results evaluated the example PH method and calculated the concise PH that approximates the actual vehicle path within an error tolerance of 1 m. Figure A8 shows in red circles the concise PH data elements representing the actual vehicle path. The actual vehicle path is shown in blue. The example method shows that the number of concise PH data elements needed to represent the vehicle path is 37; however, this document requires that no more than the most recent 15 PH points be included in a BSM (see 6.3.6.16).

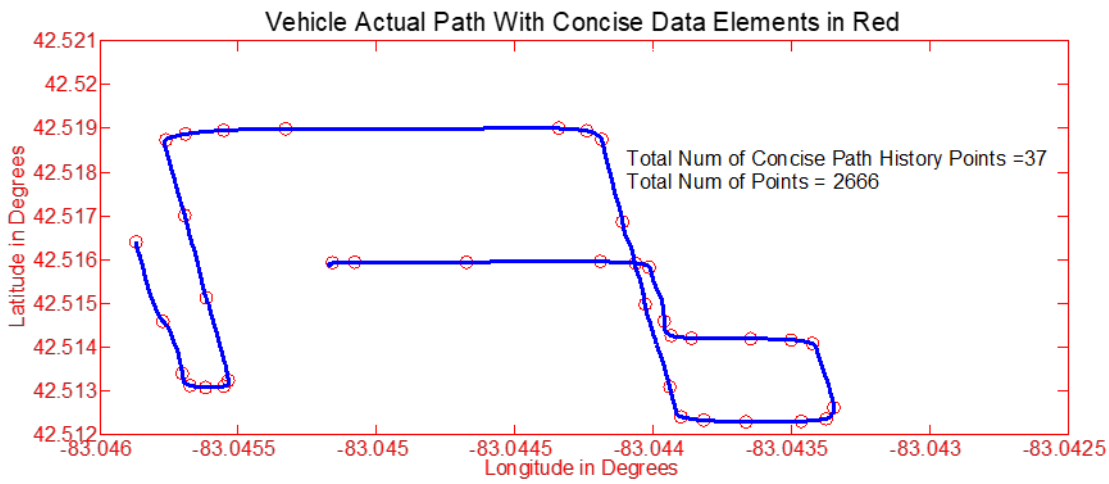

Figure A8 - Example method - representation of vehicle path

A.5.5.2 Radii of Curvature for a Curved Road

Figure A9 shows the radii of curvature (in meters) between successive, concise PH data points for a curved road segment of the vehicle path. The radii of curvature clearly indicate the curved nature of the road segment. Notice that the road segment also includes a reasonably straight section represented with a larger radius of curvature. In Figure A8, the solid line indicates the actual vehicle path and the dotted lines indicate the concise PH representation.

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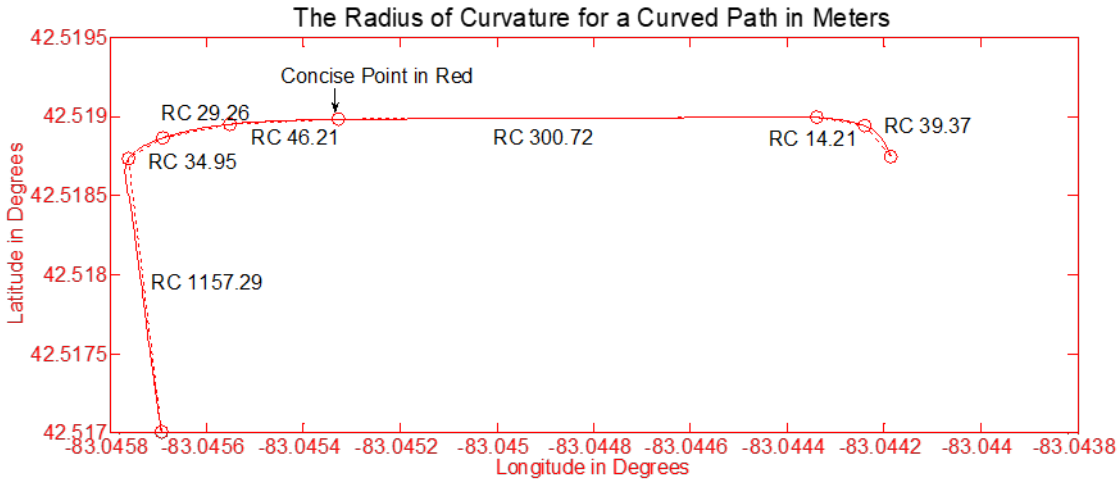

Figure A9 - Radii of curvature for a curved road

A.5.5.3 Radii of Curvature for a Straight Road

Figure A10 shows the radii of curvature (in meters) between successive concise data points for a straight road segment. The numbers indicate that the radius of curvature for a straight road segment is large. By examining these numbers, it is clear that the straight road segments are easily identified by using a certain threshold for radius of curvature. In Figure A10 the solid line indicates the actual vehicle path and the dotted lines indicate the concise PH representation.

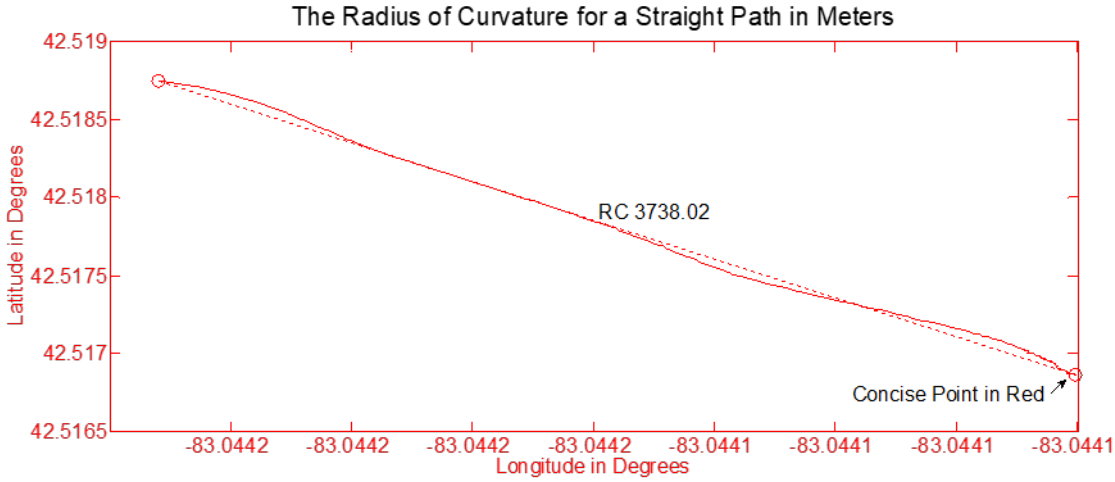

Figure A10 - Radii of curvature for a straight road

A.5.5.4 PH Concise Points and Distances Between Them for a Curved Road

Figure A11 shows the snapshot of PH data points on a curved road segment after concise data points have been computed to maintain the PH distance of at least 200 m. The PH distance is 354.66 m since no additional PH points can be dropped without violating the requirement of a minimum 200 m PH distance. It is clear the example method requires only a few PH points to represent a vehicle PH over a curved roadway segment as shown.

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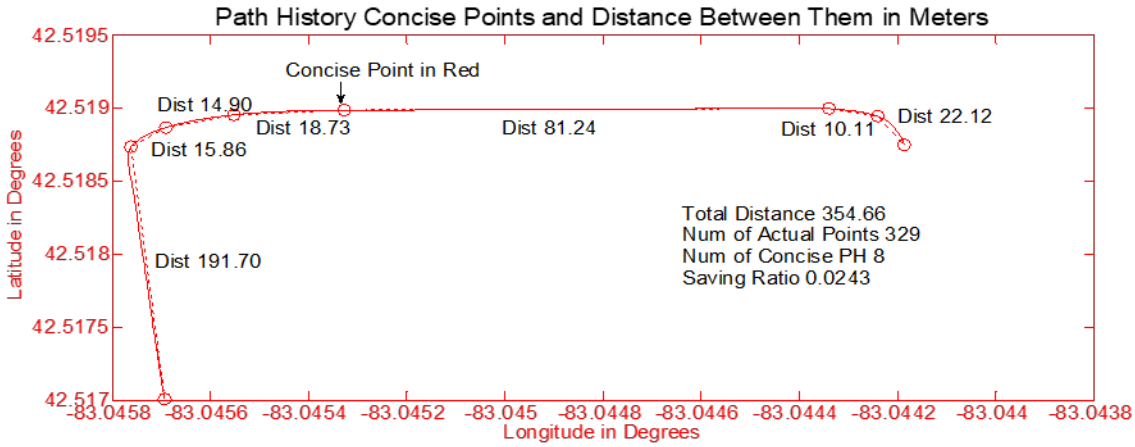

Figure A11 - PH representation of a curved road

The saving ratio shown in Figure A11 indicates the ratio of concise data elements to the actual data elements. The ratio indicates the saving in the representation of the actual path when using a concise PH representation for the example method. In Figure A11, the solid line indicates the actual vehicle path and the dotted lines indicate the concise PH representation.

A.5.5.5 PH Concise Points and Distances Between Them for a Straight Road

Figure A12 shows the snapshot of PH data points on a straight road segment after concise data points have been computed to maintain the PH distance of at least 200 m. The PH distance is 209.4 m since no additional PH points can be dropped without violating the requirement of a minimum PH distance. This also shows that maximum distance between two successive PH concise points never exceeds the stated threshold distance of K_PH_CHORDLENGTHTHRESHOLD. The saving ratio shown in Figure A11 indicates the ratio of concise data elements to the actual data elements. The ratio indicates the saving in the representation of the actual path when using a concise PH representation for the example method. In Figure A12 the solid line indicates the actual vehicle path and the dotted lines indicate the concise PH representation.

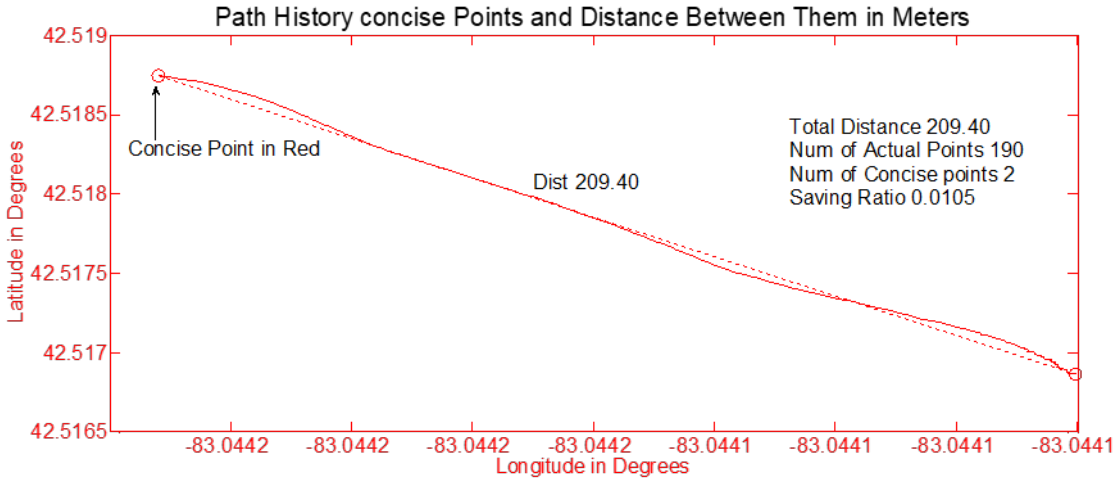

Figure A12 - PH representation of a straight road

Figure A13 shows that the PH distance of 200 m is maintained throughout the vehicle travel after an initial travel distance of 200 m is achieved.

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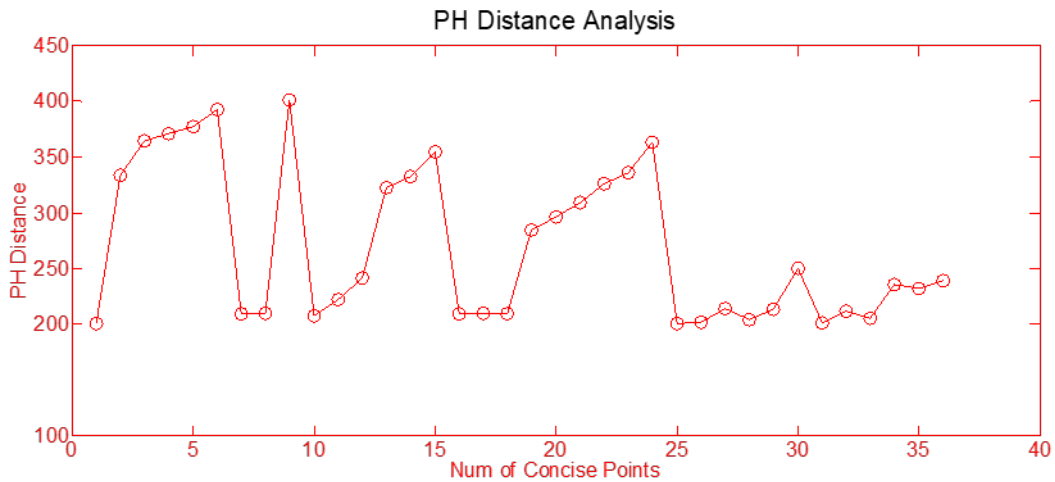

Figure A13 - PH distance analysis

A.5.5.6 PH Requirement Analysis

Figure A14 shows the actual error between concise PH data elements. Since the concise data points are chosen based on the fact that they do not violate the actual error criterion of 1 m, it is clearly shown and verified in the figure that the actual error is always less than 1 m. Similar results are generated for a straight path. The significance of these results is that the concise PH data points can be used reliably to represent the actual vehicle PH. In Figure A14, the solid line indicates the actual vehicle path and the dotted lines indicate the concise PH representation.

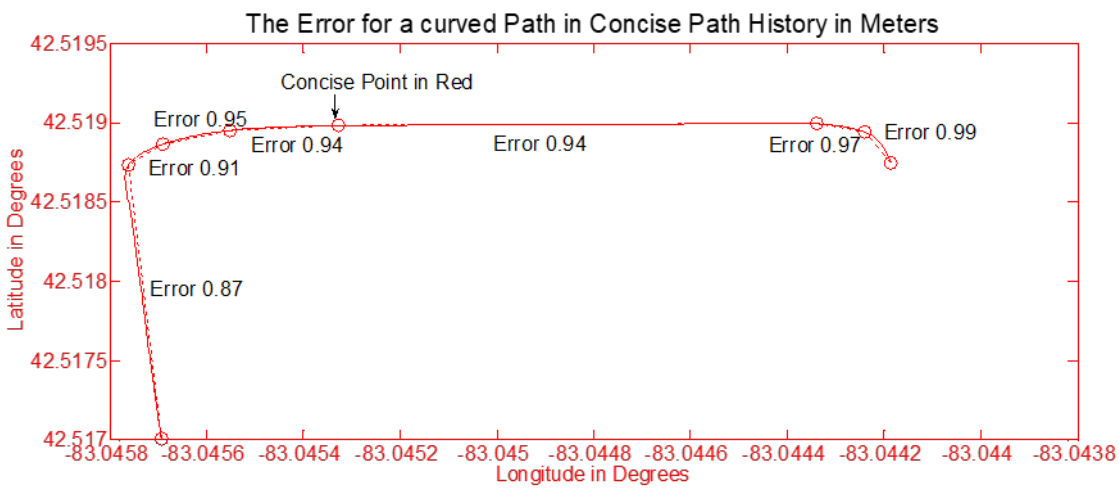

Figure A14 - PH error analysis

Figure A15 shows that the PH actual error between all the concise PH data points is in the tolerance limit of 1 m.

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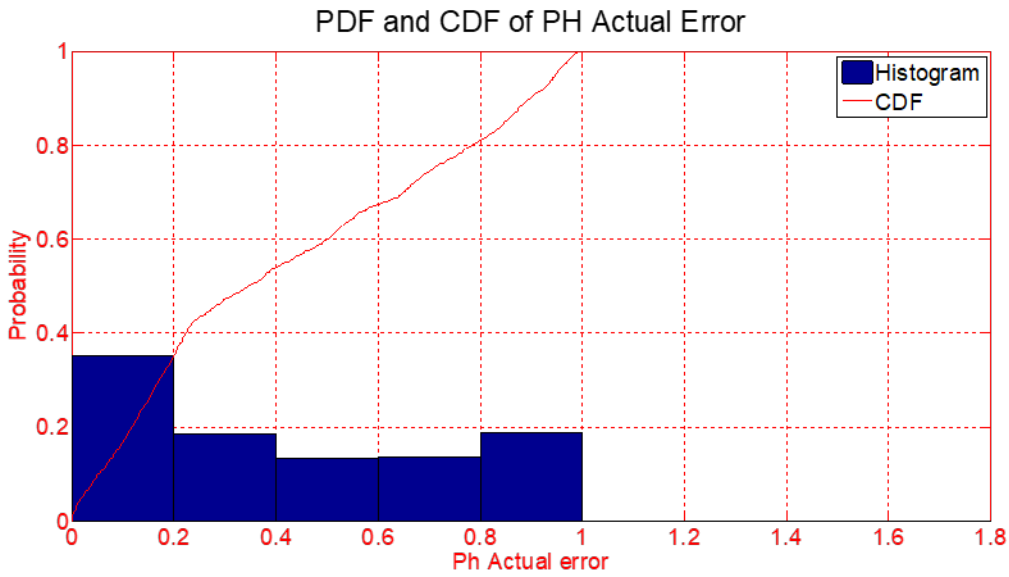

Figure A15 - PDF and CDF for the PH error analysis

A.5.6 Summary

This document has presented the PH module for a vehicle safety communications system. The module uses a history of the past GNSS locations traversed by the HV and computes an adaptable concise PH representation of recent vehicle movement over a certain distance. The PH communicated by a vehicle provides other vehicles with important information needed for predicting the roadway geometry. It plays an important role in target vehicle classification in vehicle safety communications. The example method for design and implementation of the PH module has been presented. The method has also been implemented and its performance has been evaluated. Extensive testing has shown that the concise representations of the vehicle PH computed by the example method offers significant savings in OTA wireless bandwidth when transmitting the PH information to other vehicles wirelessly, while guaranteeing that the PH error remains within the allowable tolerance of 1 m. The objective testing of VSC-A applications has also shown that the PH error tolerance of 1 m that was chosen as default satisfies the needed accuracy and meets the performance requirements of target classification and the safety applications that were developed and demonstrated in the VSC-A project.

A.6 PATH PREDICTION REFERENCE DESIGN (INFORMATIVE)

A.6.1 Introduction

Path prediction (PP) for the V2V safety communications system utilizes dynamics information provided by the vehicle to estimate the driver's intended future path. The estimate is provided without dependence on future road geometry information obtained from outside sources (e.g., map databases, vehicle probes).

PP carries out the following basic operations:

- Gathers vehicle dynamics information.
- Computes path radius using dynamics information to represent the driver's intended future path.
- Radius = 1/curvature (ρ).
- Computes confidence of the predicted path based upon the rate of change of the vehicle dynamics to infer transient conditions (i.e., non-steady-state conditions).

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A.6.2 PP Design Approach

The primary PP design approach for the V2V safety communications system uses vehicle dynamics information to calculate a (continuous) radius of curvature representing the vehicle's estimated future path. This is accomplished by using simple physics equations to compute curvature based on the vehicle speed and the rate of change of heading (yaw rate). This curvature can be extrapolated forward to provide an estimate of the likely future path of the vehicle (Figure A16).

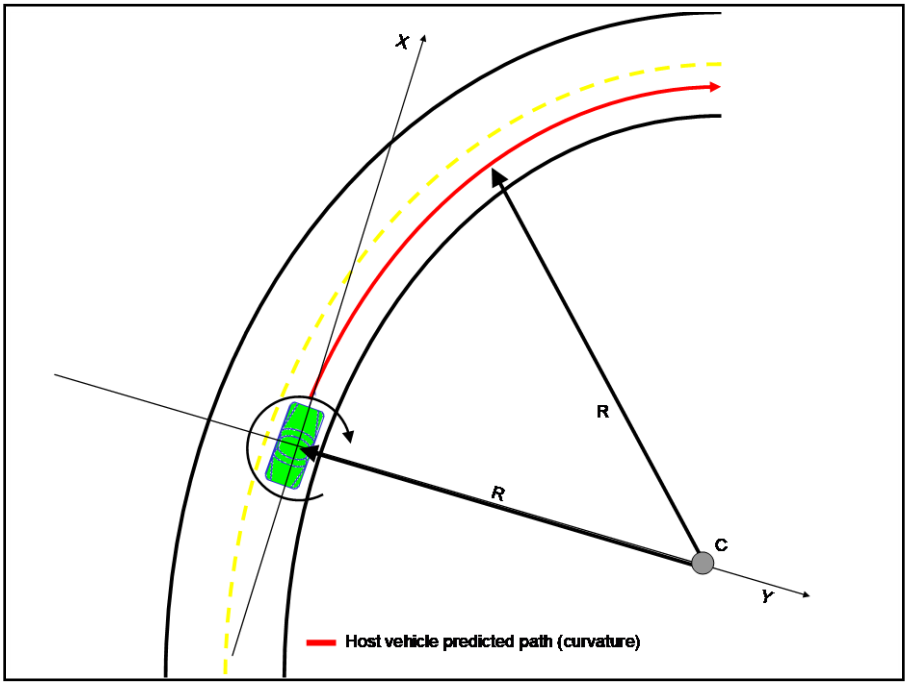

Figure A16 - Vehicle projected path

The PP module requires the following input signals:

- Vehicle speed (meters per second [m/s] used in this example)
- Yaw rate (degrees per second [degrees/s] used in this example)

Developers should pay careful attention to divide-by-zero conditions and appropriately cap intermediate and output calculations to prevent data type overflow. This is particularly important as vehicle speed approaches zero (see Simulink diagrams in subsequent sections).

A.6.3 Radius Calculation

In order to effectively filter a PP radius in meters, a reciprocal is computed in order to perform the filter operations on curvature (1/r). This prevents large discontinuities in the filter input signal when the radius oscillates between positive and negative values approaching infinity. Once the curvature has been computed, the signal is filtered in order to attenuate unwanted high-frequency noise. The filter is designed and calibrated to greatly reduce the following effects:

- Road noise
- Sensor noise
- Driver noise (in-lane wandering)

For the PP module, a second order low-pass filter is used to remove these unwanted components from the yaw rate signal. The design is a discretized version of a standard second order unity-gain filter characterized by the following equations.

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Discretized (unity-gain) second order low-pass filter:
(Filtered Curvature)

$$\frac{\omega_0^2}{s^2 + 2\omega_0 \zeta s + \omega_0^2} = \frac{\omega_0^2 T_s^2}{z^{-2} - (2+2\omega_0 \zeta T_s)z^{-1} + (\omega_0^2 T_s^2 + 2\omega_0 \zeta T_s + 1)}$$

Continuous (1/s = integrator) Discrete (1/z = unit delay)

$$y_n \text{ (for } n \geq 3) = \frac{-y_{n-2} + (2+2\omega_0 \zeta T_s)y_{n-1} + \omega_0^2 T_s^2 u_n}{(1 + 2\omega_0 \zeta T_s + \omega_0^2 T_s^2)}$$

Initialization: $y_1 = u_1$, $y_2 = u_2$

Figure A17 - Discretized second order low-pass filter

In Figure A17, $\omega_0 = 2\pi f_0$, f_0 = cutoff frequency, ζ = damping factor, and T_s = sampling time. Note: $\zeta = 1$ (default) for a critically damped system.

The vehicle radius calculation follows the basic formula:

- radius (m) = vehicle speed (m/s)/yaw rate (rad/s)

In preparation of the radius calculation, the yaw rate is converted from degrees/s to rad/s. To prevent division by zero when the vehicle is stationary and to eliminate large discontinuities in the filter input signal, the reciprocal of radius is calculated to provide curvature input.

- curvature (1/m) = yaw rate (rad/s)/vehicle speed (m/s)

Upon calculation of the curvature, the signal is passed through a discretized second order low-pass filter (Figure A17) that has been calibrated to the appropriate cutoff frequency, damping factor, and sampling rate. Once the curvature calculation has been filtered, it is converted back to a radius in meters using the reciprocal. Special care must be taken to prevent overflow of the radius calculation when curvature is zero (or near zero). Radius calculations follow the SAE sign convention for rotation, where a positive sign represents a clockwise curvature about the vehicle boresight and a negative sign represents counter-clockwise.

A final set of logic checks two conditions to determine if the path should be considered “straight”:

- Vehicle speed is less than a calibrated threshold.
- Radius calculation is greater than a calibrated threshold.

If either of these conditions exists, the filtered radius output is set to a default value identified in SAE J2735 [1] (32767 m as of revision 35). Figure A18 shows the logic flow for the host vehicle path radius calculation:

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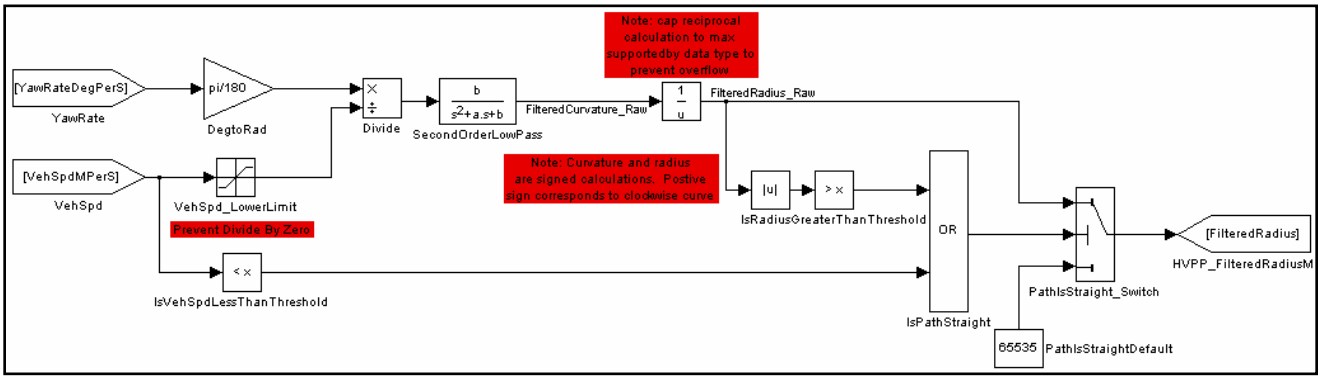

Figure A18 - Vehicle path radius calculation

A.6.4 Confidence Calculation

The preceding technique for calculating PP radii is highly effective when applied during “steady state” driving conditions; however, dynamic driving conditions can prove to be a challenge. Therefore a method must exist for identifying and communicating dynamic situations when path estimations may be largely inaccurate. Identifying “steady state” is accomplished by applying a confidence interval to a differentiated and filtered version of the yaw rate signal. The confidence indicator is calibrated to report low confidence when large changes in the vehicle yaw rate are detected over a short period of time. These conditions may include one or more of the following:

- Lane changes
- Curve entry and exit points
- Curve transitions
- Obstacle avoidance and other highly dynamic driving situations

For the PP module, a second order low-pass filter with differentiator is used to identify that the vehicle is likely in “steady state” based steering input. The design is a discretized version of a standard second order unity-gain filter characterized by the following equations.

Discretized (unity-gain) second order low-pass filter with differentiator:

$$\frac{s\omega_0^2}{s^2 + 2\omega_0\zeta s + \omega_0^2} = \frac{\omega_0^2 T_s - \omega_0^2 T_s z^{-1}}{z^{-2} - (2+2\omega_0\zeta T_s)z^{-1} + (\omega_0^2 T_s^2 + 2\omega_0\zeta T_s + 1)}$$

Continuous (1/s = integrator) Discrete (1/z = unit delay)

$$y_n = \frac{-y_{n-2} + (2+2\omega_0\zeta T_s)y_{n-1} + \omega_0^2 T_s u_n - \omega_0^2 T_s u_{n-1}}{(\omega_0^2 T_s^2 + 2\omega_0\zeta T_s + 1)}$$

(for $n \geq 3$)

Initialization: $y_1 = 0$, $y_2 = 0$

Figure A19 - Discretized second order low-pass filter with differentiator

Again, $\omega_0 = 2\pi f_0$, f_0 = cutoff frequency, ζ = damping factor, and T_s = sampling time. Note: $\zeta = 1$ (default) for a critically damped system.

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In order for the PP module to provide the highest accuracy future path estimations, the vehicle must be at or near "steady state" conditions. Determining when the host vehicle is in "steady state" is accomplished by a second calculation using the yaw rate sensor input. The PP module monitors the rate of change of the host vehicle yaw rate to determine when "steady state" conditions are most likely to exist. This is accomplished using a discretized second order low-pass filter with differentiator (Figure A19). The confidence filter is tuned with a higher cutoff frequency in order for the indicator to lead the radius calculation during dynamic driving conditions. This ensures that the confidence indicator is capable of reporting changes in confidence prior to the change in radius output.

After filtering and differentiating the yaw rate, the output is applied to a tunable lookup table that provides confidence levels ranging from 0 to 100%.

Figure A20 shows the logic flow for the host vehicle "steady state" confidence calculation:

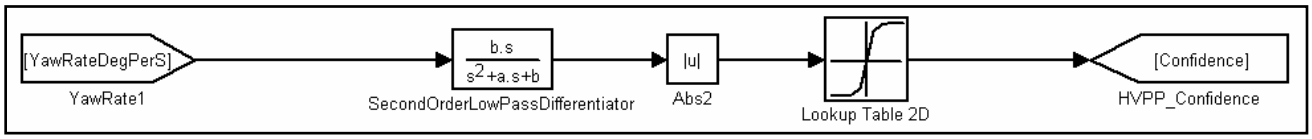

Figure A20 - Vehicle predicted path confidence calculation

A.6.5 Calibration

Table A2 contains a list of the PP calibration parameters referenced throughout this standard along with their default, minimum, and maximum configurable parameters.

Table A2 - PP calibration parameters

Calibration Parameter	Description	Default Value	Minimum Value	Maximum Value
Curvature Cutoff Frequency	Low-pass cutoff frequency for curvature filter.	0.33 Hz	0.32 Hz	0.34 Hz
Curvature Damping Factor	Curvature filter dampening factor.	1	0	2
Curvature Sampling Period	Sample time for discrete curvature filter.	100 ms	100 ms	400 ms
Minimum Vehicle Speed	Vehicle speed lower limit for curvature calculation.	1 m/s	0 m/s (straight path only)	2 m/s
Maximum Radius	Radius upper limit beyond which the path is considered "straight."	2500 m	2000 m	5000 m
Straight Path	When radius is greater than the maximum radius, the reported radius is set to this value to indicate a "straight" path. Based on SAE J2735 reserved value for path prediction radius.	32767	32767	32767
Confidence Cutoff Frequency	Low-pass cutoff frequency for confidence filter.	1 Hz	0.33 Hz	1 Hz
Confidence Damping Factor	Confidence filter dampening factor.	1	0	2
Confidence Sampling Period	Sample time for discrete confidence filter.	100 ms	100 ms	400 ms
Confidence Values	Two-dimensional table accepts filtered/differentiated yaw rate and outputs a confidence from 0 to 100%.	See Table A3 for values		

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Table A3 - Confidence lookup table

Input: Filtered/Differentiated Yaw Rate (degrees/s^2)	25	20	15	10	5	2.5	2	1.5	1	0.5	0
Output: Confidence (%)	0	10	20	30	40	50	60	70	80	90	100

A.7 OPEN SKY TEST CONDITIONS (NORMATIVE)

The designation “open sky” environment is intended to describe an environment in which there are minimal obstructions to the device’s view of the sky as used for testing in [14]. Open sky test conditions are defined, for the purposes of this standard, to be present when the following are all true:

- No view obstructions external to the vehicle can be seen, from the point of view of the GNSS antennas of reference device and device under test, starting from 5 degrees above the horizontal plane (the elevation mask) containing the antenna phase center, in all directions around the antenna.
- The number of healthy satellites used, as reported by the reference device, for GPS satellites only, is greater than or equal to seven.
- The HDOP, as reported by the reference device, for GPS satellites, is less than or equal to 1.5, and VDOP is less than or equal to three.

A.8 ADDITIONAL CONGESTION CONTROL ALGORITHM DETAILS (NORMATIVE)

A.8.1 Assumption of Latest HV State Information at RVs

For each BSM generated by the HV, use a Bernoulli trial with the channel quality indicator $\Pi(k)$ to estimate whether the BSM was successfully received by the RVs.

- If the outcome of the Bernoulli trial is positive, assume that the BSM was successfully received by the RVs. Update the latest information assumed for the RVs with the state information contained in the BSM.
- Else, if the outcome of this Bernoulli trial is negative, assume the BSM transmission was not received by the RVs (TxFailed) and do not update the latest HV state information assumed for the RVs.
- If the Bernoulli trials result in more than *vMaxSuccessiveFail* consecutive negative outcomes, assume the most recent BSM transmission was successfully received by the RVs, and update the latest information assumed for the RVs with the state information contained in the most recent BSM.

Let $\begin{bmatrix} t \\ x \\ y \\ v \\ \theta \end{bmatrix}_{\text{Latest}}$ be the HV’s assumed latest state information received by RVs and $\begin{bmatrix} t \\ x \\ y \\ v \\ \theta \end{bmatrix}_{\text{Pre-Tx}}$ be the HV’s state information contained in the message of its previous transmission (where t is the time in milliseconds when the longitudinal position x (in degrees), lateral position y (in degrees), speed v (in m/s), and heading $\theta(t)$ (in degrees) are measured. The HV’s assumed latest state information received by RVs is updated after each transmission as follows:

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If $rand() < \Pi(k)$
$TxFailed = TxFailed + 1$
$Else$
$TxFailed = 0$

$$\begin{bmatrix} t \\ x \\ y \\ v \\ \theta \end{bmatrix}_{\text{Latest}} = \begin{cases} \begin{bmatrix} t \\ x \\ y \\ v \\ \theta \end{bmatrix}_{\text{Pre-Tx}} & \text{TxFailed} > 0 \text{ and TxFailed} <= \text{vMaxSuccessiveFail} \\ \begin{bmatrix} t \\ x \\ y \\ v \\ \theta \end{bmatrix}_{\text{Latest}} & \text{otherwise set TxFailed} = 0 \end{cases}$$

where $rand()$ is a uniform random number generator with output between zero and one, and $\Pi(k)$ is the estimated channel quality indicator.

Note that this only helps an HV predict whether or not the RVs received the HV's latest state information. The actual latest state information received by RVs might be different for each RV due to different scales of fading and packet collisions in the wireless channel and they might also be different from this HV's assumed latest state information received by RVs. The derived latest state information only serves as the "expected" state information perceived by an HV and is used to adapt its transmission rate.

A.8.2 Tracking Error

Calculate the tracking error as the distance between HV local estimate position $(\hat{x}(k), (\hat{y}(k))$ and output of the HV remote estimate position, $(\tilde{x}(k), \tilde{y}(k))$ using the great circle formula, i.e.,

$$e(k) = R(\hat{x}(k)) \times (\cos^{-1}(\sin(\hat{x}(k)) \times \sin(\tilde{x}(k)) + \cos(\hat{x}(k)) \\ \times \cos(\tilde{x}(k)) \times \cos(\hat{y}(k) - \tilde{y}(k))))$$

where

$$R(\hat{x}(k)) = a \times (1 - f_1^2) / (1 - f_1^2 \times \sin^2(\hat{x}(k)))^{1.5}$$

is the Meridian Radius of the Earth in meters, at latitude $\hat{x}(k)$, a = 6378137 is the mean radius of earth in meters, $f_1 = (f \times (2 - f))^{0.5}$ is the eccentricity, and f = 0.003353 is earth's flattening.

Here $(\hat{x}(k), \hat{y}(k))$ are the latitude and longitude from the HV local estimate, converted to radians, and $(\tilde{x}(k), \tilde{y}(k))$ are the latitude and longitude from the HV remote estimate, converted to radians.

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A.8.3 PER Calculation Special Cases

Assume that the *vPERInterval* is divided into five subintervals as shown in Figures A21 and A22. Some special cases should be considered during PER calculations as follows:

Figure A21 (a) shows that the node received two packets in the first subinterval (the arrows with msg_cnt = 1 and msg_cnt = 8 represent the received packets) and zero packets in the remaining subintervals. The PER at the end of the subinterval is reported next to Figure A21 (a). Figure A21 (b) shows the status of the node after one sliding window. The PER is reported next to Figure A21 (b).

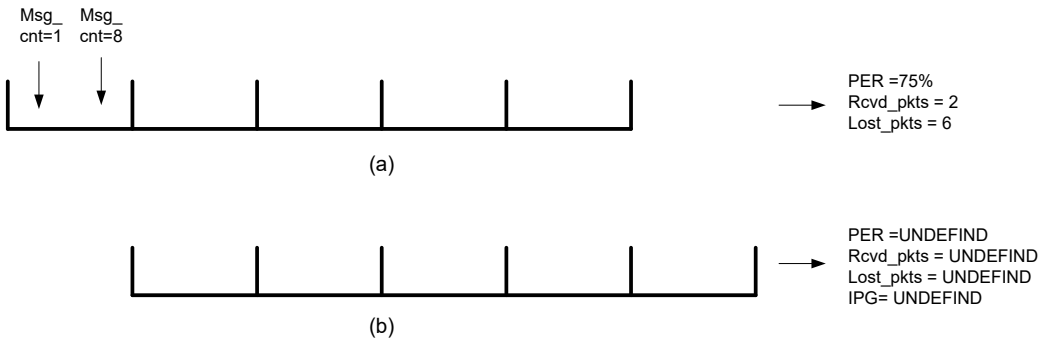

Figure A22 - PER calculation example 1

Figure A22 (a) shows that the node received one packet in the first subinterval (the arrow with msg_cnt = 1) and another packet in the second subinterval (the arrow with msg_cnt = 11) and zero packets in the remaining subntervals. The PER at the end of this subinterval is reported next to Figure A22 (a). Figure A22 (b) shows the status of the node after one sliding window. The PER is reported next to Figure A22 (b).

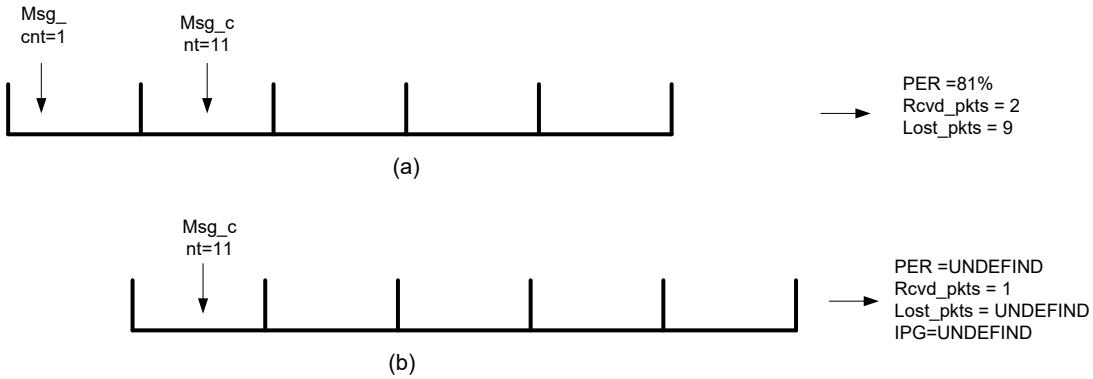

Figure A23 - PER calculation example 2

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A.9 EXAMPLE SIGNED MESSAGE WITH CERTIFICATE AND CORRESPONDING ASN.1 (INFORMATIVE)

Below is an example of a generic signed message with certificate and the corresponding ASN.1. The 1609.2 [3] profile in 6.1.2 was used to generate this sample. The payload of this security envelope (the message) is the ASCII text string “This is a BSM.”

```
03 81 00 40 03 80 0F 54 68 69 73 20 69 73 20 61
20 42 53 4D 0D 0A 40 01 20 00 00 0A 35 23 77 2A
85 00 81 01 01 00 03 01 80 00 11 22 33 44 55 66
77 50 80 80 00 C8 00 11 22 33 44 55 66 77 88 56
70 AB 00 11 22 33 44 55 66 77 88 99 00 11 22 00
01 00 11 22 33 84 00 A9 83 01 03 80 00 7C 80 01
E4 80 03 48 01 02 00 01 20 00 01 26 81 82 00 11
22 33 44 55 66 77 88 99 AA BB CC DD EE FF 10 11
12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 80 82
00 11 22 33 44 55 66 77 88 99 AA BB CC DD EE FF
10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
FF 11 22 33 44 55 66 77 88 99 AA BB CC DD EE FF
10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F

value1 TestIeee1609Dot2Data ::= {
  protocolVersion 3,
  content signedData : {
    hashId sha256,
    tbsData {
      payload {
        data {
          protocolVersion 3,
          content unsecuredData : ‘54686973206973206120425340D0D0A’H
        }
      },
      headerInfo {
        psid 32,
        generationTime {
          time 11223344556677,
          logStdDev 0
        }
      }
    },
    signer certificate : {
      {
        version 3,
        type implicit,
        issuer ecdsaNistP256AndDigest : ‘0011223344556677’H,
        toBeSigned {
          id linkageData : {
            iCert 200,
            linkage-value ‘001122334455667788’H,
            group-linkage-value {
              jValue ‘5670AB’H,
              value ‘00112233445566778899’H
            }
          },
          cracaId ‘001122’H,
          crlSeries 1,
          validityPeriod {
            start 1122867,
            duration hours : 169
          },
          region identifiedRegion : {
            countryOnly : 124,
```

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```
            countryOnly : 484,
            countryOnly : 840
          },
          appPermissions {
            {
              psid 32
            },
            {
              psid 38
            }
          },
          verifyKeyIndicator reconstructionValue : compressed-y-0 :
'00112233445566778899AABBCCDDEEFF101112131415161718191A1B1C1D1E1F'H
        }
      }
    },
    signature ecdsa256Signature : {
      r compressed-y-0 :
'00112233445566778899AABBCCDDEEFF101112131415161718191A1B1C1D1E1F'H,
      s 'FF112233445566778899AABBCCDDEEFF101112131415161718191A1B1C1D1E1F'H
    }
  }
}
```

A.10 REQUIREMENTS TRACEABILITY (NORMATIVE)

Figure A23 describes the traceability of the requirements in this standard to the scenarios defined for the vehicle-to-vehicle communications in Sections 4 and 5 of this standard.

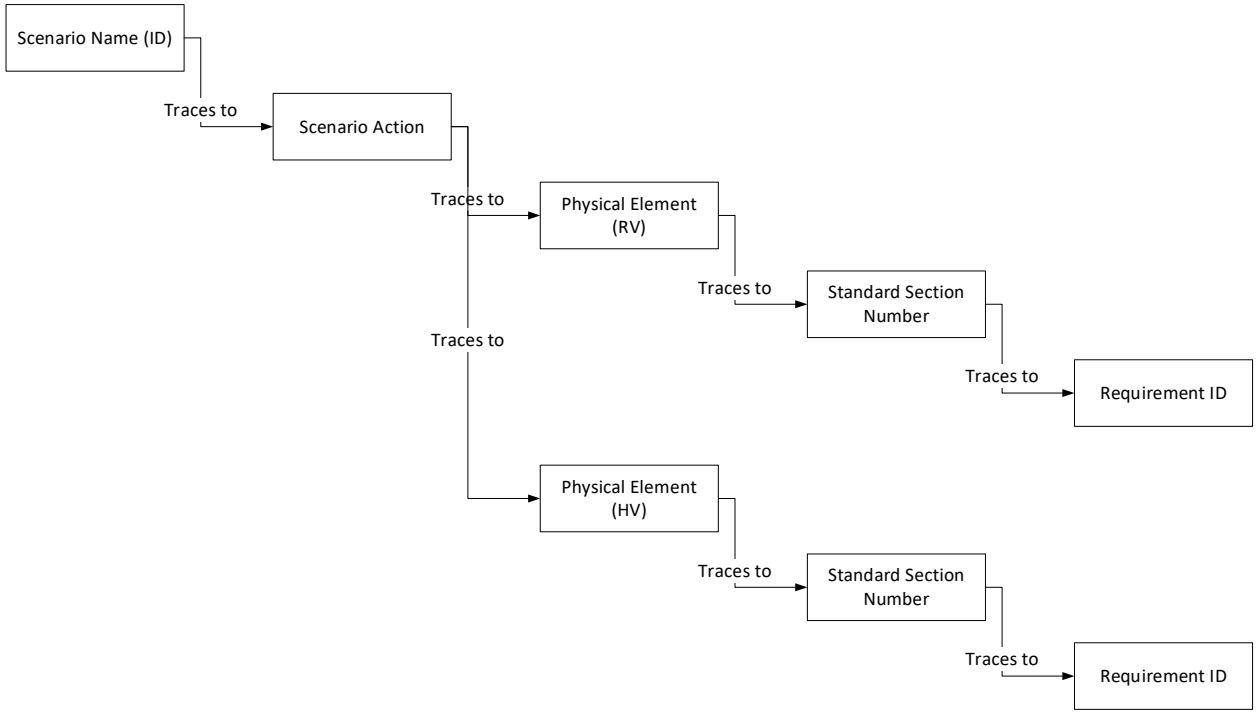

Figure A24 - Traceability requirements of this standard

The scenario is the title of a scenario and reference paragraph defined in Section 4 of this standard. For example: EEBL - lead vehicle decelerating (4.2.3), etc.

The scenario action is the specific action that applies or triggers the scenario(s). For example, RV-1 hard braking triggers the EEBL scenario (4.2.3).

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The physical element (RV or HV) refers to the subsystem(s) in Figure 1 that is the candidate(s) for implementing the function. When there is a slash ("/") mark, e.g., DSRC/ECU, it means the implementation could be in one or the other or both subsystems.

The standard section number is the section in this standard that contains the requirements for the scenario.

The requirement category corresponds to the requirements in Section 6.

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Table A4

Scenario	Scenario Action	Physical Element RV-1 (BSM Transmitter)	Standard Section (for the RV)	Requirement Category (for the RV)	Physical Element HV (BSM receiver)	Standard Section (for the HV)	Requirement Category (for the HV)
Startup (5.1.4)	Power-on	OBE Control Processor ECU			OBE Control Processor ECU		
	Retrieve stored data	OBE Control Processor ECU	6.3.7.1 - Heading 6.3.7.2 - Path History	V2V-BSMTX-DATAPERSIST	OBE Control Processor ECU	6.3.7.1 - Heading 6.3.7.2 - Path History	V2V-BSMTX-DATAPERSIST
	Randomize ID	OBE Control Processor ECU	6.5.1 - Identification Randomization	V2V-SECPRIV-IDRAND	OBE Control Processor ECU	6.5.1 - Identification Randomization	V2V-SECPRIV-IDRAND
Shutdown (5.1.4)	Power-off		6.3.7.1 - Heading 6.3.7.2 - Path History	V2V-BSMTX-DATAPERSIST			V2V-BSMTX-DATAPERSIST
Security (5.1.3)	Initial equipping of certificates and CRL	OBE Control Processor ECU	6.6.1 -Bootstrap: Initialization and Enrollment Processing		OBE Control Processor ECU	6.6.1 -Bootstrap: Initialization and Enrollment Processing	
			6.6.1.1 - Initialization Process			6.6.1.1 - Initialization Process	
			6.6.1.2 - Enrollment Process			6.6.1.2 - Enrollment Process	
			6.6.2 - Certificate Loading	V2V-SECMGMT-CERTLOAD		6.6.2 - Certificate Loading	V2V-SECMGMT-CERTLOAD
	Store certificates	OBE Control Processor ECU	6.6.3 - Certificate Storage	V2V-SECMGMT-CERTSTORE	OBE Control Processor ECU	6.6.3 - Certificate Storage	V2V-SECMGMT-CERTSTORE
	CRL	OBE Control Processor ECU	6.6.4 - Certificate Revocation List Loading	V2V-SECMGMT-CERTLOAD	OBE Control Processor ECU	6.6.4 - Certificate Revocation List Loading	V2V-SECMGMT-CERTLOAD
	Signing and verification algorithm	OBE Control Processor ECU	6.6.5 - Secure Hardware	V2V-SECMGMT-SECHW	OBE Control Processor ECU	6.6.5 - Secure Hardware	V2V-SECMGMT-SECHW
	Sign BSM and attach certificate	OBE Control Processor ECU	6.5.2 - BSM Signing	V2V-SECPRIV-BSMSIGN	OBE Control Processor ECU	6.5.4 - BSM Verification	V2V-SECPRIV-BSMVERIFY

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Scenario	Scenario Action	Physical Element RV-1 (BSM Transmitter)	Standard Section (for the RV)	Requirement Category (for the RV)	Physical Element HV (BSM receiver)	Standard Section (for the HV)	Requirement Category (for the HV)
BSM Exchange (5.1.1)	Transmit BSM	OBE Control Processor ECU	6.3.3 - First BSM after Startup and Generation Timing	V2V-BSMTX-GENTIM			
			6.3.8 - BSM Scheduling and Congestion Control	V2V-BSMTX-CONGCTRL			
Privacy (5.1.3)	5-minute certificate rotation	OBE Control Processor ECU	6.5.3 - Certificate Change	V2V-SECPRIV-CERTCHG			
	Randomize ID		6.5.1 - Identification Randomization	V2V-SECPRIV-IDRAND			
Positioning (5.1.2)	Determine position	Positioning Subsystem /OBE Control Processor ECU	6.2.1 - Position Determination	V2V-POSTIM-POSDETER	Positioning Subsystem /OBE Control Processor ECU	6.2.1 - Position Determination	V2V-POSTIM-POSDETER
EEBL - Lead Vehicle Decelerating (4.2.3)	RV-1 abruptly brakes hard	Positioning Subsystem /OBE Control Processor ECU	6.2.1 - Position Determination	V2V-POSTIM-POSDETER			
			6.2.2 - Wide Area Augmentation	V2V-POSTIM-WAAS			
		OBE Control Processor ECU	6.2.3 - Coordinate System and Reference	V2V-POSTIM-COORDSYSREF			
		Positioning Subsystem /OBE Control Processor ECU	6.2.4 - System Time Coordination	V2V-POSTIM-SYSTIMCOORD			

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Scenario	Scenario Action	Physical Element RV-1 (BSM Transmitter)	Standard Section (for the RV)	Requirement Category (for the RV)	Physical Element HV (BSM receiver)	Standard Section (for the HV)	Requirement Category (for the HV)
		OBE Control Processor ECU	6.3.1 - BSM Content	BSMTX-BSMCONT			
			6.3.6.1 - DE_DSRC_MessageID	V2V-BSMTX-DATAACC			
			6.3.6.2 - DE_MsgCount				
			6.3.6.3 - DE_TemporaryID				
			6.3.6.4 - DE_DSecond				
			6.3.6.5 - DE_Latitude & DE_Longitude				
			6.3.6.6 - DE_Elevation				
			6.3.6.7 - DF_Positional Accuracy				
			6.3.6.8 - DE_Speed				
			6.3.6.9 - DE_TransmissionState				
			6.3.6.10 - DE_Heading				
			6.3.6.11 - DE_SteeringWheelAngle				
			6.3.6.12 - DF_AccelerationSet4Way				
			6.3.6.13 - DF_BrakeSystemStatus				
			6.3.6.14 - DF_VehicleSize				
			6.3.6.15 - DE_EventFlags				
			6.3.6.16 - DF_PathHistory				
			6.3.6.17 - DF_PathPrediction				
			6.3.6.18 - DE_ExteriorLights				
			6.3.6.19 - Additional Data Elements				
	Check CRL	OBE Control Processor ECU	6.5.5. - Certificate Revocation				

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Scenario	Scenario Action	Physical Element RV-1 (BSM Transmitter)	Standard Section (for the RV)	Requirement Category (for the RV)	Physical Element HV (BSM receiver)	Standard Section (for the HV)	Requirement Category (for the HV)
	Signs the data - using the RV certificate	OBE Control Processor ECU	6.5.2 - BSM Signing	V2V-SECPRIV-BSMSIGN			
	Transmit BSM	DSRC Radio Subsystem	6.3.2 - Channel and Data Rate	V2V-BSMTX-CHDATARATE			
		OBE Control Processor ECU	6.3.8 - BSM Scheduling and Congestion Control	V2V-BSMTX-CONGCTRL			
		DSRC Radio Subsystem	6.3.4 - User Priority EDCA settings	V2V-BSMTX-UPEDCA			
		OBE Control Processor ECU	6.3.5 - Minimum Transmission Criteria	V2V-BSMTX-MINTX			
		DSRC Radio Subsystem	6.4.1.1 - Transmit Power Accuracy	V2V-RFPERF-DSRCTX			
		Antennas	6.4.1 - Radiated Power	V2V-RFPERF-DSRCTX			
	Verify signature - using host certificate				OBE Control Processor ECU	6.5.4- BSM Verification	V2V-SECPRIV-BSMVERIFY
					Positioning Subsystem/OBE Control Processor ECU	6.2.1 - Position Determination	V2V-POSTIM-POSDETER
						6.2.2 - Wide Area Augmentation*	V2V-POSTIM-WAAS
					OBE Control Processor ECU	6.2.3 - Coordinate System & Ref	V2V-POSTIM-COORDSYSREF
					Positioning Subsystem /OBE Control Processor ECU	6.2.4 - System Time Coordination	V2V-POSTIM-SYSTIMCOORD
					DSRC Radio Subsystem	6.4.2 - DSRC Receive Sensitivity	V2V-RFPERF-DSRCRXSENS

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Scenario	Scenario Action	Physical Element RV-1 (BSM Transmitter)	Standard Section (for the RV)	Requirement Category (for the RV)	Physical Element HV (BSM receiver)	Standard Section (for the HV)	Requirement Category (for the HV)
FCW-Forward Crash Warning (4.2.4)	Vehicle stopped (schedule BSM)	OBE Control Processor ECU	6.3.8 - BSM Scheduling and Congestion Control	V2V-BSMTX-CONGCTRL			
BSW /LCW-Blind Spot Warning/Lane Change Warning (4.2.5)	Schedule BSM		6.3.8 - BSM Scheduling and Congestion Control				
IMA-Intersection Movement Assist (4.2.6)	Schedule BSM		6.3.8 - BSM Scheduling and Congestion Control				
LTA - Left Turn Assist (4.2.7)	Schedule BSM		6.3.8 - BSM Scheduling and Congestion Control				
CLW - Control Loss Warning (4.2.8)	Schedule BSM		6.3.8 - BSM Scheduling and Congestion Control				
Scenarios 4.2.4 to 4.2.8		Positioning Subsystem /OBE Control Processor ECU	6.2.1 - Position Determination	V2V-POSTIM-POSDETER			
			6.2.2 - Wide Area Augmentation*	V2V-POSTIM-WAAS			
		OBE Control Processor ECU	6.2.3 - Coordinate System & Ref	V2V-POSTIM-COORDSYSREF			
		Positioning Subsystem /OBE Control Processor ECU	6.2.4 - System Time Coordination	V2V-POSTIM-SYSTIMCOORD			

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Scenario	Scenario Action	Physical Element RV-1 (BSM Transmitter)	Standard Section (for the RV)	Requirement Category (for the RV)	Physical Element HV (BSM receiver)	Standard Section (for the HV)	Requirement Category (for the HV)
		OBE Control Processor ECU	6.3.1 - BSM Content	BSMTX-BSMCONT			
			6.3.6.1 - DE_DSRC_MessageID	V2V-BSMTX-DATAACC			
			6.3.6.2 - DE_MsgCount				
			6.3.6.3 - DE_TemporaryID				
			6.3.6.4 - DE_DSecond				
			6.3.6.5 - DE_Latitude & DE_Longitude				
			6.3.6.6 - DE_Elevation				
			6.3.6.7 - DE_Positional Accuracy				
			6.3.6.8 - DE_Speed				
			6.3.6.9 - DE_TransmissionState				
			6.3.6.10 - DE_Heading				
			6.3.6.11 - DE_SteeringWheelAngle				
			6.3.6.12 - DF_AccelerationSet4Way				
			6.3.6.13 - DF_BrakeSystemStatus				
			6.3.6.14 - DF_VehicleSize				
			6.3.6.16 - DF_PathHistory				
			6.3.6.17 - DF_PathPrediction				
			6.3.6.18 - DE_ExteriorLights				
			6.3.6.19 - Additional Data Elements				
	Check CRL	OBE Control Processor ECU	6.5.5. - Certificate Revocation				
	Signs the data - using the RV certificate	OBE Control Processor ECU	6.5.2 - BSM Signing	V2V-SECPRIV-BSMSIGN			

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Scenario	Scenario Action	Physical Element RV-1 (BSM Transmitter)	Standard Section (for the RV)	Requirement Category (for the RV)	Physical Element HV (BSM receiver)	Standard Section (for the HV)	Requirement Category (for the HV)
		DSRC Radio Subsystem	6.3.2 - Channel and Data Rate	V2V-BSMTX-CHDATARATE			
		OBE Control Processor ECU	6.3.8 - BSM Scheduling and Congestion Control	V2V-BSMTX-CONGCTRL			
		DSRC Radio Subsystem	6.3.4 - User Priority EDCA settings	V2V-BSMTX-UPEDCA			
		OBE Control Processor ECU	6.3.5 - Minimum Transmission Criteria	V2V-BSMTX-MINTX			
		DSRC Radio Subsystem	6.4.1.1 - Transmit Power Accuracy	V2V-RFPERF-DSRCTX			
		Antennas	6.4.1 - Radiated Power	V2V-RFPERF-DSRCTX			
	Verify signature - using host certificate				OBE Control Processor ECU	6.5.3 - BSM Verification	V2V-SECPRIV-BSMVERIFY
					Positioning Subsystem/OBE Control Processor ECU	6.2.1 - Position Determination	V2V-POSTIM-POSDETER
						6.2.2 - Wide Area Augmentation*	V2V-POSTIM-WAAS
					OBE Control Processor ECU	6.2.3 - Coordinate System and Reference	V2V-POSTIM-COORDSYSREF
					Positioning Subsystem /OBE Control Processor ECU	6.2.4 - System Time Coordination	V2V-POSTIM-SYSTIMCOORD
					DSRC Radio Subsystem	6.4.2 - DSRC Receive Sensitivity	V2V-RFPERF-DSRCRXSENS
All Scenarios	Mac access control and physical layer	DSRC Radio Subsystem	6.1.1 - 802.11 Requirements - Table 5	V2V-STD-802.11	DSRC Radio Subsystem	6.1.1 - 802.11 Requirements - Table 5	V2V-STD-802.11

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Scenario	Scenario Action	Physical Element RV-1 (BSM Transmitter)	Standard Section (for the RV)	Requirement Category (for the RV)	Physical Element HV (BSM receiver)	Standard Section (for the HV)	Requirement Category (for the HV)
All Scenarios	Security services		6.1.2 - IEEE 1609.2 Requirements - Table 9, 10, 12	V2V-STD-1609.2		6.1.2 - IEEE 1609.2 Requirements - Table 9, 11, 12	V2V-STD-1609.2
All Scenarios	Networking services		6.1.3 - IEEE 1609.3 Requirements - Table 14	V2V-STD-1609.3		6.1.3 - IEEE 1609.3 Requirements - Table 14	V2V-STD-1609.3
All Scenarios	Multi-channel operations		6.1.4 - IEEE 1609.4 Requirements - Table 15	V2V-STD-1609.4		6.1.4 - IEEE 1609.4 Requirements - Table 15	V2V-STD-1609.4
All Scenarios	Identifier allocations	OBE Control Processor ECU	6.1.5 - IEEE 1609.12 Requirements - Table 16	V2V-STD-1609.12	OBE Control Processor ECU	6.1.5 - IEEE 1609.12 Requirements - Table 16	V2V-STD-1609.12
All Scenarios	Message encoding		6.1.6 - SAE J2735 Requirements - Table 17	V2V-STD-J2735		6.1.6 - SAE J2735 Requirements - Table 17	V2V-STD-J2735